M.Sc. Engg. (CSE) Thesis

Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models

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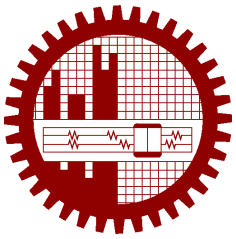

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Candidate's Declaration

I, do, hereby, certify that the work presented in this thesis, titled, "Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models", is the outcome of the investigation and research carried out by me under the supervision of Dr. Sadia Sharmin, Associate Professor, Department of CSE, BUET.

I also declare that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma or other qualifications.

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The thesis titled "**Agentic Graph Reasoning: Autonomous Knowledge Graph Navigation for Fact Verification and Hallucination Mitigation in Large Language Models**", submitted by Md. Sakif Khan, Student ID 0421052099, Session April 2021, to the Department of Computer Science and Engineering, Bangladesh University of Engineering and Technology, has been accepted as satisfactory in partial fulfilment of the requirements for the degree of Master of Science in Computer Science and Engineering and approved as to its style and contents on August 29, 2026.

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Abstract

Large language models answer factual questions fluently. But, they tend to sound just as fluent when they don't hold the facts. To fix this, researchers introduced Retrieval-Augmented Generation (RAG), which connects the LLM to an external data source. This grounds the model's answers in retrieved text. But, the retrieval happens once and up front, before any reasoning starts. So, a question that spans several facts has to be answered from whatever that first query brought back. Systems that navigate a knowledge graph do better by interleaving retrieval with reasoning. Even they, however, send out whatever their final generation call produces, and nothing checks the answer's claims before it reaches the user.

To close that gap we propose Agentic Graph Reasoning (AGR): a knowledge-graph question-answering framework. Here, an agent plans sub-objectives and walks the graph through a constrained, deterministic tool API. Its distinguishing component is a Structural Verification Layer. Before the answer is emitted, it splits the draft into atomic claims and checks each one against the triples the agent actually traversed. Claims it cannot ground are re-explored or withdrawn. So, AGR is more likely to hedge than to assert. And the claims its traversal grounds come back paired with the triples that support them.

We evaluate AGR on a Freebase-derived environment of 2.59 million entities and 8.31 million triples. We test it on 400 certified questions from each of WebQSP and ComplexWebQuestions. We compare it against four baselines: parametric answering, vector retrieval, GraphRAG, and an agentic navigator. All five systems share one frozen backbone and the same call budget. So, any difference between them comes from architecture, not model capacity.

AGR reaches a Hits@1 of 0.755 on WebQSP and 0.522 on ComplexWebQuestions. It beats every baseline on both datasets, at half the language-model calls. This includes the agentic navigator, which actually leads AGR on the questions its own budget lets it finish. But that budget runs out on 29% and 44% of questions on the two datasets; AGR's own budget never does. AGR also never asserts an entity it cannot ground, across $1{,}709$ assertions, against 22.1% of the $1{,}001$ the parametric control asserts. And on ComplexWebQuestions, AGR is the only system whose accuracy rises with hop count; every other system ends up below where it started.

A four-condition ablation, with paired significance testing, finds an effect in only one component. The other three — including claim verification, the component this work sets out to contribute — show no detectable accuracy effect. And the one effect it does find runs the opposite way from what we expected: removing the planner *improves* WebQSP accuracy by 0.083 F1 ($p = 0.006$) and cuts token use by 31%, while it trends the other way on ComplexWebQuestions. So, decomposition

should be gated on question structure, not applied unconditionally. We read all 259 remaining failures by hand. The characteristic error is not invention; it is what we call the *echo attractor*: a real, grounded entity one hop from the answer, which any grounding check will pass because it is genuinely true, and which different systems fall into together. The same reading confirms 57 questions where the benchmark, not the system, was wrong.

Chapter 1

Introduction

Large language models answer factual questions fluently. But often enough they get them wrong. The errors are not random. They are systematic and confident. They are also notoriously hard to catch from the output alone. This is because a fabricated answer reads in exactly the same register as a correct one. The responses get worse when an answer calls for composing several facts rather than recalling a single one. This thesis takes on that gap in two steps. First, we give the model a symbolic knowledge graph to walk through. Then, before it asserts anything, we check what it is about to say.

The rest of this chapter motivates the problem and explains why the obvious remedies still fall short. From there it states the problem and the research questions, summarises our contributions, and sets out how the rest of the document is organised.

1.1 Hallucination in Large Language Models

A *hallucination* is a generated statement that is fluent and syntactically well formed but backed by no source the model can point to. Very often it is simply false. Huang et al. [1] survey the phenomenon at length. They argue that it is an intrinsic consequence of how these models are built, not an incidental defect to be patched away. That framing matters here. This is because it determines what kind of remedy is on the table at all. If hallucination were a bug, we would fix it. But it is structural. So the response has to be architectural.

1.1.1 Where Hallucination Originates: Training Data, Model, and Prompt

Hallucinations arise at three distinct points. It is worth separating them. Each calls for a very different kind of intervention.

One is the *training data*. A model trained on a corpus that carries errors, staleness, or thin coverage of a domain will reproduce those defects faithfully. Fixing them means curating the

corpus and retraining. That is something only the model's producer can do.

A second is the *model* itself: its parameterisation and its decoding procedure. What a model knows is stored diffusely across its weights, with no explicit index. So it holds no reliable internal signal that separates a fact it has memorised from a plausible continuation it is simply constructing. Intervening here means changing the architecture or the training objective. That, again, is largely out of reach.

The third is the *prompt*. The prompt is the one this thesis takes on. When the context supplied alongside a question already carries the facts needed to answer it, the model does not have to fall back on parametric recall. Structured context can override defects inherited from the training data without touching the model at all. Retrieval augmentation [2] rests on that premise, and so does this work. Our contribution is in *how* that context is assembled, and in *what gets checked* before the answer goes out.

1.1.2 Why Multi-Hop Questions Are the Hard Case

Single-fact questions are comparatively benign. "Where was Ada Lovelace born?" takes one lookup. The model either knows it or it does not.

Multi-hop questions, such as "Who directed the film that won Best Picture in 1998?", call for an ordered chain of lookups. Each step depends on the result of the last. Two things make them the genuinely hard case. The first is that errors compound: a wrong resolution at the first hop rarely surfaces downstream. The rest of the chain then proceeds confidently from a false premise. The second is that a question-level correctness check never sees the intermediate results at all. That blindness leaves a system that answers correctly by accident indistinguishable from one that answers correctly by sound reasoning.

These questions also tend to carry temporal, categorical, or ordinal *constraints* that have to be satisfied together rather than one at a time. Dropping one silently is one of the commonest ways a system gets such a question wrong instead of simply declining to answer it.

1.2 From Retrieval Augmentation to Graph-Structured Grounding

1.2.1 Vector Retrieval and Its Structural Blind Spot

Retrieval-Augmented Generation [2] attaches the model to an external corpus. The system embeds the question and retrieves passages that are semantically similar to it. It lets the model answer from those. Where the answer lives in a single retrievable passage, this works well.

Its weakness, though, is structural. Vector retrieval treats a corpus as flat chunks of text and ranks them by similarity to the question. For a multi-hop question, that signal is actively misleading. The passage holding the *intermediate* entity may bear little lexical or semantic resemblance to the original question. At that point, the system does not yet know which entity to look for. Flat retrieval also has no way to represent the relationships between entities. They survive, at best, as adjacency inside a retrieved chunk. But relationships are exactly what a compositional question has to traverse. Surveys of the intersection between graphs and language models single this out as the central limitation of flat retrieval [3].

1.2.2 Knowledge Graphs and Static GraphRAG

A *knowledge graph* makes those relationships explicit. It stores facts as typed edges between identified entities. So composing two facts becomes a traversal rather than an inference over prose. GraphRAG [4] exploits exactly this: it builds a graph over a corpus and retrieves structured context in place of raw passages. Related systems carry the idea into long-term memory [5] and temporally-qualified facts [6].

What these systems still do is retrieve *statically*: they fetch a subgraph around the linked entities in one shot and hand it to the model. Retrieval runs first and runs only once. So it cannot know what the next reasoning step will need. Take a question whose second hop depends on the answer to its first. A fixed-radius neighbourhood is then either too small to contain that answer, or so large that the relevant edges are buried among thousands of irrelevant ones.

1.2.3 Agentic Retrieval: Reasoning as Navigation

The natural response is to interleave retrieval with reasoning: let the model take one step, look at what it found, and decide where to look next. Retrieval then turns into *navigation*. The system becomes an agent operating on the graph rather than a pipeline reading from it.

Think-on-Graph [7] set the pattern with LLM-guided beam search over the graph. Plan-on-Graph [8] added sub-objective decomposition and reflection-driven backtracking. Reasoning-on-Graphs [9] generates grounded relation paths as explicit plans. Other systems add tool abstractions [10], cope with incomplete graphs [11], or generalise across structured sources [12]. Recent surveys treat this meeting of language models and knowledge graphs as a research programme in its own right [13, 14].

Agentic navigation is the right shape for the problem. This thesis adopts it. What remains open is what it still leaves unsolved.

1.3 Limitations of Existing Agentic KGQA Systems

Nothing checks what the final answer asserts. Agentic systems ground their answers in the subgraph they traverse. Our own measurements confirm they do this effectively: the entities they assert exist in the graph and are reachable along the paths the systems walked (Section 5.10). Fabrication is not the deficit. The deficit is *precision*. Every system surveyed in Chapter 2 emits whatever its final generation call produces from the traversed context. Nothing checks that each asserted entity actually answers the question asked. An entity can sit in the retrieved subgraph, have been retrieved correctly, and still be the wrong answer — an intermediate node from the reasoning chain, a container where the question asked for a member, or one of several plausible neighbours picked more or less at random. A system can equally assert a crowd of entities where the question admits only a few, inflating recall at the cost of precision. Both show up as a measurable precision gap between systems (Section 5.7.2). That gap motivates a check without being evidence that the check works. Section 5.12 is where this thesis asks whether the check proposed here is what produces it. Chain-of-Verification [15] and multi-agent debate [16] show that post-hoc self-checking reduces factual error in free-form generation. Hu et al. [17] demonstrate a self-correcting agentic graph pipeline in a clinical setting. What none provides is a claim-level check against the retrieved triples, carried out *before* the answer is emitted, inside a graph-navigating agent.

The contribution of individual agentic mechanisms is unquantified. Agentic systems are assemblies of mechanisms — decomposition, scoring, backtracking, self-correction. The literature tends to report them as wholes. When one beats a static baseline, the published work rarely establishes which mechanism produced the gain, or whether some contribute nothing at all. So later work inherits the entire assembly on faith along with its cost. Component-level ablation is the standard remedy. This literature conspicuously underuses it.

Accuracy is reported without cost. Navigation is expensive: every exploration step costs at least one model call. Agentic systems routinely burn an order of magnitude more tokens per question than a single-shot baseline. An accuracy table that leaves this out makes agentic methods look uniformly preferable. The honest picture is a trade-off. That trade-off is exactly what a practitioner needs before deciding whether to deploy such a system.

1.4 Problem Statement

We address the following problem. Given a natural-language question q and a knowledge graph $G = (E, R, T)$ with entities E, relation types R, and triples $T \subseteq E \times R \times E$, produce an answer set $A \subseteq E$ together with a supporting triple set $S \subseteq T$, such that S justifies every entity in A and contains only triples the system actually traversed in reaching A.

Three properties set this apart from standard knowledge graph question answering. The output is

a *pair*, not an answer on its own: the supporting triples are part of the contract. That inclusion makes the reasoning auditable rather than merely asserted. That support also has to be *traversed*, not just retrievable after the fact. That requirement closes off the possibility of justifying, post hoc, an answer the system reached some other way. And the system checks the justification *before* it emits the answer. So it can drop or hedge unsupported content instead of merely flagging it.

1.5 Research Questions

RQ1: does agentic navigation improve multi-hop factual accuracy? Does interleaving retrieval with reasoning over a knowledge graph yield more accurate answers than static retrieval, and does the advantage grow with the number of hops a question requires? Answering it calls for a no-retrieval control. The standard benchmarks predate current models and may be partly memorised.

RQ2: what does pre-generation verification contribute beyond graph navigation? Given a system that already navigates the graph, what does a claim-level check against the traversed triples add? This is posed as *what does it contribute* rather than *does it reduce hallucination*. The answer turns out to have several parts that a yes-or-no framing would obscure (Section 5.15.1).

RQ3: which components contribute what, at what token cost? Decomposition, backtracking, verification, and learned candidate scoring each cost model calls. Which of them measurably improve accuracy or groundedness, by how much, and at what price in tokens and latency?

1.6 Our Contribution

We present Agentic Graph Reasoning (AGR), a knowledge-graph question-answering framework built as an explicit state machine over a constrained, deterministic graph tool API. It carries a *Structural Verification Layer* that decomposes the draft answer into atomic claims and checks each one against the traversed subgraph before the system emits anything. *Structural* names what that check establishes and, in the same word, bounds it: the two graph routes ask whether an edge joins a claim's endpoints, not whether it is the edge the claim names, and not in which direction. So a claim that X is Y's mother is certified by an edge recording that X is Y's child. That is the price of admitting free-text relations. Section 4.15 reports both the reason and the exposure. Where a claim is unsupported, the design either routes to targeted re-exploration or drops the claim from the answer. AGR returns what survives paired with the triples that support it.

Both repair routes are reported at the frequency we actually measured rather than the one we intended. That frequency is low: the re-exploration route never fired on the development set. The

depth budget is already exhausted by the time the verifier is reached. And the entity filter ran on three of eighty questions (Section 4.13.4). On the vast majority of questions the layer passes through and certifies a draft at no extra model call. That is a finding about when verification can act rather than a defect in it. Section 5.15.1 treats it as one.

Four further contributions follow from evaluating that framework. **A component-level ablation** of the planner, the backtracking mechanism, the verification layer, and the learned component of candidate scoring reports accuracy, groundedness, and token cost for each with paired significance testing, closing for one architecture the attribution gap of Section 1.3. **Stratum-dependent decomposition** is what that ablation finds: the literature treats question decomposition as straightforwardly beneficial. Removing the planner *improves* accuracy on the shallower benchmark while trending toward harming it on the deeper one. The asymmetry, not the pooled average, is the result. Only the improving arm reaches significance ($p = 0.006$). The harming arm does not ($p = 0.088$). So it is reported as a direction rather than an effect. **The echo attractor** names a recurring failure in which several independent systems converge on the same wrong answer — typically the question's own topic entity, an intermediate node from the reasoning chain, or a coarser-granularity container. No evaluation treating those systems as independent can see that convergence. It accounts for 13 of the 259 failures read in Section 5.14.5, though the contribution is the named mechanism and what it means for consensus-based evaluation rather than the frequency. And **quantified benchmark defect rates** for WebQSP and CWQ come from a consensus pre-pass followed by per-item adjudication, separating the two kinds of label error by their opposite evidence signatures and reporting results both with and without the affected items.

The evaluation protocol itself fixed four decisions before any test data was touched — the scoring hyperparameters, the sample size, the baseline-certification diagnostic, and the judge-agreement bar — and reports how each turned out, including the one narrowly missed.

1.7 Scope and Delimitations

Our knowledge environment is a static, curated subset of Freebase [18], specifically the union of per-question subgraphs, each pre-extracted so as to contain its own question's answer. That makes it a friendlier place to search than the full knowledge base. It also puts answer reachability above 97% by construction (Section 3.3). The absolute accuracies we report are therefore not comparable to evaluations over full Freebase. The comparisons between systems are comparable, though, because every system navigates the very same environment. Section 5.15.1 states the limit in full. We do not take on graph construction from unstructured text, knowledge-base completion, or the temporal evolution of facts. And we draw only English factoid questions, from WebQSP [19] and ComplexWebQuestions [20]. Long-form generation and conversational settings are outside our scope.

All systems share one backbone model. We hold it constant so that none of the differences we report can be credited to model capability rather than to architecture. What we establish, then, is how these architectures compare on a single backbone, not how they scale across backbones. Finally, AGR is a research prototype that we evaluate offline. We treat latency as a cost measure rather than as evidence of production readiness.

How to read the title. This thesis is submitted under the title registered with CASR at the proposal stage. That title is fixed at registration and is not revised here. Three of its terms are broader than what the chapters below actually establish. They are better read against the findings than taken as claims in their own right. *Hallucination mitigation* does occur. But it is not this work's contribution: Section 5.10 attributes the elimination of structural hallucination to graph navigation, with the agentic baseline reaching the same zero without any verification layer at all. *Autonomous* names a design this thesis argues against. Under the control pattern of Section 4.1, the program orchestrates the loop. The model is consulted only at constrained decision points. And *fact verification* is narrower here than the phrase suggests: the check establishes structural adjacency rather than the relation a claim asserts. It never reaches answer relevance (Section 4.15). Put in the terms the measurements actually support, what follows is a claim-level structural check placed before emission, together with a component-level attribution of the mechanisms around it.

1.8 Thesis Organization

Chapter 2 establishes the background the technical chapters assume. It also surveys prior work: knowledge graphs and their Freebase-specific idiosyncrasies, knowledge graph question answering and its evaluation conventions, language models as reasoning engines, retrieval augmentation, graph search, and the statistical tools used throughout. It ends with a comparison of agentic knowledge graph question answering systems that pinpoints the gap this thesis addresses and justifies the choice of baselines.

Chapter 3 describes how the knowledge environment is built and validated, including the answer-reachability gate that bounds the accuracy attainable inside it. Chapter 4 specifies the AGR framework — the tool API, the state machine, the planner, the scoring function, the backtracking policy, and the budgets that guarantee termination. Section 4.10 treats the Structural Verification Layer in detail. That is the component this work set out to contribute. It is also, as it happens, the one whose measured effect turns out to be the narrowest of the three reported in Section 5.15.1. Chapter 5 reports the evaluation in three parts: the experimental design fixed in advance (Section 5.1), the measurements and the answers to the three research questions (Section 5.7), and a hand reading of every failure that remained (Section 5.13). Chapter 6 summarises the contributions, states the limitations, and points to future work.

Chapter 2

Background and Related Work

This chapter establishes the concepts and notation the technical chapters assume. It also surveys the work AGR builds on and competes with. It is deliberately compact. Only material actually used later is included. Standard results are stated rather than derived. The background comes first: knowledge graphs, question answering over them, language models as reasoning engines, retrieval augmentation, and search. The survey follows, from static graph-augmented generation, through path-retrieval and agentic exploration, to verification and factuality measurement. It closes with a structured comparison of the seven closest systems. That comparison serves two purposes. It locates the gap this thesis addresses. And it justifies the choice of baselines in Section 5.4.

2.1 Knowledge Graphs

2.1.1 Triples, Entities, and Relations

A *knowledge graph* $G = (E, R, T)$ consists of a set of entities E, a set of relation types R, and a set of triples $T \subseteq E \times R \times E$. A triple (h, r, t) asserts that relation r holds from head entity h to tail entity t — for example, (Ada Lovelace, place_of_birth, London). Relations are directed and typed. Both directions carry meaning: the inverse of place_of_birth is a legitimate but distinct relation.

Two properties matter for what follows. Triples are *atomic*. Each asserts exactly one fact. So a claim can be checked against the graph by testing for the existence of specific edges. They are also *composable*: a chain $(e_0, r_1, e_1), (e_1, r_2, e_2), \ldots$ expresses a multi-step relationship that no single triple states. That composability is what makes a knowledge graph the natural substrate for multi-hop questions.

2.1.2 Freebase: MIDs, Schema, and Mediator Nodes

Freebase [18] is a large, collaboratively-built knowledge base, frozen since 2016. It is also the substrate for the standard knowledge graph question answering benchmarks. Three of its characteristics affect any system built on it. Entities are identified by opaque *machine identifiers* (MIDs) such as m.02mjmr, with human-readable names attached as a separate property. Names are neither unique nor stable. So entity identity and entity surface form must be handled separately. Relations are namespaced paths such as people.person.place_of_birth, encoding domain, type, and property. They are drawn from a vocabulary of tens of thousands that includes many carrying no factual content, such as internal type and key predicates.

Most consequentially, Freebase represents n-ary facts through *mediator* or *compound value type* (CVT) nodes. A fact with more than two participants — an actor playing a particular character in a particular film — is not a single edge but an anonymous intermediate node linking the participants. So a relationship a user would describe as one hop is two edges in the graph. Any system reporting hop counts over Freebase must state how it treats these nodes. The choice changes every depth measurement. Systems that ignore them lose the facts they encode [11].

2.1.3 Property Graphs, Neo4j, and Cypher

The triple model above is the RDF view. The alternative is the *property graph* model, in which nodes and edges are both typed and may carry arbitrary key–value properties. Neo4j [21] implements it. Cypher [22] queries it, with a MATCH clause that expresses graph patterns directly. A triple store maps onto it naturally — entities become nodes carrying their MID and name, relations become typed edges. Traversal patterns that would require recursive joins in a relational schema are expressed as literal path patterns instead. That is the practical reason for choosing a graph store here.

2.1.4 Knowledge Graph Question Answering

Multi-Hop Questions and Constraint Satisfaction Knowledge graph question answering (KGQA) takes a natural-language question and a graph and returns the entity or entities that answer it. A question is k-*hop* if answering it requires traversing k edges from the entities named in the question — its *topic entities* — to the answer. Beyond hop count, questions vary in *constraints*: conditions that filter candidates rather than extending the chain. “Which of Tolstoy’s novels was published after 1875?” is one hop plus a temporal constraint. Constraints must be satisfied jointly with the chain. So a system that traverses correctly while silently dropping one returns a superset of the answer — a failure hop-count analysis alone does not expose.

Two families of approach have historically divided this field. *Semantic parsing* methods translate the question into a formal query and execute it. That approach is exact and interpretable, but it

requires the parse to be exactly right [19, 23]. *Information retrieval* methods instead retrieve a relevant subgraph and rank candidate answers within it, degrading more gracefully at the cost of transparency. The agentic systems considered here are closer to the second family. But they recover much of the first family's transparency by logging the traversal itself.

Evaluation Conventions: Hits@1 and F1 Two metrics are conventional. *Hits@1* scores a question 1 if any entity the system asserts is in the gold answer set, and 0 otherwise. It is a per-question binary correctness measure. It is also the headline number in most of this literature. The name is historical and slightly misleading. It was defined for systems that emitted a *ranked* list, where only the top entry counted. It has since been carried over to systems that emit an unordered set, where it becomes the set-intersection test just stated. This thesis uses the set-intersection form throughout, because that is what the systems it compares against report. Section 5.5 fixes the string-matching rule exactly.

Because many questions admit multiple correct answers, Hits@1 alone rewards a system that names one correct entity and ignores the rest. *F1* therefore complements it, computed per question between the predicted answer set A and the gold set A^*:

$$P = \frac{|A \cap A^*|}{|A|}, \qquad R = \frac{|A \cap A^*|}{|A^*|}, \qquad F_1 = \frac{2PR}{P+R},$$

and then averaged over questions. The distinction between P and R is load-bearing in this thesis: a system that asserts many entities can achieve high recall while its precision falls. Section 1.3 argues that precisely this failure goes unmeasured when only Hits@1 is reported.

2.2 Large Language Models as Reasoning Engines

2.2.1 In-Context Learning, Chain-of-Thought, and the ReAct Loop

Sufficiently large language models perform tasks specified entirely in their prompt, without gradient updates — *in-context learning* [24]. Prompting the model to produce intermediate reasoning steps before its final answer, *chain-of-thought* prompting [25], substantially improves performance on multi-step problems. Both are used here: the agent's planner, scorer, evaluator, and verifier are prompted components of a single frozen model. No component is fine-tuned. That matters for comparability, since several systems surveyed in Section 2.5 are fine-tuned and therefore not directly comparable to prompting methods.

A language model can also be given *tools*: named operations it may invoke, whose results re-enter its context [26]. The *ReAct* pattern [27] interleaves reasoning and action in a loop — the model reasons about what to do, acts, observes the result, and reasons again. That loop is the control structure underlying agentic graph navigation. Where the tools are graph operations,

each iteration extends a traversal. The sequence of tool calls is a complete record of how the answer was reached.

2.2.2 A Working Definition of Hallucination for This Thesis

"Hallucination" is used loosely in the literature. This thesis adopts a narrow, operational definition tied to the knowledge environment: an asserted entity is *ungrounded* if it does not exist in the graph, or exists but is not reachable from the question's topic entities within the traversal budget. A claim is *unsupported* if the traversed triples do not entail it.

Two consequences of this definition should be stated plainly. The definition is relative to the environment: an entity absent from the graph is ungrounded even if true in the world. And groundedness is not correctness — an answer can be perfectly grounded, every entity real and reachable, and still be the wrong answer to the question. Section 5.7 shows that this gap is not hypothetical. Section 1.3 is built on it.

2.2.3 Retrieval-Augmented Generation

Dense retrieval encodes queries and documents into a shared vector space and retrieves by nearest neighbour, capturing semantic similarity where lexical matching fails [28]. Sentence-level encoders such as Sentence-BERT [29] produce embeddings suitable for direct cosine comparison. In this thesis embeddings serve two bounded purposes — linking question phrases to graph entities, and pre-ranking relation candidates during exploration. They never establish that a fact is true. That role belongs exclusively to the symbolic triples. Graph-based retrieval replaces flat passages with structured context [4, 5]. The retrieved unit is a subgraph rather than a document. That preserves relationships explicitly. The distinction that matters here is not graph versus vector but *static versus interleaved*: whether retrieval happens once before reasoning begins, or repeatedly as reasoning proceeds.

2.2.4 Search over Graphs

Best-first search maintains a frontier of candidate nodes ordered by a heuristic estimate of promise and expands the most promising one first [30, 31]. *Beam search* bounds the frontier to the b highest-scoring candidates at each depth, trading completeness for a fixed memory and computation budget. With an expensive scoring function — here, a language model call — the beam width is also the principal cost control. Bounded search commits to choices that may prove wrong. *Backtracking* recovers by abandoning the current branch and resuming from a previously unexplored alternative. That recovery requires maintaining a stack of such alternatives and a trigger condition for when to pop it. Both are specified in Section 4.5.

Relation to Monte Carlo Tree Search: A Terminological Clarification The original proposal for this work described its navigation strategy as "inspired by Monte Carlo Tree Search". That description is withdrawn here. The reason is worth stating explicitly. Monte Carlo Tree Search [32, 33] is defined by four phases: selection, expansion, *simulation* (a rollout to a terminal state), and *backpropagation* of the rollout's value to the ancestors of the expanded node. The two phases that give MCTS its statistical character are simulation and backpropagation. Together, they let node values be estimated from sampled outcomes rather than from a fixed heuristic.

The method used in this thesis has neither. There are no rollouts. No value is propagated back up the search tree: each candidate is scored once, by a fixed function combining embedding similarity with a language-model judgement. That score is never revised in light of what is found later. What the method is, is *guided best-first search with a bounded beam and backtracking*. It is described that way throughout. The contrast is not hypothetical. Genuine tree search has been applied to this exact task: KBQA-o1 [34] drives an agentic knowledge base question answering process with full MCTS — policy and reward models, rollouts, and value backpropagation — over Freebase. Reasoning with Trees [35] likewise builds an explicit search tree. That those systems exist is precisely why the terminology matters here: describing a single-pass scoring heuristic as MCTS-inspired would invite comparison with machinery this framework does not implement.

2.2.5 Statistical Tools Used in This Thesis

Three tools recur, each chosen for a specific property of the data.

Bootstrap confidence intervals [36] quantify sampling uncertainty by resampling the evaluated questions with replacement and recomputing the metric. The 2.5th and 97.5th percentiles of the resulting distribution give a 95% interval. That is the *percentile* method, used here in preference to bias-corrected variants. The intervals are read as a scale on the reported difference rather than as a hypothesis test. They also require no distributional assumption about metrics such as per-question F1.

McNemar's test [37] assesses whether two systems differ in accuracy when both are evaluated on the *same* questions. It considers only the discordant pairs — questions one system answers correctly and the other does not — and tests whether the split between the two discordant counts departs from chance. Paired testing is essential here, because all systems are run on identical question samples. An unpaired test would discard the pairing and lose power.

Cohen's κ [38] measures agreement between two annotators on a categorical judgement, correcting for agreement expected by chance:

$$\kappa = \frac{p_o - p_e}{1 - p_e},$$

where p_o is observed agreement and p_e is the agreement expected if both annotators labelled independently at their observed marginal rates. It is used to validate the automatic groundedness judge against human annotation. Conventional interpretation follows Landis and Koch [39], on whose scale values between 0.61 and 0.80 denote substantial agreement.

2.3 Static Graph-Augmented Generation

2.3.1 GraphRAG and Query-Focused Summarization

GraphRAG [4] constructs a knowledge graph from a text corpus using a language model. It partitions the graph into communities and pre-computes summaries at each level. So a query can be answered from structured context rather than retrieved passages. Its strength is global questions — those whose answer depends on the corpus as a whole rather than any single passage. Flat retrieval handles these poorly.

The relevant limitation is that retrieval remains a single step preceding generation. The context is assembled before the model has begun reasoning. So it cannot be conditioned on intermediate findings. GraphRAG also builds its graph by language-model extraction. This is appropriate for unstructured corpora, but it introduces extraction error into the reference itself — a consideration that shapes the environment design in Section 3.1.

2.3.2 HippoRAG and Memory-Structured Retrieval

HippoRAG [5] draws on the hippocampal indexing theory of human memory. It combines a knowledge graph with a personalised PageRank retrieval step, so that a single query can integrate evidence distributed across many passages. It substantially improves multi-hop retrieval over flat baselines at low cost.

Its retrieval is nonetheless a single graph-propagation step, not an agent-directed traversal: the propagation is fixed. The model does not choose where to look next. Temporal extensions such as T-GRAG [6] address a different axis — conflicting and redundant facts across time. This is orthogonal to the navigation question. So it is noted here for completeness rather than compared against.

2.3.3 Path-Retrieval and Semantic-Parsing KGQA

Reasoning-on-Graphs Reasoning-on-Graphs (RoG) [9] adopts a planning–retrieval–reasoning pipeline. A fine-tuned model generates relation paths grounded in the graph schema. Those paths are used to retrieve matching instantiations. The model then reasons over the retrieved

paths. Because answers are produced from retrieved paths, RoG is faithful by construction and can present the path as an explanation.

Two properties distinguish it from the present work. It is *fine-tuned*. This places its reported numbers outside direct comparison with prompting methods. And its plan is generated in one shot before retrieval: if the generated relation path does not instantiate in the graph, there is no mechanism to revise it mid-course. The pre-generation verification question does not arise here, because grounding is guaranteed structurally rather than checked.

RoG is nonetheless the one surveyed system this thesis shares a data distribution with: the knowledge environment of Section 3.1.3 is built from its published subgraphs. So the comparison is taken up directly, figures and all, in Section 5.8.1.

StructGPT and KG-Agent StructGPT [12] provides a general interface for reasoning over structured data — knowledge graphs, tables, and databases — through an invoke–linearise–generate loop. Specialised interfaces extract evidence that is linearised into text for the model. StructGPT is deliberately general rather than graph-specific. It performs no open-ended graph search.

KG-Agent [10] is closer to the agentic setting: a multifunctional toolbox, a knowledge-graph executor, and a dynamic memory, driven by an instruction-tuned model that autonomously selects tools across iterations. KG-Agent demonstrates that a small fine-tuned model with good tool abstractions is competitive with much larger prompted ones. It plans implicitly through tool selection rather than committing to explicit sub-objectives. It provides no backtracking mechanism. And it performs no verification of the final answer.

2.4 Agentic Graph Exploration

2.4.1 Think-on-Graph

Think-on-Graph (ToG) [7] established the template for training-free agentic KGQA. The model performs iterative beam search over the graph. At each step it is shown candidate relations and entities and prunes them itself. Beam width and maximum depth are both set to three. Exploration terminates when the model judges the retrieved paths sufficient. At that point it answers directly.

ToG is the most directly comparable prior system to AGR — training-free, prompting-based, operating on Freebase, evaluated on the same benchmarks. It is therefore the agentic baseline in Section 5.4. Its limitations are instructive. There is no explicit decomposition. So multi-constraint questions are handled implicitly. Low-scoring branches are pruned within the beam. But there is no mechanism to return to an abandoned node once the beam has moved on. And

the transition from exploration to answering is a single model judgement: nothing checks the answer that follows.

2.4.2 Plan-on-Graph

Plan-on-Graph (PoG) [8] is the closest system to AGR in ambition. It decomposes the question into sub-objectives, including constraint conditions. It explores adaptively, with a breadth that varies by sub-objective. And it adds Guidance, Memory, and Reflection mechanisms. Reflection decides both whether to self-correct and which entity to backtrack to. That dual role makes PoG the one surveyed system with genuine backtracking.

PoG's self-correction operates on the *search process*: it reconsiders where to explore. It does not decompose the draft answer into claims and check each against the retrieved triples before responding. The distinction is the gap this thesis occupies. It is stated precisely in Section 2.5.1.

A naming hazard must be recorded here. A second, unrelated system is also abbreviated PoG: Paths-over-Graph [40]. It reports considerably higher figures. Survey tables that cite "PoG" without disambiguation are a known source of contaminated comparisons. This thesis therefore always writes Plan-on-Graph or Paths-over-Graph in full.

2.4.3 Generate-on-Graph and Reasoning with Trees

Generate-on-Graph (GoG) [11] targets the incomplete-graph setting through a thinking–searching–generating loop: when the graph lacks a required fact, the model generates candidate triples from its parametric knowledge and continues. It also expands subgraphs dynamically to handle the mediator nodes described in Section 2.1. ToG tends to skip these. The trade-off is directly opposed to this thesis's: GoG deliberately admits model-generated facts into the evidence, whereas AGR admits only graph triples.

Reasoning with Trees [35] applies explicit tree search to knowledge graph question answering. It is cited in Section 2.2.4 as an example of genuine tree-search machinery. That machinery contrasts with the guided best-first search implemented here.

2.4.4 KBQA-o1 and Genuine Tree Search

KBQA-o1 [34] is the closest system to the search strategy the present work's proposal originally described, and therefore the most important one to distinguish from it. It performs agentic knowledge base question answering with full Monte Carlo Tree Search — a policy model proposing actions, a reward model scoring them, and rollouts whose values propagate back through the tree — wrapped around a ReAct-style agent that generates logical forms step by step

against a Freebase environment. The exploration additionally generates annotations used for incremental fine-tuning. So the search improves the model that drives it.

Two aspects bear on this thesis. First, KBQA-o1 is what the withdrawn "MCTS-inspired" description of Section 2.2.4 would have claimed: it has the rollouts and value backpropagation that AGR does not. Naming the difference honestly is what keeps the two separable.

Second, its ablation is directly relevant to the methodological argument of Section 2.5.1. Removing MCTS from KBQA-o1 while leaving every other component intact drops overall GrailQA F1 from 78.5 to 48.5: a thirty-point attribution to a single mechanism, obtained precisely because the authors ablated components rather than reporting the assembly as a whole. That is the same instrument this thesis applies in Section 5.12. The agreement between the two is evidence that the instrument is informative.

KBQA-o1 is not a baseline here. It is fine-tuned rather than prompted. It evaluates under a low-resource few-shot protocol. And it does not report ComplexWebQuestions. So no controlled comparison against it is available in this environment.

2.4.5 Verification and Self-Correction in Language Models

Chain-of-Verification Chain-of-Verification (CoVe) [15] has the model draft a response, plan verification questions about it, answer those independently, and regenerate a corrected response. It establishes the core premise of this thesis's verification layer: that decomposing a draft and checking its parts reduces factual error. But it performs the checks against the model's own knowledge rather than an external symbolic source. Multi-agent debate [16] achieves a related effect by having several model instances critique one another.

This distinction matters given evidence that language models cannot reliably self-correct reasoning without external feedback [41]. A verification step whose ground truth is the same model that produced the error is limited by construction. AGR's verification is grounded in traversed triples. That grounding makes the check external to the generator.

Self-Correcting Graph RAG A self-correcting agentic graph RAG system for clinical decision support in hepatology [17] demonstrates the combination of agentic retrieval with a correction loop in a high-stakes domain. It substantiates the practical motivation for verification, though in a domain-specific setting rather than on the open-domain benchmarks used here. Reflexion-style verbal self-improvement [42] provides the general agentic pattern of which it is a graph-specific instance.

2.4.6 Measuring Factuality

Claim-Decomposition Metrics FActScore [43] evaluates long-form generation by decomposing text into atomic facts and verifying each against a reference source. It reports the supported proportion. This decomposition-then-verify protocol is the basis for the groundedness metric in Section 5.5.3, with the knowledge graph serving as the reference. The essential methodological requirement, adopted here, is that the verifying judge must be independent of the system that produced the text.

Factuality Benchmarks and Their Limits FactBench [44] provides a dynamic benchmark for in-the-wild factuality evaluation. The original proposal for this thesis intended to use it both as a source for the knowledge environment and as the hallucination evaluator. That plan was abandoned. The reason is methodological rather than practical: if the evaluator's facts are contained in the graph the system retrieves from, a low hallucination rate follows by construction rather than by merit. The evaluation in Section 5.1 therefore measures claim grounding against the graph with a human-validated judge. This keeps the evaluator's ground truth outside the system's evidence.

2.5 Comparative Summary of Agentic KGQA Systems

Table 2.1 compares the seven closest systems on the mechanisms relevant to this thesis. Table 2.2 records their evaluation settings and published accuracy.

Three caveats attach to Table 2.2 and are carried into every comparison in this thesis.

First, the figures are drawn from the original publications and reflect different backbones. RoG and KG-Agent are fine-tuned. Their numbers are not comparable with prompting methods at all. Among the prompting methods, the backbone accounts for much of the spread: the same methods drop sharply when run on smaller open-weight models. That sensitivity is the direct motivation for holding the backbone constant across every system in Section 5.1. The literature does not hold it constant. So a cross-paper table cannot settle which architecture is better.

Second, evaluation protocols differ. Some papers evaluate on sampled test subsets. And answer-matching conventions are not uniform across implementations.

Third, and most concretely, two papers disagree about a third. Plan-on-Graph's own comparison table measures Think-on-Graph at 57.1 and 67.6 on CWQ, while Think-on-Graph reports 58.9 and 69.5 for itself. Neither is wrong: one is an original result and the other a reimplementation. This thesis is in exactly that situation with respect to its own agentic baseline (Section 5.4.3). The gap is 1.8 and 1.9 points, the same order as many improvements claimed in this literature. A cross-paper table cannot distinguish a real advance from a reproduction difference of that size. That is why no claim in this thesis rests on one.

Table 2.1: Mechanism comparison of the seven closest prior systems. No system performs claim-level verification of the final answer against retrieved triples before responding.

System	Exploration strategy	Decomposes query?	Backtracks?	Verifies before answering?
ToG [7]	Training-free LLM-guided iterative beam search; beam width and depth both 3	No	No — prunes within beam, cannot return	No
PoG [8]	Self-correcting adaptive planning with Guidance, Memory, Reflection	Yes, into sub-objectives with constraints	Yes — Reflection selects the entity to return to	Partial — corrects the search, not the answer
RoG [9]	Planning–retrieval–reasoning over generated relation paths	Partial — relation-path plans	No	Faithful by construction; no explicit check
KG-Agent [10]	Toolbox with KG executor and dynamic memory; autonomous tool selection	Implicit, via tool calls	No	No
GoG [11]	Thinking–searching–generating; generates triples when the KG is incomplete	Yes	No — expands subgraph instead	No
StructGPT [12]	Iterative reading-then-reasoning via invoke–linearise–generate interfaces	No	No	No
KBQA-o1 [34]	Monte Carlo Tree Search with policy and reward models over a ReAct agent; fine-tuned	Implicit, stepwise logical form	Via MCTS tree revisiting	No
AGR (this work)	Guided best-first beam search with explicit backtracking	Yes, into ordered sub-objectives	Yes — threshold, dead end, or evaluator veto	**Yes — claim-level, pre-generation**

2.5.1 Positioning of This Work

Table 2.1 makes the gap explicit. Decomposition is well established: PoG and GoG both do it. Backtracking exists, in PoG. Faithful grounding by construction exists, in RoG. But in the final column, *no surveyed system decomposes its draft answer into atomic claims and checks each against the retrieved triples before responding*. PoG's Reflection corrects the search while it is running. RoG's answers are grounded because of how they are produced, not because anything examined them. The slot is open. Section 1.6 states what this thesis puts in it: the Structural Verification Layer (Section 4.10) together with a measurement of what it buys, a component ablation pricing each agentic mechanism in accuracy and tokens (Section 5.12), and the controlled agentic comparison itself (Section 5.15.1).

Two features of that positioning belong here rather than there. Both are claims about the literature. The first is that these mechanisms are typically reported as bundles. So which of them earns

Table 2.2: Reported evaluation settings and Hits@1 as published. These figures are *not* directly comparable to one another: backbones differ, some papers evaluate on sampled subsets, and the backbone accounts for much of the variation.

System	Backbone	Datasets	WebQSP	CWQ
ToG	GPT-3.5	CWQ, WebQSP, GrailQA [45], QALD10-en, SimpleQuestions, WebQuestions, T-REx, Zero-Shot RE, Creak	76.2	58.9
	GPT-4		82.6	69.5
PoG	GPT-3.5	CWQ, WebQSP, GrailQA	82.0	63.2
	GPT-4		87.3	75.0
RoG	Fine-tuned LLaMA2-7B	WebQSP, CWQ	85.7	62.6
KG-Agent	Fine-tuned 7B	WebQSP, CWQ, plus other KBs	83.3	72.2
GoG	GPT-3.5	WebQSP, CWQ (complete and incomplete KG)	78.7	55.7
	GPT-4		84.4	75.2
StructGPT	GPT-3.5	WebQSP, MetaQA, TableQA, Text-to-SQL	72.6	—
KBQA-o1	Llama-3.1-8B	GrailQA, WebQSP, GraphQ; 100-shot low-resource protocol	59.8†	—
	Llama-3.3-70B		67.0†	—

† KBQA-o1 reports **F1**, not Hits@1, and evaluates under a 100-shot low-resource protocol. These cells are therefore *not* comparable with the rest of the column and are shown only to situate the system. It does not evaluate CWQ.

Every figure is the one its own authors published. That this is not the same as a comparison is visible inside the table: PoG's paper reproduces Think-on-Graph and measures it at 57.1 and 67.6 on CWQ, against the 58.9 and 69.5 Think-on-Graph reports for itself. Two published papers disagree by 1.8 and 1.9 points on the same system and benchmark. That gap is the order of many claimed improvements in this literature. It is also the reason Section 5.1 reimplements its baseline rather than citing a number.

its cost is not established. That is the gap the ablation fills, and where the planner's stratum dependence appears. The second is that the thesis does not depend on beating prior systems on raw Hits@1. The comparison that matters is against the same architecture with the verification layer removed, on the same graph, with the same backbone. That is why Section 5.12 rather than Table 2.2 carries the argument. Where AGR is compared against a prior system, that comparison is run under controlled conditions in this thesis's own environment rather than quoted across papers.

Chapter 3

The Knowledge Environment

Everything the agent can know, it knows from the graph. The environment therefore bounds the accuracy any system in this thesis can attain. A result reported without that bound is uninterpretable. This chapter describes how the environment was built, what it contains, and — equally important — what it cannot express.

3.1 Design Goals and Source Data

3.1.1 Decoupling the Graph Source from the Question Sets

The original proposal for this work described ingesting WebQSP and FactBench into a graph database. That description conflated two distinct roles. The correction is worth stating. It shaped everything downstream.

WebQSP [19] is not a knowledge graph. It is a set of questions annotated with semantic parses over Freebase [18]. The graph is Freebase. WebQSP supplies questions and gold answers against it. The two roles are kept strictly separate here: Freebase-derived triples constitute the environment, while WebQSP and ComplexWebQuestions [20] supply questions and gold answers and contribute no facts to it.

FactBench [44] was removed entirely. It had been intended both as a source of graph content and as the hallucination evaluator. That dual role is circular: if the evaluator's facts are inside the graph the system retrieves from, a low hallucination rate follows by construction. Groundedness is instead measured against the graph by an independent judge, as described in Section 5.5.5.

3.1.2 Why a Curated Subgraph Rather Than LLM-Extracted Triples

The proposal also specified a language-model entity-relationship extractor to build the graph. That approach is appropriate when the source is unstructured text. It is what GraphRAG does [4].

Using an extractor here would be self-defeating. The graph is the reference against which model output is verified. Building that reference with a language model would introduce the very error class the thesis measures. And no claim about hallucination reduction would survive the observation that the ground truth was itself generated.

The environment is therefore assembled exclusively from curated Freebase triples. Language models are used to *navigate* the graph and to *judge* answers, never to populate it.

3.1.3 Source Datasets

Every dataset this thesis consumes is public, standard, and third-party. None of it was collected, authored, or annotated here. Three artifacts are used. The subsections below describe each in turn: a Freebase snapshot as the knowledge source, and the WebQSP and ComplexWebQuestions question sets as the benchmarks. All three arrive through one distribution: the per-question subgraphs published with Reasoning-on-Graphs [9], released as rmanluo/RoG-webqsp and rmanluo/RoG-cwq. So the graph and the questions evaluated over it are guaranteed to be mutually consistent. And any system that has been measured on those benchmarks has been measured on these questions.

What *is* built here is the knowledge environment derived from them: the union graph of Section 3.2, its property-graph store, and its vector index. That derivation is deterministic and scripted. The environment it produces is recorded in Table 3.1. The distinction is worth stating plainly. The two are easily conflated: the datasets are standard and unmodified, the environment assembled from them is this thesis's own, and Section 3.3 measures what the assembly costs. Which questions are actually scored, and how they are sampled from these datasets, is a separate matter specified in Section 5.2.

The Freebase Snapshot Freebase has been frozen since 2016. That freeze makes it stable and reproducible. But it also dates it: a property that cuts both ways and motivates the no-retrieval control in Section 5.4. Two routes to a working Freebase environment were considered.

The first is a full SPARQL endpoint: the community-standard preprocessed Virtuoso image. It ships a database of over 53 GB. Its maintainers recommend on the order of 100 GB of RAM for acceptable query latency. Its advantage is a complete and realistic search space, including the hub entities of very high degree that make naive traversal explode.

The second is the per-question subgraph distribution published alongside Reasoning-on-Graphs [9], in which each record pairs a question with its topic entities, gold answers, and the triples of a multi-hop neighbourhood around those topic entities.

The second route was chosen, for a reason that should be recorded plainly: the hardware for the first was not available. The consequence is a scoped environment rather than all of Freebase. Section 3.3 quantifies what that costs.

WebQSP and ComplexWebQuestions as Question Sets WebQSP [19] contains questions whose answers lie predominantly one or two hops from the topic entity. WebQSP descends from WebQuestions [23]. ComplexWebQuestions [20] is constructed by composing WebQSP queries into deeper chains with conjunctions, superlatives, and comparatives. CWQ is the genuine multi-hop benchmark of the two. Reporting both is deliberate: WebQSP establishes comparability with prior work, and CWQ is where the multi-hop claim is actually tested.

3.2 Graph Construction

3.2.1 Union of Per-Question Subgraphs from the RoG Distribution

The environment is the *union* of every per-question subgraph in the RoG distribution, across both datasets and all three splits. Each record's triple list is read, whitespace-normalised, and inserted into a global deduplicated set. Entity and relation strings are interned to integers, so that the union and its adjacency structure fit in memory.

Unioning is not incidental. A per-question subgraph in isolation is a near-oracle: it was extracted to contain the answer, so an agent searching it faces almost no distractors. Merging every question's neighbourhood into one global graph restores distractor mass, since the agent exploring one question's topic entity encounters edges contributed by unrelated questions. WebQSP contributes 3,791,302 unique triples over 1,298,304 entities. Adding CWQ brings the union to 8,309,194 triples over 2,592,892 entities.

Two consequences must be disclosed. First, the RoG subgraphs were pre-extracted so as to guarantee answer reachability. That guarantee makes the resulting environment friendlier than raw Freebase: fewer irrelevant hub explosions, and a higher answer-reachability rate than a BFS extraction from the full store would produce. That friendliness is a threat to external validity, revisited in Section 5.15.1. The mitigating argument is that *every system compared in this thesis navigates this identical environment*. Relative comparisons therefore remain sound even where absolute numbers are optimistic.

Second, the distribution's triples are keyed by human-readable entity names rather than by Freebase machine identifiers. The node identifier is therefore the name string. As a result, distinct real-world entities sharing a surface form are merged into a single node. That merging is inherent to name-keying and is also revisited in Section 5.15.1.

3.2.2 Relation Filtering and Noise Removal

An extraction from raw Freebase would require aggressive predicate filtering: administrative namespaces, type and key predicates, and above all type.type.instance, which contributes

millions of edges per type node. Degree capping would likewise be mandatory, since uncapped breadth-first expansion from a hub entity explodes at the second hop.

No such filtering was applied here. The reason is structural rather than an oversight: the RoG distribution ships subgraphs that have already been scoped around topic entities. So the administrative predicates and hub explosions that filtering exists to remove are largely absent from the source. The only transformation applied to relation names is sanitisation into Cypher-legal identifiers, described in Section 3.2.4. What survives is a vocabulary of 7,058 distinct relation types. That vocabulary is the relation search space the agent faces at every exploration step.

3.2.3 Treatment of Mediator Nodes and Its Effect on Hop Counts

Freebase reifies n-ary facts through mediator nodes, as described in Section 2.1. In the name-keyed distribution these surface distinctively: named entities appear as their names, while unnamed mediators survive as raw machine identifiers of the form m.0d05w3 or g.11b62t84jq. Nodes are flagged as mediators by matching that pattern. The flag is stored as an is_cvt property.

Mediators are *retained*, not collapsed into hyper-edges. Retaining them preserves the facts they encode — systems that skip mediators lose exactly those facts [11] — at the cost of inflating raw hop counts, since one logical hop through a mediator is two graph edges. The conversion is roughly two raw hops per logical hop.

Two conventions follow. They are not the same one. Every statement about *distance in the graph* — the reachability gate of Section 3.3, the hop strata of Section 5.2, the path-length bounds quoted anywhere in this thesis — is in *raw* hops. That is what the graph actually contains. The agent's depth budget is not: max_depth counts explorer expansions. One expansion crosses a mediator transparently. So it may consume two raw edges while costing one unit of depth. That discount is deliberate — the agent should not be charged twice for an artifact of the encoding (Section 4.2.3). But the discount also means max_depth $= 4$ does not bound the search at four raw hops. The two numbers should not be compared without the conversion.

The scale of this decision is easy to understand: **1,652,618 of the 2,592,892 nodes — 63.7% — are mediator-form nodes carrying no human-readable name.** Nearly two thirds of the environment is connective tissue rather than nameable entities. That imbalance is why the tool layer in Section 4.2 must traverse through mediators transparently rather than present them to the agent as candidate answers.

3.2.4 Graph Store Construction

Property-Graph Schema Entities become :Entity nodes carrying an integer id, the name string, and the boolean is_cvt flag. Triples become typed relationships whose type is the sanitised

Table 3.1: The knowledge environment as constructed and imported.

Property	Value
Entities (nodes)	2,592,892
Triples (relationships)	8,309,194
Node and relationship properties	16,087,870
Distinct relation types	7,058
Mediator-form nodes (is_cvt)	1,652,618 (63.7%)
Triples contributed by WebQSP subgraphs	3,791,302
Entities after WebQSP alone	1,298,304
Rows skipped during import	0
Import wall-clock time	36.4 s
Peak import memory	1.09 GiB
Neo4j version	Community 5.26.28

relation name — non-alphanumeric characters replaced by underscores, since Freebase predicates contain dots that are illegal in unquoted Cypher relationship types — with the original dotted string retained as an fb_name property.

Native relationship types were chosen over a single generic type carrying a name property. The alternative simplifies import but slows filtered expansion. Filtered expansion is the agent's hottest operation: every exploration step retrieves the relations of an entity and then the neighbours along one of them.

Integer identifiers were chosen over names as the import key. Entity names contain commas, quotation marks, and unicode, all of which invite escaping bugs and silent collisions in bulk-import CSVs. The name survives as an ordinary property and is what the entity resolver matches against.

Import used Neo4j's offline bulk importer, with the uniqueness and index constraints created afterwards. Appendix E.18 records the versions and the constraint set.

Graph Statistics Table 3.1 records the environment as built. These figures, together with the dataset versions and the hop cap, constitute the reproducibility record for the environment.

The zero skipped rows deserve emphasis. The import was deliberately configured to tolerate skips. Node and relationship counts in the database match the CSV row counts exactly. So the count-reconciliation check of Section 3.3 passes without any unexplained gap to account for.

3.2.5 The Vector Index

Entity names and the relation vocabulary are both embedded with all-MiniLM-L6-v2 [29], a 384-dimensional sentence encoder. Appendix E.17 records how the two indexes were built and checked. One property of the relation index is load-bearing for Section 4.4: relation names are

verbalised before embedding. So people.person.place_of_birth is encoded as the phrase it means rather than as a namespaced path. That verbalisation is what lets a question phrase score against a schema identifier at all.

The role of embeddings in this thesis is deliberately narrow. They link question phrases to graph entities and pre-rank relation candidates during exploration. They never establish that a fact is true — that role belongs exclusively to the symbolic triples. Every claim the verification layer certifies is certified against an edge, not against a cosine.

3.2.6 Entity Resolution and Surface-Form Linking

The Exact, Lexical, and Vector Cascade Mapping a question's surface mention to a graph node uses a three-stage cascade, each stage attempted only if the previous one fails.

Exact matching queries for a node whose name equals the normalised mention. Because the environment is name-keyed and the benchmarks supply topic entities as names, this resolves nearly everything.

Lexical matching queries the full-text index, handling typographical variation and partial names. Mentions are escaped for Lucene syntax before the query: benchmark entity names contain parentheses, colons, and slashes, all of which are Lucene operators that would otherwise crash or silently distort the query.

Vector matching encodes the mention and queries the entity vector index, handling paraphrases and misspellings that survive the first two stages.

The cascade is referred to by these names throughout, never as "tiers". This thesis reserves "tiers" for the two-tier groundedness metric of Section 5.5.3. That naming rule is about this prose, not about the code. The code does call them tiers: the resolver returns a tier field taking the values 1, 2 and 3. A reader moving between the two will meet the collision. Appendix B declares it where the field is specified, rather than leaving the disclaimer above to look like a claim about the implementation.

Threshold Selection on Development Data The lexical stage accepts its top hit only when the full-text score exceeds 1.0, falling through to vector matching otherwise. The threshold exists because the full-text index returns a ranked list for almost any input, including inputs that match nothing meaningfully. Without a floor, the lexical stage would confidently return the least-bad token overlap. The vector stage would never be reached.

Resolution was measured over every topic-entity mention in both test sets. Table 3.2 reports the outcome.

Surface forms supplied by the benchmarks therefore resolve to nodes in this environment essentially without loss. The limits of that measurement should be read alongside it: the graph

Table 3.2: Entity resolution over all topic-entity mentions in both test sets. A mention counts as resolved only when the returned node's name equals the mention exactly.

Dataset	Mentions	Exact stage	Unresolved
WebQSP	1,661	1,661 (100%)	0
CWQ	5,488	5,478 (99.8%)	10
Total	**7,149**	**7,139 (99.86%)**	**10 (0.14%)**

is keyed on the same name strings the benchmarks supply, both coming from the same RoG distribution. The measurement therefore establishes only that the *resolver* does not lose entities the environment contains — not that entity linking would be easy against an independently-built graph, and not that linking from free text would work at all. Topic entities are given rather than linked throughout (Section 5.15.1). What follows for the error analysis of Section 5.13 is the narrow version: a failure in this environment cannot be attributed to the resolver having failed to find the question's topic.

3.3 Environment Validation Gate

No experiment was run until the environment was certified. The gate has three parts of increasing strength.

3.3.1 Answer-Reachability Protocol

For every test question, two properties are computed over the union graph. *Existence* asks whether the gold answer entities appear as nodes at all. *Reachability* asks whether at least one gold answer is connected to at least one topic entity within a bounded number of hops.

Reachability is computed by breadth-first search treating edges as *undirected*. The agent's neighbour-retrieval tool looks in both directions. The gate must model the agent's actual traversal semantics. The bound is four raw hops. By Section 3.2, that bound is approximately two logical hops through mediators: enough for all of WebQSP and most of CWQ. A question whose topic entity is itself a gold answer counts as reachable at zero hops.

This number is the **accuracy ceiling**: if a gold answer is unreachable, no navigating system can find it, however well designed.

3.3.2 Coverage Results and the Induced Accuracy Ceiling

Table 3.3 reports the gate.

Three observations follow. Every question in both datasets has at least one topic entity present in the graph. So no question is lost at the entry point.

Table 3.3: Environment coverage over the full test sets, at a bound of four raw hops. The reachability column is the Hits@1 ceiling this environment induces over the whole benchmark. The evaluation draws 400-question samples from these splits, whose own ceilings differ slightly by sampling and are the operative bound on the reported results (Table 5.1).

Dataset	Test questions	All answers exist	≥1 exists	≥1 reachable
WebQSP	1,628	94.7%	97.3%	**97.3%**
CWQ	3,531	99.4%	99.7%	**99.7%**

More strikingly, *existence and reachability are identical* in the ≥ 1 column — 1,584 of 1,628 for WebQSP and 3,519 of 3,531 for CWQ. Every question with at least one gold answer present in this environment has at least one reachable within four hops. The two columns agree question for question, not merely in total. The claim is deliberately the weaker of the two available: the *all answers exist* column is lower — 94.7% against 97.3% on WebQSP. So on 42 WebQSP questions some but not all gold entities are present, on top of the 44 where none are. And this measurement says nothing about whether the ones present are all reachable. What the measurement does say is that reachability is never the binding constraint on *answerability*. When a question is unanswerable here, it is because entities are absent altogether. That pattern is a direct consequence of the construction in Section 3.2: the subgraphs were extracted around topic entities precisely to guarantee reachability. The measurement confirms that the property survived unioning.

Finally, CWQ's coverage exceeds WebQSP's. That result inverts the expected ordering: the deeper benchmark has the better-covered answers. Two things account for this inversion. They are not competing. The larger is constructional: CWQ contributes far more questions and therefore far more subgraphs to the union. So its own answers are better represented. What that leaves — the residual WebQSP shortfall — is semantic. The residual shortfall is dominated by answers that are not entities at all, a gap Section 3.3.3 takes up.

The ten unresolved mentions of Table 3.2 are not a random residue. Every one is a bare machine identifier of the form g.123852dg that the benchmark carries in place of a name. So no surface form exists for the resolver to match. Appendix E.23 reads them. The strongest check the gate makes is not the linking rate but the re-verification of reachability directly in Cypher, independently of the Python that computed it. That independent re-check is what makes Table 3.3 a certification rather than a self-report.

3.3.3 What the Environment Cannot Express

Certifying reachability establishes that answers which *are* entities can be found. That certification says nothing about answers that are not entities. This is where the environment's real limits lie.

The import schema of Section 3.2.4 gives every node a single :Entity label. There is no typed

literal node, no datatype, and no distinction between an entity and a value. A date or a number can therefore exist in the graph only incidentally, as an entity whose *name* happens to be a numeric string. Measured over all 2,592,892 nodes by scripts/count_literals.py, which writes results/phase1/literal_census.json:

- **Dates** are effectively absent. Only 77 nodes carry a full YYYY-MM-DD name and 175 a bare four-digit year — together under 0.01% of the graph, and untyped. So these dates cannot be compared, ordered, or filtered by range.
- **Numeric values** appear on 12,799 nodes (0.49%) under the strict rule that the entire name is a single number. But the population is dominated by identifiers such as ISBNs rather than by measurable quantities. That count is sensitive to its own definition. That sensitivity is why the definition is committed alongside it: admitting names that merely carry no letters ("(1955)", "12-4") raises it to 13,888. The argument does not turn on which is used — both are under 0.6%. But the figure should not be read as a property of the graph independent of how it was counted.
- **Ordinals and superlatives** are not expressible at all. That gap is a property of the tool layer rather than the data: Section 4.2 exposes no aggregation, sorting, or comparison operator. So "the smallest" or "the earliest" cannot be computed even when every candidate is present in the graph.

Together, these three gaps define a class of questions that is *unanswerable in this environment*: any question whose gold answer is a date, a measured quantity, or a superlative selection over candidates. Such a question is not a reasoning failure when the system misses it — the environment could not have supplied the answer. The class is referenced by name in Section 5.14.4, where it accounts for a substantial share of the residual failures. The class is also a known consequence of a design decision: modelling literals as typed nodes was considered and not adopted, in exchange for a uniform schema and a simpler tool API.

Chapter 4

The AGR Framework: Architecture, Navigation, and Verification

This chapter specifies the Agentic Graph Reasoning framework: how the agent is structured, what operations it may perform on the graph, how it decides where to look next, how it recovers from wrong turns, and what guarantees bound its cost. The verification layer is specified in Section 4.10. Everything before it is the machinery that produces the traversal it verifies. That layer is the component this thesis set out to contribute. Section 5.7 measures its contribution as narrower than intended: auditability rather than accuracy, with even the precision advantage over the agentic baseline belonging to the architecture as a whole rather than to the layer. So the layer is introduced here as one part of the architecture rather than as the part the rest exists to serve.

4.1 Architectural Overview

4.1.1 The Controlled-Agent Pattern

There are two ways to build a graph-navigating agent. The first hands the model a set of tools through native function-calling and lets it drive the loop itself, ReAct-style [27]. The second inverts control: the *program* orchestrates the loop, calls the graph tools deterministically, and consults the model only at decision points, through constrained prompts that must return JSON. AGR uses the second. The choice is deliberate enough to be worth defending. It determines what the word "agentic" means in this thesis. Four consequences follow:

- **Every graph operation is logged by construction.** The traversal is not reconstructed from the model's narration of what it did. It is the literal sequence of tool calls the program made.

- **Budgets are enforced in code, not requested in prompts.** A prompt asking the model to stop after four hops is a suggestion. A counter checked in a router is a guarantee (Section 4.7).
- **Failures are reproducible.** A malformed model response degrades a decision. It cannot corrupt the control flow. Every call goes through one wrapper that parses the response as JSON and, if that fails, spends a second metered call asking the model to repair its own output. Each per-node statement below of the form "makes one model call" is therefore the normal case rather than a bound. The second call is charged to the budget like any other. So the call counts of Section 5.11 already contain whatever repairs occurred.
- **Each model call is small, single-purpose, and cacheable.** That combination is what makes the response cache of Section 4.8 viable.

The agency in AGR therefore lives in the *decision policy* — which sub-objectives to pursue, which edges look promising, whether the current path is a dead end, whether the draft answer is supported — and not in unconstrained tool invocation. Prior systems in this space make the same choice [7, 8]. Stating it explicitly forecloses the objection that AGR is merely a prompt around a function-calling loop.

4.1.2 The State Machine

The framework is a state machine: a directed graph of six nodes over a shared typed state, built with LangGraph. Figure 4.1 gives the topology.

Two properties of this topology matter. The topology contains *cycles*. That is why a state-machine framework is used rather than a linear pipeline: exploration loops until the evaluator is satisfied. Verification can send the agent back to explore. The verifier also sits *between* the decision to answer and the answer itself. That placement is the structural expression of this thesis's contribution — there is no path from evaluator to answerer that does not pass through verification.

4.2 The Graph Tool API

The five operations are specified in full, with the Cypher each executes, in Appendix B.

4.2.1 Rationale: Constrained Tools Instead of Free-Form Cypher

An obvious alternative is to let the model write Cypher directly. That alternative was rejected. Generated-query approaches introduce a failure mode that is indistinguishable, in the results,

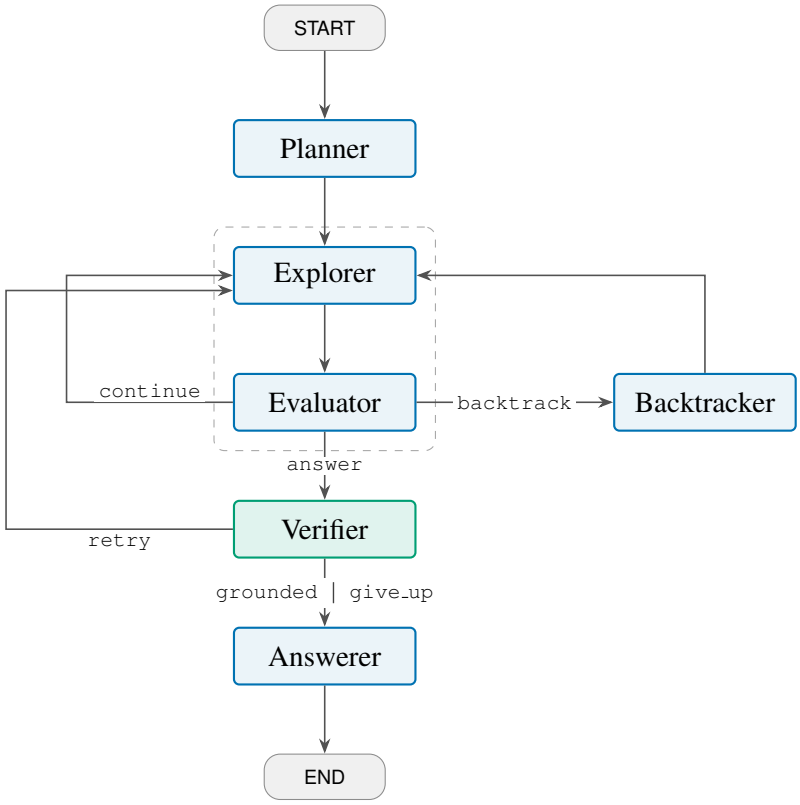

Figure 4.1: The AGR state machine. Three cycles exist: the dashed region marks the Explorer ↔ Evaluator search loop; Evaluator → Backtracker → Explorer restores an earlier frontier; and Verifier → Explorer is verification-driven re-exploration. All three are bounded by the budgets of Section 4.7 rather than by the graph structure. Every path from Evaluator to Answerer passes through the Verifier. That routing is the structural expression of this thesis's contribution.

from a reasoning failure: when a question is answered incorrectly, one cannot tell whether the agent reasoned badly or merely emitted invalid syntax. Since the thesis's central claims are about reasoning and verification, a confound that contaminates every error category is unacceptable.

The tools are therefore parameterised Cypher templates (Section 5.6.2), written once and unit-tested. They accept entity identifiers and return names alongside them. So the model always sees human-readable text while the program tracks identity. Every invocation is appended to a JSONL log with its arguments, result, and latency.

4.2.2 The Five Operations, of Which Four Are Live

Table 4.1 lists the operations. There are five, not the four originally planned. Only four are reachable from a node in the final design. verify_triple was built first as the claim checker. It did not survive contact with generated text. Claim decomposition produces relations phrased in English, while Freebase predicates are neither phrased that way nor reliably one edge long. So requiring the relation to match rejected claims that were true. verify_connection was added in

Table 4.1: The graph tool API. All operations are deterministic parameterised Cypher; none consults an embedding except search_entity, which delegates to the resolver of Section 3.2.6.

Operation	Returns	Role
search_entity(*s*, k)	up to k candidate entities with id, name, score, and resolver stage	anchor seeding; the only entry point from text to graph
get_relations(*e*)	relation types incident to e in *both* directions, each with its fanout count	candidate generation for the scorer
get_neighbors(*e*, *r*, *d*)	entities reached from e along r in direction d, each with the via chain that reached it	frontier expansion
verify_triple(*h*, *r*, *t*)	{direct, via_cvt, supported}	*not called by any node*; built for claim checking and superseded by verify_connection (Section 4.2.2)
verify_connection(*a*, *b*)	{direct, via_cvt, connected}	relation-agnostic claim checking — the check the verifier actually uses (Section 4.2.3)

response. It drops the relation, asking only about adjacency. It superseded verify_triple rather than complementing it (Appendix E.28). It is left in the API as the natural primitive for the relation-aware check Section 6.3 proposes. But nothing in the running loop calls it.

Three design decisions inside these operations are load-bearing.

Relations are returned with fanout counts, in both directions. The count is free signal: a relation with a fanout of forty thousand is almost never the edge that answers a specific question. The scorer can use this signal without an extra query. Retrieving incoming as well as outgoing edges matters. Freebase's relation direction is a modelling artefact — a question about where someone was born may be answerable through people.person.place_of_birth outward or location.location.people_born_here inward.

Administrative predicates are filtered at the tool layer. Relations whose names begin with common., freebase., type., kg., user., dataworld., rdf-schema#, or owl# are dropped before the candidate list is built. These carry no factual content. Section 4.9 shows what happens when they are not filtered: they consume beam slots and drag the agent into the schema rather than the data.

The relation list is capped at 300 entries, sorted by fanout. An earlier cap of 80 was raised after it was found to hide precisely the low-fanout, high-precision relations that answer specific questions — again, Section 4.9.

4.2.3 Mediator Traversal and Relation-Agnostic Adjacency

Section 3.2 established that 63.7% of the environment's nodes are unnamed mediators. If the agent saw these as candidate entities, most of its frontier would consist of nodes with no surface form and no meaning to a language model.

get_neighbors therefore hops through them transparently. When an expansion reaches a node

Table 4.2: The agent state. Fields marked *append-only* accumulate across node executions through reducers; all others are overwritten by whichever node returns them.

Group	Fields
Question	qid, question, gold_q_entities
Plan	plan: ordered sub-objectives, each with a status (pending/active/done) and its resolved entities
Search	anchors (current frontier), traversed (*append-only* triple log), backtrack_stack, banned, last_expanded
Answer	candidate_answers, draft, answer, answer_entities, supporting_triples
Verification	supported_claims, unsupported_claims, unsupported_claim_objs
Control	budget (meter, mutated in place), trace (*append-only*), and the router scratch keys _eval_decision, _last_n_new, _frontier_max_score

flagged is_cvt, the tool immediately expands that node one further step, excluding the origin and excluding other mediators, and returns the entities on the far side. Each returned neighbour carries a via chain — one relation for a direct edge, two for a mediator-mediated one. The model reasons about logical neighbours. The log retains the full raw chain. So hop accounting stays exact.

verify_triple applied the same principle to checking: it matched *undirected* and accepted a mediator in the middle. Direction-strictness would have generated false rejections. The claims it was given came from generated text that has no notion of which way a Freebase edge points. That relaxation was necessary but not sufficient, for the reason Section 4.2.2 records: relaxing direction does not help when the relation itself has no Freebase counterpart.

verify_connection goes further and drops the relation entirely, asking only whether two entities are adjacent directly or through one mediator. It exists because claim decomposition frequently produces a claim whose relation does not correspond to any single Freebase predicate. "X played for Y" may be realised through a roster mediator with no edge literally named for playing. Requiring an exact relation match in such cases rejects true claims. verify_connection is the adjacency test the verification layer uses. Section 4.12.1 gives the two routes it sits in. The cost of dropping the relation is stated there and not here: an adjacency test cannot tell a true claim from a false one that happens to join the same two entities. That gap is the boundary case Section 5.14.3 analyses.

4.2.4 Agent State

The state is a typed dictionary threaded through every node. Table 4.2 groups its fields by purpose.

Two disciplines govern this structure. Both are worth stating. Violating either produces bugs that

are extremely hard to localise.

Nodes write, routers read. A router is a pure function that inspects state and returns an edge label. It never mutates. Any signal a router needs must therefore be deposited in state by the node preceding it. That requirement is what the three underscore-prefixed scratch keys are for. The explorer records how many triples it added and the best score it saw. The evaluator records its decision. The routers read those and decide.

The budget meter is mutated in place and never returned as a delta. The meter is a mutable object shared by reference across all nodes. Counters that are merged rather than mutated can be silently lost when two nodes return overlapping state. Enforcement must not be able to drift.

4.3 The Planner

4.3.1 Decomposition into Ordered Sub-Objectives

The planner makes one model call, converting the question into an ordered chain of sub-objectives plus the entity mentions to seed the search. The chain is strictly sequential. Later objectives reference earlier results with a #N back-reference. A general dependency-graph planner was not built. A sequential chain covers the composition structure that dominates CWQ. A DAG planner is a research problem in its own right.

The prompt is reproduced in full as Appendix A.1. It is few-shot with one example per question archetype: single hop, composition, conjunction, superlative. The single-hop example exists to demonstrate that *not* decomposing is a valid output. Over-decomposition of simple questions is a real failure mode. Section 5.12 shows it is not hypothetical: on the shallower benchmark, removing the planner improves accuracy.

The prompt also instructs the model to extract only specific named entities as topic mentions, and explicitly not generic type words such as “celebrity”, “university”, “country”, or “senator”. That instruction was added in response to evidence discussed in Section 4.9: type words resolve successfully to type nodes in the graph, consuming anchor slots with entities that lead nowhere.

4.3.2 Plan Validation and Degenerate-Plan Handling

The planner’s output is validated, not trusted. Four checks run: the plan must contain a non-empty list of sub-objectives; it must contain no more than four; every #N reference must point strictly backwards; and topic mentions must be present. The back-reference check catches the planner’s most common structural error, a step that references itself or a future step.

If any check fails — or if the model call itself raises — the planner falls back to a single sub-objective consisting of the whole question, seeded with the dataset’s topic entities. That fallback

is a deliberate design principle: **planning can degrade but can never abort a run.** A degraded plan is a data point that appears in the results. A crash is a lost question that biases them.

Anchor seeding has two modes. In the default benchmark configuration, anchors are seeded from the dataset's annotated topic entities. That choice isolates reasoning from entity linking and is the standard protocol in this literature [7, 9]. The alternative seeds from the planner's own extracted mentions, giving the end-to-end number. Because Section 3.2.6 showed resolution succeeds on 99.86% of mentions, the two modes differ very little in this environment.

Setting the planner ablation flag bypasses the model call entirely and installs the fallback plan directly. That substitution is what makes "no planner" a one-configuration-line change rather than a separate code path.

4.4 The Explorer

The explorer performs one expansion of the frontier. It is the only node that touches the graph in bulk. Its structure is the heart of the navigation policy.

4.4.1 Frontier Construction and the Single Scoring Pass

The node proceeds in five steps:

1. **Checkpoint.** The current frontier, the current depth, and the best score seen at this point are pushed onto the backtrack stack.
2. **Gather.** For every anchor, get_relations is called and the results are collected into a *single* candidate list, each row tagged with the anchor it came from.
3. **Score.** The whole list is scored in one pass (Section 4.4.2). Pooling candidates across anchors before scoring is what allows the hybrid scorer to consult the model *once* per expansion rather than once per anchor.
4. **Beam.** Candidates are taken in descending score order into per-anchor buckets of width three, skipping any (anchor, relation, direction) triple on the ban list (Section 4.5.3).
5. **Expand.** Each selected edge is expanded with get_neighbors; the resulting triples are appended to the traversal log, and the neighbours become frontier candidates.

The new frontier is deduplicated by entity identifier, sorted by the score of the edge that produced each entity, and truncated to the anchor cap. Sorting by score rather than by insertion order matters: an insertion-ordered frontier keeps whichever entities happened to be produced first. That bias causes the agent to drift away from the anchor that mattered.

4.4.2 The Hybrid Scoring Function

Each candidate edge receives a hybrid score blending semantic similarity with a model judgement; the prompt carrying the second is Appendix A.2:

$$\text{score}(c) \;=\; \alpha \cdot \cos\big(\mathbf{v}_{\text{rel}(c)},\, \mathbf{v}_{\text{obj}}\big) \;+\; (1-\alpha)\cdot \text{LLM}(c),$$

where $\mathbf{v}_{\text{rel}(c)}$ is the precomputed embedding of the candidate's relation name (Section 3.2.5), $\mathbf{v}_{\text{obj}}$ is the embedding of the *rendered active sub-objective*, and $\text{LLM}(c) \in [0, 1]$ rates how likely following that edge from that anchor is to advance the sub-objective.

Three properties of this design deserve emphasis.

The objective, not the question, is embedded. Hop two of a two-hop question is scored against "find the director of Titanic", with the back-reference already substituted by the resolved entity name, rather than against the original question text. That substitution is the concrete mechanism by which decomposition is supposed to help. Isolating it is what makes the planner ablation interpretable.

The model scores only a shortlist. Candidates are pre-ranked by embedding similarity and only the top twenty are shown to the model, in one batched call. Without the pre-filter, an expansion from five anchors with hundreds of relations each would be unaffordable.

The equation above understates what that shortlist does. The gap is worth closing here. A candidate outside the top twenty is not scored with the model term *omitted*. It is scored with $\text{LLM}(c) = 0$. That is the lowest rating the model could have given it. The embedding pre-filter is therefore a hard gate rather than a soft ranking aid: at $\alpha = 0.7$ an unrated candidate is penalised by up to 0.30 against an identically-similar rated one. So it can only re-enter the beam if its embedding similarity exceeds the rated candidate's by that margin. That penalty is defensible: the pre-filter is an embedding ranking, and a candidate below the cut is one the embedding already judged worse. But it means the twenty-candidate cut interacts with α rather than sitting orthogonally to it. A larger α softens the gate as a side effect of weighting the model less.

The blend is applied after the call, not inside it. The scoring prompt contains no configuration value — no α, no τ. That omission is not cosmetic: because the response cache (Section 4.8) keys on prompt content, a prompt free of configuration means every condition in an α sweep shares the same cached model responses. The sweep therefore costs one set of calls instead of one per condition. Embedding a configuration value in that prompt would multiply sweep cost by the number of conditions without changing any result.

If the model call fails or the budget is exhausted, the scorer treats every model score as zero rather than aborting the expansion. So each candidate scores $\alpha \cdot \text{emb}$. That is the embedding ordering scaled by α, not the embedding score itself. The distinction is worth stating precisely. It does not matter here, though it could be assumed to: $\alpha > 0$ is a positive scaling, so the ranking

is identical. The backtracking trigger compares τ against σ. That threshold takes the unscaled embedding maximum as one of its terms (Section 4.5.3). No reported number changes on this path. Setting $\alpha = 1$ disables the model term entirely. That configuration is the embedding-only ablation.

4.5 The Evaluator and Routing Policy

4.5.1 Sub-Objective Completion Judgment

The evaluator makes one model call per expansion. It is shown the active sub-objective, a window of retrieved facts, and the remaining budget. It returns a decision (continue, backtrack, or answer), whether the objective is complete, and which entities satisfy it.

The fact window is *relevance-ranked, not recent*. Traversed triples are verbalised, embedded, and the thirty most similar to the current objective are shown, in traversal order. The distinction is not incidental: a large expansion can add hundreds of triples. A window showing the most recent thirty will then push the answer out of view in exactly the situations where the expansion succeeded. Section 4.9 documents the case that forced this change.

4.5.2 The Groundedness Filter on Resolutions

The evaluator's resolved entity names are not accepted as given. Each is looked up against a table built from the tail entities of the *traversed* triples. Any name that does not appear there is discarded.

The groundedness filter is small in code and significant in effect. A language model asked which entities satisfy an objective will readily list entities it knows from training but never retrieved. The filter refuses to launder that parametric recall into the agent's evidence. Section 4.9 records a case in which the evaluator confidently resolved nine correct countries that appeared in no retrieved triple. The filter rejected all nine — costing a point on that question while preserving the property that every entity the system proposes came from the graph.

The table is built from tail names only. An entity that appeared solely as the head of a retrieved triple is therefore not in it. A resolution naming one is dropped even though the traversal did reach it. Appendix E.27 explains why that asymmetry is deliberate in the evaluator and where it is compensated for elsewhere.

Resolutions that survive the filter are accumulated into candidate_answers on *every* evaluation, whether or not the objective was judged complete. An earlier design discarded resolutions when the objective was incomplete. That earlier design threw away correct answers found mid-run.

When an objective completes and further objectives remain, the plan advances, the decision is

forced to continue, and the resolved entities become the next frontier, truncated to the anchor cap of Section 4.7. When the last objective completes, the decision is forced to answer.

4.5.3 Backtracking Triggers and the Backtrack Stack

Backtracking fires on the first of three conditions, evaluated in this order:

1. **Dead end** — the last expansion produced no new triples.
2. **Low score** — the frontier's *signal maximum* fell below the threshold τ.
3. **Evaluator veto** — the model judged the direction a dead end.

The order matters. The first two are deterministic and cost nothing. Checking them before deferring to the model's judgement means a mechanically hopeless frontier is abandoned without an argument about it.

The second trigger deserves a precise statement, because it is easy to assume it tests the blended score and it does not. The quantity compared against τ is

$$\sigma \;=\; \max\Big(\; \underbrace{\max_{c\in E} \operatorname{score}(c)}_{\text{blended, expanded edges}} \;,\; \underbrace{\max_{c\in R} \cos\big(v(r_c), v(o)\big)}_{\text{embedding, all candidates}},\; \underbrace{\max_{c\in R_K} s_{\mathrm{LLM}}(c, o)}_{\text{model, rated candidates}} \Big),$$

the largest of three signals: the best blended score among the edges actually expanded, the best raw embedding similarity over every candidate relation the anchors offered, and the best model score over the top-K candidates the model was asked to rate. The trigger therefore fires only when *all three* signals are weak, which makes it deliberately conservative: it abandons a branch only on unanimous evidence of hopelessness. Section 5.7.1 shows the consequence, that τ barely fires at the values swept.

The backtracker then does three things. It pops the *highest-scoring* snapshot from the stack — not the most recent one, which would make backtracking a simple undo. It restores that snapshot's anchors and depth. And, critically, it adds (anchor, relation, direction) triples to a **ban list** that the explorer consults on its next pass. The ban list is what makes backtracking a search operation rather than a loop: without it, a deterministic scorer facing an identical restored frontier selects identical edges, producing identical facts and an identical decision to backtrack until the budget is exhausted. Section 4.9 shows both behaviours in the logs.

The implementation departs from what this section specifies in three places. All three are recorded rather than repaired, under the standing rule of Section 4.9. Repairing them would change the frozen results this thesis reports. Appendix E.12 gives all three: two concern the σ formula and are masked in practice by the dead-end trigger being evaluated first. The third is the one with a measurable consequence.

That third is a misalignment between what the backtracker bans and what it restores. It bans the *most recent* explorer pass. It restores the *highest-scoring* snapshot, frequently older than that pass. So everything expanded in between stays unbanned. The explorer can be handed back a frontier with all of its previously selected edges still available. It then re-selects them. The scorer is deterministic. Over the main matrix, 53 explorer passes — 15 on WebQSP and 38 on CWQ — expanded an (anchor, relation) set identical to one the same question had already expanded. That is exactly the repetition the ban list exists to prevent. The misalignment that produces it is visible in at least 65 of the 262 backtracks, where the restored depth is not the depth the most recent snapshot carried.

That last distinction is deliberate. An earlier version caught all exceptions as budget exhaustion. That conflation silently misreported genuine errors as expected behaviour — exactly the kind of defect that inflates apparent robustness. What now carries the distinction is the recorded exception rather than the handler: the handler catches (BudgetExhausted, Exception), whose first arm is unreachable, since the former subclasses the latter. It is the trace entry, which stores the exception itself, that tells the two apart. Whether it does so correctly is untested by anything in this work: across all 6,960 committed runs no handler of this kind fired even once. The third path is entered on a missing draft rather than on an exception. So it needs its own check. It gets the same answer, though — every one of the 3,690 committed records that ran this graph carries a non-empty draft. Every degraded path described in this section is therefore specified and never exercised. That is also why its fact window is described here and nowhere in Section 5.7.

4.6 The Answerer

The answerer is a finalizer rather than a generator. Which of its three paths runs is decided entirely by what the verification layer left in state. Two of the three are specified in Section 4.10: the *verified* path, where no claim was unsupported and the draft is emitted with no further model call, and the *repair* path, where unsupported claims remain and one rewrite call restates the answer around a deterministically filtered entity list (Section 4.13.4). On the development set the verified path handled 77 of 80 questions at zero additional cost.

The third path exists for the case where the verifier raised before producing a draft at all. It makes one model call over the traversed facts and the accumulated candidates. If the model budget is exhausted, it falls back without a call to the resolved entities of completed plan steps, or failing that to the accumulated candidates, producing a degraded but non-empty answer. If the call raises for any other reason, the error is recorded in the trace. The answer is an explicit failure string — a distinction that matters: an earlier version caught all exceptions as budget exhaustion and so misreported genuine errors as expected behaviour.

Every degraded path described here is specified and never exercised. Across all 6,960 committed runs no handler of this kind fired once. Every one of the 3,690 committed records that ran this

Table 4.3: Budget configuration, and the two tool-layer caps that bound retrieval alongside it. The rows above the rule are the budget proper: they are hashed. The hash is recorded in every run record. So results can always be attributed to the configuration that produced them. The two rows below the rule are constructor arguments of KGTools and are *not* in that hash — they are listed here because they bound what the agent can see. That bound makes them budgets in effect if not in implementation. Four budgets carry two enforcement sites rather than one. The paragraph after this table says which and why.

Parameter	Value	Enforced in
max_depth	4	route_after_eval *and* route_after_verify
beam_width	3	explorer (slice)
max_anchors	5	explorer *and* evaluator (slices)
max_backtracks	3	route_after_eval
max_llm_calls	25	LLM client
max_verify_iters	2	route_after_verify *and* the verifier
max_seconds	300	route_after_eval *and* route_after_verify
max_fanout	200	get_neighbors — neighbours returned per expansion; sets truncated when it binds
max_relations	300	get_relations — candidate relations offered to the scorer

graph carries a non-empty draft. So the third path is never entered either. That is also why its fact window is described here and nowhere in Section 5.7. Appendix E.21 records what the untested handler actually distinguishes.

4.7 Budgets and Termination Guarantees

Budgets do three jobs: they bound cost, they guarantee termination, and they produce the efficiency data of Section 5.11. Table 4.3 lists them.

The two tool-layer caps deserve a sentence of their own. They bound recall rather than cost. max_fanout caps neighbours returned from a single expansion at 200. On a high-degree entity the true neighbour set exceeds that: the frontier sees a truncated view. An answer sitting outside the first 200 is unreachable through that edge no matter how well the scorer ranks the relation. The hub-noise failures of Section 5.14.5 are where this would be expected to matter. The tool returns a truncated flag alongside the neighbours. Two defects in it are recorded rather than repaired, under the rule of Section 4.9. First, the flag does not reach the tool log. That log records the returned neighbour count instead, and cannot distinguish a capped expansion from one that returned exactly 200. Second, it is computed before mediator traversal, on the raw row count, while the final list is truncated after mediators have been expanded into it. So an expansion

returning few raw rows but many entities through their mediators is cut without the flag being set. Both make it a lower bound on truncation rather than a record of it. No analysis in this thesis reads it — the cap's effect is measured from the returned counts instead, as in Section 5.4.3.

Enforcement is confined to routers and the client. That confinement is what keeps it auditable, though the mapping is not one site per budget: Appendix E.11 gives it. Three of the budgets are checked in two places rather than one. In every case the second site exists because the first does not cover the verify–explore cycle — the retry edge re-enters the explorer without passing through the post-evaluation router. So without a second check a repair would silently buy an expansion past the depth budget. Section 4.13 records the development trace that exposed exactly that.

Together these give a termination guarantee that does not depend on model behaviour: every cycle in Figure 4.1 passes through a router that decrements or checks a monotone counter. So no sequence of model outputs, however adversarial, can produce a non-terminating run. That guarantee is verified directly by the mechanics tests of Section 5.6.2. Those tests script an evaluator that always says continue and assert that the depth budget halts it.

4.8 Instrumentation and Reproducibility

Two logs are written per run. The *tool log* records every graph operation with its arguments, result size, and latency. The *run log* records, per question: the answer and its extracted entities, the draft, the verification outcome and any unsupported claims, the count of supporting triples, every backtrack with its trigger reason and restored depth, the full decision trace, the budget snapshot, wall-clock time, and — for provenance — the backbone description, the budget configuration hash, the run configuration, and the current git commit.

The response cache underwrites the thesis's reproducibility claims. Every model call is keyed by a hash over the model identifier, temperature, reasoning-effort setting, the full prompt text, and the schema hint. Hits are served from disk. The subtle part is that a cache hit must be *indistinguishable* from a live call inside the agent: the budget check still runs, the call counter still increments, and the original token counts are replayed into the meter from the stored record. So a fully cached rerun reproduces the original run's budget snapshots exactly rather than reporting zero tokens. One exception is a property of the technique rather than of this run — the replay restores prompt and completion tokens but not reasoning_tokens. The cache stores that field and does not add it back. That field is zero throughout this thesis, since the backbone runs at reasoning effort none (Section 5.3). So the snapshots do reproduce exactly here. Anyone reusing the design under a reasoning-effort setting would find they do not. A separate counter records cache hits so that live and replayed runs can be told apart, repeated experiments cost nothing, and a code change that is not supposed to alter behaviour can be verified by confirming that a rerun is served entirely from cache. Because the key includes the model string, and the backbone

configured here is a dated snapshot rather than an alias, a silent server-side change cannot poison previously cached results.

4.9 Design Validation on the Smoke Set

Two tuning sets appear in this thesis. They are not the same. This section uses the *smoke set*: twenty questions drawn from the training and validation splits, never from test, used once to validate the assembled framework's mechanics. The 80-question *development set* of Section 5.6.1, on which the hyperparameters were swept, is a separate and larger construction. Only it carries any reported number.

The first end-to-end run answered roughly three to four questions correctly. The final configuration answers thirteen to fourteen. That interval is not incidental tuning — it is where several of the mechanisms described above were shown to be necessary. Appendix E.7 gives the record: the five systematic root causes the first run's failures collapsed into, the mechanism each one motivated, and three episodes that bear on later chapters. Two of those episodes are the evidence Section 4.10.1 rests on — a confident answer no retrieved triple supported, and nine entities the evaluator asserted without retrieving. The third is where the standing rule against analysing an unchecked metric comes from.

This is also the one place in this thesis whose *evidence* is not fully archived. What survives is one record file, results/phase3/smoke20_a0.5_t0.2.jsonl, scoring thirteen correct, six hedged and one wrong at $\alpha = 0.5$ rather than the frozen 0.7. What does not is the before-state, read from a console log and never committed. So the three-to-four figure and the root causes rest on that reading alone. Neither number enters any result. That is why the omission was tolerable and why the figures are given as ranges.

4.10 The Structural Verification Layer

Section 4.1 described a navigation loop that ends with a set of traversed triples and a set of accumulated candidate entities. Nothing in that loop constrains what the final answer text *asserts*. The sections that follow specify the layer that does, together with the repair, filtering, and evidence-reporting policies built on it. Except where a sentence names the test sets explicitly, every quantity reported here is recomputed from the archived development-set run at the frozen configuration ($\alpha = 0.7$, $\tau = 0.20$, verify_claims = true, $n = 80$). Where a behaviour is established on a handful of questions rather than a distribution, the count is given in the sentence that makes the claim. The exceptions are in Section 4.13, where what the development set showed and what the 800 test questions show are not the same thing. Reporting only the first would leave this account contradicting Section 5.13.

4.10.1 Motivation: Verifying Before Emitting, Not After

Section 1.3 argued that agentic KGQA systems are typically evaluated on whether the correct answer is *among* what they assert, leaving the precision of the assertion unmeasured. The navigation loop makes that concrete: its final act is a language-model call over a fact window and a candidate list. The model is free to name an entity that appeared in neither, with no downstream component to object. The first end-to-end run over the smoke set (Section 4.9) produced both shapes of that failure — an answer no retrieved triple supported, and a list of nine entities that appeared in no retrieved triple at all. Neither was caught by anything examining the answer.

The design response is to check the draft against the environment *before* it is emitted, rather than to fact-check an emitted answer afterwards. That ordering is where the layer parts company with the post-hoc verification literature. Chain-of-Verification [15] improves a response by drafting it, planning verification questions, answering them, and revising. FActScore [43] decomposes generated text into atomic facts and scores each against a corpus. Both are placed after generation. Both take the model's own answers to the verification questions as evidence. The layer specified here borrows their decomposition step (Section 4.11) and rejects their placement and their evidence source: the check runs before emission and adjudicates against traversed triples rather than against further generation. A model asked to check its own output is a weak detector of its own errors [41]. Three properties follow from placing the check inside the loop. The checker sees the same retrieved evidence the drafter saw. So a disagreement between them is informative rather than an artifact of differing context. The check happens while budget remains. So a rejected claim can trigger further exploration rather than only a retraction. And the arbiter of whether a fact holds is the environment, not a second opinion from the model that produced the draft.

What the layer promises should be stated precisely. The terms are easy to inflate. It certifies that the atomic claims a draft asserts are *supported* by the environment in the sense fixed in Section 2.2.2: the traversed triples, or the graph in the retrieved neighbourhood, exhibit a connection consistent with the claim. It does not certify that the answer is correct. An answer every one of whose claims is supported can still be the wrong answer to the question. Section 4.15 shows that this is a structural gap rather than a hypothetical one.

4.11 Claim Decomposition of the Draft Answer

Verification begins with a single model call that produces three things at once: a prose draft, the list of entities that draft asserts as the answer, and a decomposition of the draft into atomic claims.

The call sees a fact window of at most sixty traversed triples — ranked by embedding similarity between the triple's verbalisation and the *question*, with traversal order preserved among the

survivors — together with up to twenty-five accumulated candidate entities. Each claim is emitted as a triple of strings $\langle h, r, t \rangle$ over entity *names*, not identifiers, since the model has never seen an identifier.

Four provisions in the prompt are load-bearing. Each exists because its absence produced a specific defect.

Names are to be copied exactly as they appear in the facts. The graph's surface forms are frequently not the ones a language model would generate unprompted — Freebase renders Yale University as University Yale. Every downstream check is keyed on the name string.

answer_entities names what the draft asserts as the answer, not every entity it mentions. Without this the list drifts toward a bag of mentioned names, including the question's own subject.

A hedged draft has zero claims. This makes abstention explicit rather than something inferred from the absence of an answer.

Answer entities must be specific entities of the kind the question asks for, never a restatement of the question or an entity type. This provision was added after development runs produced answers of the form "the senators from New Jersey are the United States Senate members associated with New Jersey" — grammatically an answer, semantically empty, and, as Section 4.15 explains, entirely invisible to a claim-level check. It is a drafting-side mitigation of a failure the checking side cannot address.

The relation slot r is deliberately free text. The model is not asked to name a Freebase predicate. It has no reliable way to do so. A wrong guess would be indistinguishable from a wrong claim. The consequence is that the structural checks below cannot match on the relation at all. The cost of that choice is analysed in Section 4.12.1 and again in Section 4.15.

Decomposition on the development set was sparse. Across 80 questions the drafter emitted 121 claims — a mean of 1.51 and a median of 1 per question, with a maximum of 5. Ten questions produced no claims at all. For those ten the layer certified nothing. Section 4.15 returns to what that means for the outcome label they received.

4.12 Two-Stage Claim Checking

Each claim is routed through at most three tests: two structural, one model-based. Algorithm 1 states the procedure. Figure 4.2 draws the same routing as a single claim's path through it. The subsections that follow explain each test and the reason for their order.

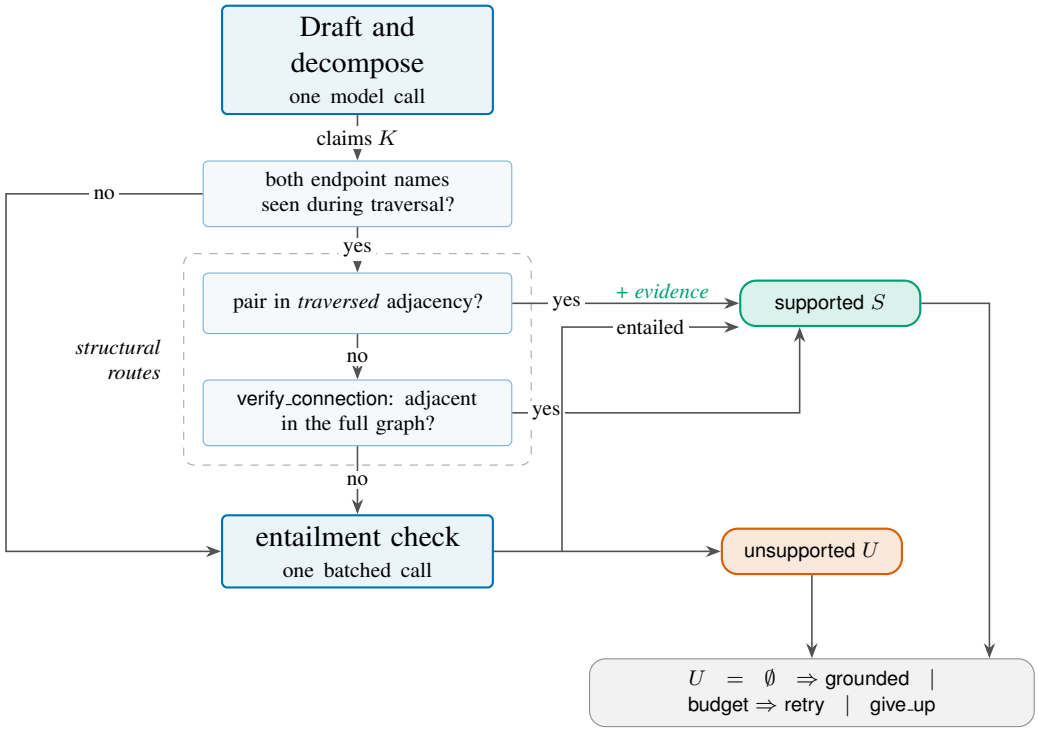

Figure 4.2: The path of a single atomic claim through the layer, mirroring Algorithm 1. Three tests are tried in a fixed order. A claim leaves at the first one that settles it. The two structural routes are boxed. Only the first of them adds to the supporting-triple list. That asymmetry is why Section 4.14 can report evidence for some grounded answers and not others. A claim naming an entity the agent never traversed skips both structural routes entirely, since the name-to-identifier map is built from traversed triples alone.

Algorithm 1 Structural verification of a draft

Require: question q, traversed triples T, candidates C, tools, budget
Ensure: draft, answer entities, supported and unsupported claim sets, supporting triples
$W \leftarrow$ TOPFACTS$(q, T, 60)$; $\mu \leftarrow \{$ name $\mapsto$ id $\}$ over T {first occurrence wins}
$(draft, A, K) \leftarrow$ DRAFTANDDECOMPOSE(q, W, C) {one call}
if $\neg$ verify_claims **then**
  return $(draft, A, \emptyset, \emptyset, \emptyset)$ {ablation condition}
end if
$adj \leftarrow \{(h,t),(t,h) : (h,r,t) \in T\}$; $S \leftarrow \emptyset$; $P \leftarrow \emptyset$; $E \leftarrow \langle\,\rangle$ {E is a sequence}
for all $c = \langle h, r, t\rangle \in K$ **do**
  if $h \notin \mu$ **or** $t \notin \mu$ **then**
    $P \leftarrow P \cup \{c\}$ {endpoint never traversed}
  else if $(\mu[h], \mu[t]) \in adj$ **then**
    $S \leftarrow S \cup \{c\}$; $E \leftarrow E +\!\!+ \langle\, \theta \in T : \{\theta_h, \theta_t\} = \{\mu[h], \mu[t]\} \,\rangle$ {appended, not unioned}
  else if verify_connection$(\mu[h], \mu[t])$ **then**
    $S \leftarrow S \cup \{c\}$ {no evidence recorded}
  else
    $P \leftarrow P \cup \{c\}$
  end if
end for
$U \leftarrow \emptyset$
if $P \neq \emptyset$ **then**
  $V \leftarrow$ ENTAIL(W, P) {one batched call; on any failure $V \equiv$ false}
  partition P into S and U by V
end if
if $U \neq \emptyset$ **and** verify budget remains **then**
  $iters \leftarrow iters + 1$; append repair objective for the first $c \in U$; re-anchor on $\mu[c_h]$
end if
route: $U = \emptyset \Rightarrow$ grounded; else verify, depth, and time budget all remain $\Rightarrow$ retry (**goto** explorer); else give_up
return $(draft, A, S, U, E)$

4.12.1 The Structural Check

The structural check operates on entity identifiers. So it first needs a mapping from the names the model produced back to graph nodes. That mapping, μ, is built from the traversed triples alone: every head and tail name encountered during navigation is registered against its identifier, first occurrence winning. Its scope is worth stating explicitly. It bounds everything that follows. **A claim naming an entity the agent never traversed cannot reach either structural route**, regardless of whether that entity exists in the graph. Such claims fall directly to the entailment check.

For claims whose endpoints both resolve, the first route is *traversed adjacency*. The traversed triples are collapsed into an undirected set of endpoint pairs. A claim is supported if its endpoint pair is in that set. Traversed adjacency is exact, costs nothing beyond a set lookup, and —

uniquely among the three — produces citable evidence: every traversed triple joining the pair is appended to the supporting-triple list. Appended, not unioned: Section 4.14 records what that costs.

The second route is the verify_connection tool (Section 4.2.3), a single Cypher query asking whether the two entities are adjacent in the *full* graph, either directly or through one mediator node. It exists because the first route tests only the *traversal*, not the graph. The agent may have reached both endpoints along different branches without ever walking the edge between them.

Both routes are relation-agnostic. Since r is free text with no reliable mapping onto a Freebase predicate, neither asks whether the edge it finds is the edge the claim names, and neither asks about direction. That is the price of admitting free-text relations. It is the layer's principal acceptance risk rather than an incidental looseness.

Appendix E.22 records how often each route fires. Two figures from it are quoted later. The second route is rare — consulted five times on the development set and on a small minority of test claims. So the first route carries almost all of the checking. And the record cannot say how many of the $2{,}008$ accepted claims were certified by a relation-blind test rather than by entailment: the answer lies somewhere in $[39, 2{,}008]$. That interval is what Section 4.15 and Section 5.15.1 both read against. Section 6.3.2 is the change that would narrow it.

4.12.2 The Entailment Check

Claims that fail both structural routes — together with those whose endpoints were never traversed — accumulate into a pending set and are checked in one batched model call. The claims are numbered, rendered as h – r – t, and presented against a fact block, with the model asked for a boolean verdict per claim. The prompt defines unsupported explicitly, to include the three cases that would otherwise be resolved charitably: plausible but absent, contradicted, and inferable only through outside knowledge.

Three properties of this check must be stated. Each bounds what its verdicts mean.

The fact block is the same window the drafter saw. It is the same sixty triples, ranked by relevance to the *question* rather than to the individual claim under test. A claim about an entity peripheral to the question may therefore be judged against facts that never mention it. The bias this introduces is directional: it disfavours peripheral claims and favours claims about material the drafter had already been shown.

Failure is conservative and lossy. If the call raises for any reason — budget exhaustion, transport error, malformed output — every pending claim is marked unsupported. That default is the right one for a system whose purpose is to avoid unsupported assertions. But it means an infrastructure failure and a genuine rejection produce the same record except in the trace.

The check can only add support, never remove it. Claims accepted structurally never reach it. A claim whose endpoints are adjacent for a reason unrelated to the asserted relation is accepted

without any semantic examination.

4.12.3 Why the Structural Check Runs First

Three reasons, in descending order of importance. *Authority*: the thesis's position, fixed in Appendix E.17, is that embeddings and language models rank while symbols decide. So the model must be consulted only where the graph is silent — it fills gaps and never overrides a finding of the environment. *Evidence*: only traversed adjacency yields triples that can be attached to the answer. So the order does not change which claims are accepted, but it does change how many can be justified. *Cost*, least importantly: a set lookup is free, a verify_connection probe is one indexed query, and the entailment call is the layer's only mandatory expense beyond drafting. So ordering cheap-to-expensive runs it on a residue rather than on every claim.

4.13 Repair Policies for Unsupported Claims

The router following the verifier is pure Python and has three outcomes: no unsupported claims routes to grounded; unsupported claims with budget remaining route to retry; otherwise give_up.

4.13.1 Targeted Re-Exploration Under Remaining Budget

On retry, the verifier has already re-targeted the agent before the router runs. It takes the *first* unsupported claim, synthesises a repair objective of the form verify whether $h\ r\ t$, marks every active plan step done, appends the repair step as the new active objective, and — if the claim's head name resolves — resets the frontier to that single entity. Control returns to the explorer, which now searches for the missing evidence specifically.

Two limitations follow. Only one claim is repaired per iteration. The one chosen is first by position in the draft's decomposition rather than by importance to the answer. Both keep the repair path cheap. Neither was revisited. The path was rarely exercised.

The retry route requires verify, depth, *and* time budget together. That conjunction is why it almost never fires: by the time the verifier is reached the depth budget is typically already spent. Appendix E.26 measures it — the development-set condition under which the route could have run, why that correlation does not survive the move to test, and the instrumentation consequence that follows for reading the repair counts.

4.13.2 Answer Rewriting and Hedging

On give_up the answerer makes one rewrite call. It receives the draft, the unsupported claims, and — crucially — the list of entities already decided to survive, with an instruction to keep

them named exactly as written. It is asked to remove or hedge the unsupported material and to respond with prose only. That last phrase is the whole of the fix described in Section 4.13.4: in the original design this call also regenerated answer_entities. That regeneration turned out to be the single largest defect in the layer.

4.13.3 Iteration Cap and the Draft-Only Fallback

Repair is capped at two iterations. That cap, together with the depth and time conditions, guarantees the verify–explore cycle terminates. On the development set the cap was never reached, for the reason just given: the cycle never began. On test it binds on 15 questions — 4 on WebQSP and 11 on CWQ. That is the cap doing the job it was specified for rather than sitting unused. It is the second place where the development set understated this route.

The grounded path costs nothing. When no claim is unsupported the answerer emits the draft verbatim with no further model call — 77 of 80 development questions took this path. The layer's aggregate cost is correspondingly small: mean tokens per question rise from 4,858 without claim checking to 4,922 with it, about 64 tokens or 1.3%, and mean model calls from 7.2 to 7.3. Whatever the case for or against the layer, it is not a cost argument.

A distinct path exists when verify_claims is false: the verifier returns the draft and its entity list untouched with outcome draft_only. That outcome is the ablation condition of Section 5.6, not a runtime fallback — the system never selects it dynamically. A third path, the legacy answerer of Section 4.6, runs only when the verifier raised before producing a draft at all.

4.13.4 Entity Filtering of the Final Answer

The rule is one line. Its justification is the most instructive result in this chapter.

The rule. On the give-up path the final entity list is computed deterministically from the draft's own list: an entity is kept if it appears in at least one supported claim, or if it appears in no unsupported claim. Strings are copied verbatim. The function can only remove entities, never add one. The disjunction's second branch is deliberately permissive — an entity the decomposer never mentioned in any claim is retained, on the principle that the layer should punish unsupported assertions and not imperfect decomposition recall.

The scope. This filter applies to the give-up path only. On the grounded path the draft's entity list passes through unchanged. There are no unsupported claims to filter against. On the development set the filter therefore ran on 3 of 80 questions.

The Attribution Census

The filter exists because of a paired comparison between the verified condition and its verify_claims = false twin at the frozen α. In aggregate, claim checking *cost* accuracy: 64 hits against draft-only's 66. Aggregate counts cannot distinguish a mechanism working as intended — turning an unsupported lucky hit into an honest hedge — from a mechanism misfiring. So the two runs were diffed per question. Exactly three questions differed. Table E.2 gives all three.

The classification is unambiguous. *No entailment verdict was involved in any of the three*. Appendix E.15 gives them question by question. The finding is worth stating in one sentence. It is the conclusion the whole subsection exists to support:

> The verification logic was sound; its repair path frequently was not.

Replacing generation with deterministic filtering removed two of the three failures. It left the third. That is a different defect.

What the Census Does and Does Not Establish

Three qualifications belong here rather than in a limitations chapter. Without them, the numbers above would be read as more than they are.

The census is $n = 3$. It is a complete enumeration of the questions on which the two conditions differed. That is exactly why it could be classified by inspection. But three questions support a mechanism claim, not a rate.

Claim checking did not reduce wrong answers on this set. Post-fix verified scores 65 / 11 / 4 against draft-only's 66 / 10 / 4. The wrong count is identical. The layer cost one correct answer and removed none. Its case therefore cannot be made on development-set accuracy. This thesis does not attempt to make it there. Neither of the two aggregate measurements that might seem to carry it survives Section 5.7 either. The groundedness result of Section 5.10 is reached by the agentic baseline as well. That result makes it a property of graph navigation rather than of this layer. Removing the layer, in turn, does not move per-question precision. So the precision margin over that baseline is not evidence for it either (Section 5.15.1). What remains is the output contract of Section 4.14 — an auditability claim rather than an accuracy or a precision one — together with the case-level evidence of Section 5.15.1, where the layer is seen removing a false assertion on the questions its filter reaches. That is a narrower case than this section originally made. It is the one the measurements support.

The residual difference is a single question. Its interpretation involves a label changed after the fact. The sole remaining diff is Harbaugh, where draft-only's "hit" asserts Baltimore Ravens — supported by no traversed triple, and not a team Jim Harbaugh played for — alongside

an unverifiable founding-date filter. Set-intersection Hits@1 scores that as a clean hit. Per-question F1 in Section 5.7 does not. The question was additionally relabelled from cvt_heavy to unanswerable_env after this analysis, on the grounds that it carries a date constraint the environment cannot express (Section 3.3.3). The relabelling is defensible on its own terms and was recorded as a decision. But it was made after seeing the result it affects, and is flagged as such here and in Section 5.15.1. The aggregate figures reported above are not rescored on account of it.

4.14 Output Contract: Answer Paired with Supporting Triples

The system pairs each answer with the triples that support it. That pairing is the explainability deliverable named in Section 1.6. Its actual coverage is narrower than the phrase suggests in two independent ways.

Only one of the three routes records evidence. Supporting triples are collected in the traversed-adjacency branch alone. A claim certified by verify_connection is accepted with nothing attached, and so is a claim certified by entailment. The contract is therefore "answer paired with whatever traversed triples happened to justify it", not "answer paired with justification for every claim".

What survives in the log is the count, not the evidence. The pairing holds inside the run — the answerer receives the verifier's triple list and emits against it. But RunLogger writes n_supporting_triples, an integer, and drops the list. No committed artifact in this work contains a single supporting triple. Every statistic in this section is computed from that counter. An examiner can confirm that these answers came from a system which tracked its evidence. They cannot inspect the evidence. That bounds the auditability this section claims. It is the same gap Section 6.3.2 approaches from the other side — one logging change repairs both. It is recorded rather than repaired late, under the standing rule of Section 4.9. Repairing it now would separate the code from the results already frozen against it.

The measured consequence on the development set: 13 of 80 answers carry no supporting triples at all. Ten of those asserted no claims — they are hedges. Zero is the honest number. Of the remaining three, one had every claim rejected and two had their single claim certified by verify_connection, which records no evidence. Across all 80 records the median is 3 and the mean 4.1, with a maximum of 16, read from the persisted counter since the triples themselves are exactly what the log does not keep.

Three instrumentation defects in this area are recorded rather than quietly fixed, in keeping with the same standing rule. Appendix E.10 gives them: a falsy-empty-list trap in the answerer, since repaired; an unrepaired counter whose name misdescribes what it counts, and which is therefore fenced off from every reported result; and a mismatch between Algorithm 1 and the code, the

algorithm accumulating E by set union where the implementation appends. The third bears on the figures above. Two claims sharing an endpoint pair each append the same matching triples. So n_supporting_triples counts matches rather than distinct triples. The exposure is bounded at 378 of the 908 verifier firings, itself an over-count. That bound makes that counter an *upper* bound on distinct supporting triples — as the median of 3 and mean of 4.1 above inherit.

4.14.1 A Worked Example

Appendix E.20 walks one question end to end — *which university, founded in 1701, did actor James Franco attend?* — through drafting, claim decomposition, both structural routes, and the entity filter, showing where each mechanism specified above fires on a single trajectory. Its instructive part is the counterfactual: before deterministic filtering replaced a second generation call, the same trajectory emitted an entity the traversal had never supported. Replacing generation with filtering is what removed it.

4.15 What the Verification Layer Establishes, and What It Does Not

The ways this layer can be wrong follow from its design, independently of how often they occur. Appendix E.8 enumerates them: five mechanisms of wrongful rejection, led by the composite claim whose unverifiable component drags down its verifiable one, and four of wrongful acceptance, led by relation-agnostic adjacency. Section 5.14.3 reports which of them actually fired. In the vocabulary of Section 5.14.3, a claim is *wrongly rejected* when the layer withholds support the environment would in fact provide, and *wrongly accepted* when it certifies a claim the environment does not support.

One of those mechanisms bounds the layer's principal risk and belongs here rather than behind the body. Both structural routes ignore the relation and ignore direction. So any edge between the endpoints certifies any claimed relation between them: a claim that X is Y's mother is accepted on an edge recording that X is Y's child. The population exposed is therefore not the rare second route but every claim the two of them accept between them. That population is, on test, somewhere in $[39, 2{,}008]$ of the $2{,}008$ accepted claims — the lower end what the log pins down, the upper end what the mechanism permits. Nothing measured in this thesis narrows that interval. Section 6.3.2 is the change that would.

Stated at the level of precision that catalogue requires: the layer establishes that the entity pairs a draft's claims relate are connected in the environment. On the repair path, it also makes the emitted entity list a deterministic function of those verdicts rather than a second generation. It does not establish that the claimed relation is the one the edge records, that an entity absent from

every claim was grounded at all, that the answer is responsive to the question, or that a claim it could not check is false. Those boundaries are properties of the design rather than defects discovered in it. Section 5.13 reports how much of the residual error they account for.

Chapter 5

Evaluation: Setup, Results, and Error Analysis

The evaluation is reported in three parts: the experimental design fixed before any test question was answered (Section 5.1), the measurements it produced (Section 5.7), and a hand reading of every failure that remained (Section 5.13).

5.1 Experimental Setup

This part of the evaluation fixes everything that had to be decided before any test question was answered: which questions, which systems, which numbers, and which rules for reading them. It records the decisions that went the other way as well as the ones that stood, each where it belongs rather than collected into a confessional at the end.

The aggregate figures quoted in this chapter are regenerated by scripts/build_thesis_numbers.py into results/phase4/thesis_numbers.json from the scoring, groundedness, judge, and census artifacts. Anything read instead from a run record or a census sheet is cited to it there. Two *reported* figures cannot be recomputed from this repository and say so where they appear: the API spend of Section 5.2, a billing record, and the second through fourth determinism rounds of Section 5.3, read from console tables that were never committed. Most tables are hand copies checked against that JSON rather than emitted from it. That is the weaker guarantee.

5.1.1 Mapping Experiments to Research Questions

Each research question of Section 1.5 is answered by a distinct experiment rather than by one table read three ways. **RQ1** is answered by the main matrix of five systems on two datasets, scored with Hits@1 and F1 and broken down by hop stratum with paired significance tests (Sections 5.7.2, 5.7.3 and 5.9). The per-stratum breakdown, not the aggregate, is what makes

the claim about *multi-hop* accuracy. **RQ2** is answered by three measurements that do not agree with one another. That disagreement is itself the finding: the two-tier groundedness metric (Section 5.10), the precision and F1 columns of the main matrix, and the verifier ablation (Section 5.12). **RQ3** is answered by four single-deviation ablations on stratified half-samples, each compared against the unmodified system restricted to the same questions (Sections 5.6 and 5.12).

5.1.2 Pre-Registration and the No-Tuning Policy

One rule governed everything from the moment the test files were built: *nothing gets tuned*. No prompt was edited, no configuration value changed, and no threshold adjusted after test data was touched. Defects discovered mid-matrix were documented rather than fixed. Four decisions were fixed in writing before the measurements they govern: the frozen configuration $\alpha = 0.7$, $\tau = 0.20$ with claim verification enabled, tuned on the development set (Section 5.6.1) and never revisited; the sample size of 400 questions per dataset, named as the fallback before the API budget was known (Section 5.2); the baseline certification threshold, that a retrieval baseline hedges with the gold answer visible in its own retrieved facts on fewer than 15% of its hedges (Section 5.4); and the judge validation threshold of Cohen's $\kappa \geq 0.7$ against blind human annotation (Section 5.5.5). The last was missed, by 0.0005, and is reported as missed. A threshold that is only pre-registered when it is met is not pre-registered at all.

The single code change Phase 4 required — a `use_planner` flag to disable decomposition — was certified as a no-op on the frozen path by re-running an already-answered development question and confirming that every language-model call was a cache hit. The reference condition in the ablation table is therefore provably the same system that produced the main matrix.

Path fidelity — agreement between the relation chain the system traversed and the chain the benchmark's gold SPARQL query specifies — was the third criterion named in the approved proposal. It is not reported. It requires the gold relation chains. The RoG distribution (Section 3.1.3) publishes name-keyed triples without the originating SPARQL. So the chains would have to be reconstructed from the upstream releases and re-aligned to this environment's relation vocabulary. That was outside the project's remaining budget. Section 6.3.5 treats it as future work.

5.2 Evaluation Datasets and Test Sets

5.2.1 WebQSP and ComplexWebQuestions

Two benchmarks are used, in the RoG distribution [9] whose per-question subgraphs also form the knowledge environment (Section 3.1.3). **WebQSP** [19] is the semantic parses of WebQuestions:

natural questions from search logs, mostly one or two hops, with multi-valued answers common. **ComplexWebQuestions** (CWQ) [20] is built by composing WebQSP questions with additional constraints through SPARQL templates, giving deeper chains, conjunctions, superlatives, and comparisons — and, as Section 5.2.4 documents, a higher rate of machine-generated label defects.

The distinction that matters operationally is visible in the gold answer sets. Counted as the metrics count them — over distinct answer strings, since Section 2.1.4 defines precision and recall over sets — WebQSP averages 7.15 gold entities per question. But its median is 1.5. Exactly 200 of its 400 questions carry a single gold answer. The mean is pulled up by a long tail whose largest question lists 237. CWQ averages 1.75 with a median of 1.[1] So the two benchmarks differ less in the typical question than the means suggest, and more in the tail. That tail is what decides how a metric rewarding one correct entity behaves. This is one of the reasons F1 is reported alongside Hits@1 (Section 5.5).

5.2.2 Stratified Sampling, Seeds, and Published Question IDs

Only the *test* splits are used. The development set of Section 5.6.1 was mined from train and validation. The builder asserts zero overlap between the two pools.

Sampling was a budget decision. The full test splits hold 1,628 and 3,531 questions. Five systems over both, at the measured token rates, projected to roughly $45–55, against a stated ceiling of $15–20. The pre-registered fallback — stratified samples of 400 per dataset (Table 5.1), giving bootstrap confidence intervals of roughly ±5 points on Hits@1 — was therefore adopted, with the arithmetic recorded before the runs fired. The whole benchmark, including the ablations, ultimately cost $11.13 in API spend. That figure is read from the provider's billing record rather than derived from the run artifacts. The reproducible proxy is the token and call accounting of Section 5.11. That accounting is recorded per question in every run record.

Sampling is proportional within hop strata at seed 42. The resulting question sets are committed as immutable artifacts (results/phase4/test_webqsp.json, results/phase4/test_cwq.json). Every question identifier is listed in Appendix C. They were never regenerated after any system ran.

A near miss worth recording. The first build read the training shards rather than the test shards. The development-overlap guard caught it — the contamination guard, built for hygiene, is what found the contamination. No system had run against the files. They were deleted and rebuilt. The rebuilt pools report 1,628 and 3,531 with skipped={}. Appendix E.25 records

[1] The distinction is not decorative. The raw answer lists repeat a string on 4 WebQSP and 6 CWQ questions. Those repeats sit in the tail. Counting them takes the WebQSP mean to 7.98 and its largest question to 382. Nothing scored in this thesis sees the repeats: the scorer deduplicates before matching. Quoting the raw counts would therefore describe a population no reported number was computed over.

Table 5.1: The two certified test samples. The unreachable stratum is retained deliberately. The ceiling is the share of questions with at least one gold answer reachable within four hops, measured over these 400 questions, and is the upper bound on Hits@1 for every result reported in Section 5.7.

Dataset	n	h1	h2	h3plus	unreach.	Ceiling
WebQSP	400	256	128	4	12	97.0%
CWQ	400	137	211	49	3	99.2%

how it was diagnosed. That near miss is the reason a cheap overlap assertion belongs in every benchmark builder.

5.2.3 Hop-Count Stratification and the Accuracy Ceiling

The stratification variable is the length of the shortest path in *this* environment between any question topic entity and any gold answer entity, bounded at four raw hops. That is the same machinery used to certify the development set, so that mediator nodes are counted consistently as the two edges they are (Section 4.2.3). Questions fall into h1, h2, h3plus, or unreachable.

Table 5.1 carries two facts the results chapter leans on repeatedly. WebQSP is 64% one-hop with a four-question h3plus tail — too small to interpret, and labelled as such wherever it appears. CWQ is majority two-hop with a genuine deep tail of 49. The samples preserve the strata proportions of their parent pools. Per-stratum numbers therefore describe the benchmark rather than the sample.

These ceilings are not the ones the validation gate reported. The difference is worth naming rather than leaving for a reader to reconcile. Table 3.3 measured the full test splits and found 97.3% and 99.7% reachable. The figures above are the same quantity over the 400 questions actually evaluated, at 97.0% and 99.2%. Neither corrects the other — one bounds what the environment could support across the whole benchmark, the other bounds what any system in Section 5.7 could have scored. It is the second that every accuracy in this thesis is read against. The gap is sampling. It stays under a point on both datasets because the draw preserved the strata proportions (Appendix E.24).

The unreachable stratum was kept rather than dropped. Dropping it would raise every system's score by the same amount and misdescribe the environment. Keeping it means the ceiling is reported honestly and, as Section 5.10 shows, that these questions become the sharpest available probe of parametric leakage: any "hit" on a question with no graph path to gold is an assertion the environment cannot support.

Table 5.2: Gold-label adjudication. Rows and distinct questions are separate units because the pass emits one row per consensus answer. Exclusions are the union of gold_wrong and ambiguous_question questions.

Dataset	Rows	Questions	gold_wrong	ambiguous	Excluded
WebQSP	89	58	12	10	22 (5.5%)
CWQ	59	47	17	2	19 (4.8%)

5.2.4 Gold-Answer Quality Control

Both benchmarks carry label defects, CWQ more. Its questions and answers are generated by SPARQL templates over the knowledge base rather than written by hand. A question whose gold answer is wrong is not a system failure. Counting it as one corrupts both the accuracy tables and the error taxonomy. Some detection procedure was therefore necessary — though a naive one is worse than none, as the results below show.

The detection principle is **independent multi-system consensus against gold**: when at least three of the five systems of the main matrix assert the same non-gold answer, the label is worth examining. The voters are those five systems, spanning two knowledge sources and four retrieval paradigms, rather than the four ablation conditions, whose agreement would describe an architecture rather than a label. Appendix D.4 gives the procedure, the three verdicts, the automatic triage that cleared 15 of 89 rows on WebQSP and 2 of 59 on CWQ, and the two *opposite* evidence signatures that make the adjudication reproducible — a world-fact error requiring mixed consensus, an annotation-pipeline error showing graph-only consensus. One row is emitted per (question, consensus answer) pair, so rows and questions are separate units throughout and question counts are always deduplicated.

Census-Based Exclusions and the Dual-Reporting Policy Two numbers must not be conflated. Table 5.2 separates them: 58 and 47 questions were *flagged* by the consensus pass, while 22 and 19 were *adjudicated* as defective. The gap is not noise. Most flagged questions are all five systems converging on the same wrong answer. That convergence is a finding about the systems rather than the labels, and it is treated as one in Section 5.14.5. A policy of rescoring on consensus alone would have corrupted the results with exactly those cases. The excluded questions are removed from the failure census of Section 5.13 as benchmark artifacts rather than system failures, ambiguous_question on the same logic, an underdetermined question being no more a system failure than a broken label. Of those excluded, 12 of 22 (WebQSP) and 15 of 19 (CWQ) were failures of the full system. Those are the ones that shrink the census pool.

The accuracy tables are not rescored. They are reported exactly as measured, with this section as the footnote, for two reasons. Rescoring on self-adjudicated labels would substitute one contestable ground truth for another. And the errors cut both ways: the remaining 10 and 4

excluded questions are ones the full system scored as *hits* against labels now judged broken or ambiguous. That symmetry is what makes the footnote credible rather than defensive. It also means the direction of the net bias is unknown rather than conveniently favourable. Two scope limits belong on the record. The rule that automatically assigned gold_wrong to questions with implausibly long gold lists is a heuristic whose outputs were spot-checked rather than individually verified. And the manual adjudication covered every mixed-evidence flag plus every graph-only flag that was not a topic echo — broad, but not the complete flag population.

5.3 Backbone Model Selection and Qualification

All systems share one frozen backbone: gpt-5.4-mini-2026-03-17 at temperature 0.0, with reasoning_effort explicitly set to "none" and the response cache enabled. The model string, both sampling parameters, and the cache mode are stamped into every one of the $4{,}000$ test-matrix records and every ablation record. So any record found anywhere states what produced it. A non-reasoning model was chosen so that sampling parameters remain available and no hidden reasoning tokens contaminate the cost metric. The second was verified rather than assumed, by a check that logs the reasoning-token count on every call and read zero across every run in this thesis.

Selection between gpt-4.1-mini-2025-04-14 and gpt-5.4-mini-2026-03-17 was settled by four rounds of a determinism probe counting how often a *decision signature* — a hash of the plan, the expanded relations, the evaluator verdicts, and the answer — repeats exactly across reruns of the same questions. The rounds disagreed with one another, because temperature-zero decoding on a hosted API is greedy rather than deterministic and a sequential agent amplifies a single divergent token. Appendix E.9 gives the four rounds, that mechanism, and the tiebreaker pre-registered before rounds three and four. That tiebreaker decided the choice on longevity grounds.

The stability figure that belongs to this thesis is the frozen model's, which is the lower of the two. gpt-5.4-mini repeated its decision signature on 8 of 12 probes, or $\approx 67\%$: roughly one trajectory in three diverging on a rerun. The $9/12$ that would read as $\approx 75\%$ is the runner-up's. None of these rates is measured tightly enough to separate from the others at $n = 12$. The thesis claims determinism at none of them. Only the first round survives as per-question artifacts, because the qualification script writes its detail file in truncating mode and later rounds overwrote it.

Two observations from those four runs matter more than the choice they settled. Each candidate carried a *persistent* ungrounded answer — 4.1-mini asserted "United States Dollar" for a Dominican-peso question in two runs of four, 5.4-mini named the wrong Beckham child in all four. Both carried *intermittent* ones, too: the same question on the same graph under the same budget moved between grounded-correct and ungrounded-wrong as the sampling weather changed. Backbone selection does not remove this class of error; it relocates it. That is the

premise of this thesis, measured over four runs and two frontier-family models. It is the reason Section 4.10 exists.

Reproducibility here therefore rests on the cache, not on the model. Its mechanism is specified in Section 4.8. Appendix E.3 gives the identity check it supports: a hit is indistinguishable from a live call. So a fully cached rerun reproduces the original run's budget snapshot rather than approximating it. One defect qualifies that — the replay restores prompt and completion tokens but not the reasoning_tokens counter. It does not bite here: that field is zero in every one of the 6,960 agent runs committed for this work. It is recorded rather than repaired, for the reason Section 5.1.2 gives. Every table in Section 5.7 comes from one recorded run per condition. That run replays byte-identically for as long as the cache survives. Fresh-run variance is a different problem. It is *not* measured here. It cannot be bounded by the bootstrap intervals. Section 5.15.1 carries it as a limitation. Latency is computed over cold-cache records only. That restriction is what keeps that column honest where a run was interrupted and restarted.

5.4 Baseline Systems

Four baselines span the space between no retrieval and agentic navigation, all implemented against the same tools, the same graph, and the same backbone.

5.4.1 The Two Non-Graph Baselines

The **no-retrieval LLM** takes one call, the question, no facts, and an instruction to return an empty entity list if the answer is unknown. WebQSP and CWQ predate current models and are plausibly memorised. So this system measures the floor above which any retrieval result must be read. Its entity strings come from parametric memory rather than the graph. So surface forms mismatch gold more often than for graph-grounded systems, and its score is a *lower bound* on parametric knowledge. That asymmetry is reported rather than corrected. Grading one system leniently and another strictly would corrupt the comparison.

Vector-RAG verbalises every triple into a sentence, embeds it with MiniLM [29], and indexes it. The question retrieves the top 30 by exact inner product, and one call answers from them — dense retrieval [28] applied to a graph. Its limitation is structural rather than incidental: a mediator-reified fact, a marriage or a film role, has no single triple expressing it. So no single retrieved sentence can express it either. Triples with a mediator endpoint are filtered at build time, since a verbalised half-edge is meaningless and leaks raw machine identifiers into answers. Filtering the noise does not repair the paradigm. Both facts are reported.

5.4.2 Static GraphRAG

The structure-without-navigation comparison [4]: resolve the question's topic entities, expand a *one-logical-hop* neighbourhood under the same relation blocklist and mediator pass-through contract every other graph-touching system uses, rerank by embedding similarity to the question, and answer from the top 30 facts in one call. It shares Vector-RAG's answering prompt and cutoff exactly. So the difference between the two isolates retrieval strategy and nothing else.

The radius deserves a precise statement. An earlier draft and the implementation's own docstring both described it as two hops. It is not. Expansion runs once from each topic entity. A mediator node is hopped through and its two relations concatenated. So a mediator path costs two graph edges while remaining one logical hop. An ordinary neighbour is emitted and never expanded again. The run records settle it: every fact is rendered as topic relation endpoint, a genuine second hop would be anchored at an intermediate entity, and all 7,431 logged facts begin with a topic entity.

The radius is not the only thing that bounds this baseline. The fanout is, and it binds harder. Expansion retrieves at most 100 edges per topic entity under a Cypher query with no ORDER BY. So on a higher-degree entity the 100 retrieved are an arbitrary sample of the neighbourhood rather than its best or its first. Measured over the 711 distinct entities the 800 test questions resolve to, 55.1% carry more edges than the cap retains. At least one topic entity is truncated on 72.5% of questions (Appendix E.14). So for a hub topic the baseline answers from a small fraction of the neighbourhood, chosen without a stated criterion. The hub-noise diagnosis of Section 5.9 has to allow that the answer may never have entered the sample the reranker saw. The full system's own get_neighbors shares the property (Appendix B.3) but not the exposure: its cap is 200 and applies per relation rather than per entity, binding on 3.3% of its 16,301 neighbour calls.

Two implementation details fixed on development data were comparability repairs rather than improvements. Neighbourhood expansion initially passed *through* ordinary entities and stopped at mediator nodes, exactly inverted for Freebase and the reason machine identifiers were reaching answers. And retrieved facts are deduplicated by endpoint pair before reranking. Cosine similarity favours redundant phrasings of one connection and was crowding the informative row out of the top 30. The rule keeps the shortest rendering. That choice resolves systematically towards the direct edge over the mediator-expanded path whenever both connect the same pair. That bias runs against this baseline on exactly the questions mediator expansion was added to serve. So it is disclosed rather than left in the code.

What this baseline licenses, and what it does not. A one-hop retriever cannot contain a two-hop chain. So its decay across hop strata (Section 5.9) is partly a property of the radius and not only of the paradigm — a distinction that has to be drawn. The alternative is to read an implementation limit as a result. The evidence that it is a radius effect is visible in the strata: this baseline scores

0.44 on WebQSP's h2, where two-hop questions are largely mediator paths a one-hop expansion *does* reach, against 0.16 on CWQ's h2, where the chains are genuine compositions it cannot reach at all. A retriever failing uniformly on depth would not show that gap. The claim this thesis therefore makes on the strength of this baseline is the narrow one: *retrieval fixed before reasoning must commit to a radius in advance, and any fixed radius is wrong for some question in the set.* It does *not* claim that a two-hop static retriever would score as this one does. That experiment was not run. The honest expectation is that it would recover much of the CWQ h2 gap while losing precision to a larger and noisier window. Re-running it is the first item in Section 6.3.5. Where the per-stratum argument needs a static comparator that is definitely not radius-limited, it uses Vector-RAG, whose one-triple context cannot represent a chain at any radius.

5.4.3 Think-on-Graph (Agentic Baseline)

The credibility-critical comparison. Think-on-Graph [7] is reimplemented on this environment's tools: beam width 3, depth 3, language-model relation pruning, language-model entity pruning, and a sufficiency check at each depth. It is a reimplementation rather than the authors' code, running on this graph with this backbone. That makes it a *stronger* comparison than citing published numbers — graph, model, budget, and scoring rule are held constant, and the algorithm is what varies. It correspondingly means its numbers are not comparable to the published ones and are not presented as such.

Holding the budget constant is not the same as making it irrelevant. A fixed cap binds harder on a method whose cost grows multiplicatively. So the comparison measures search quality and affordability together. Section 5.8 separates them by splitting the results on whether the cap actually truncated the baseline. The answer changes what the aggregate margin should be taken to mean. Under the same 25-call cap the full system operates under, Think-on-Graph is clipped on 117/400 WebQSP questions (29.2%) and 176/400 CWQ questions (44.0%). A clipped run still answers, from whatever it gathered, by the same graceful degradation the full system uses. One asymmetry there favours the baseline and is stated so the clip rates are not mistaken for a count of runs that returned nothing: the baseline stops its search one call *before* the shared limit, reserving that call for the answer it then has to compose, where AGR makes no such reservation.

A second asymmetry is not in the baseline's favour, and it bears on how Section 5.8 should be read. Both systems call the same tools but truncate the returned candidate sets at different widths: Think-on-Graph keeps the first 40 relations per entity and 20 neighbours per relation, the pruning widths of the reimplemented algorithm, against AGR's 300 and 200 (Table 4.3). Since get_relations returns rows by descending fanout, a binding cut discards the low-fanout tail — the material Section 4.2 reports raising AGR's own cap from 80 to 300 to stop discarding. It binds on 31.6% of the entities Think-on-Graph expanded against 1 of AGR's 3,097 (Appendix E.19).

The consequence is a bound on one specific claim. Section 5.8 concludes that the margin over Think-on-Graph is affordability rather than search quality. That conclusion is safe in the direction it is stated, since a narrower candidate set cannot explain why the baseline runs *out of calls*. But the residual gap on unclipped questions is measured against a baseline searching a thinner pool. So it is a lower bound on that baseline's attainable quality rather than an estimate of it. Equalising the widths is the first item in Section 6.3.5.

5.4.4 Variables Held Constant, and Certification

Every system shares the backbone and its sampling parameters, the knowledge environment and its relation blocklist, the entity resolver, the question files, the budget meter and its 25-call cap, the output schema so that scoring reads one field for all systems, and the record schema so that any record self-describes with backbone, budget hash, run configuration, and source commit. Both graph-navigating systems seed from the dataset's annotated topic entities. That choice removes entity linking as a confound. It is stated as a limitation in Section 5.15.1.

Two things are *not* held constant. Both are named rather than buried. The first is deliberate: the baselines get a plain answering prompt while the full system's drafting prompt carries the anti-vacuity and grounding provisions of Section 4.11, which are part of the system under test rather than of the shared harness. The second is a property of the algorithms and is where a reader should look first for a confound — the width of the candidate set each system considers per step, with the measured binding rates and the bound they place on Section 5.8 given in Section 5.4.3, and the static baseline's own fanout cap in Section 5.4.2.

Baselines were debugged exclusively on development data and certified before any test question ran. The pre-registered diagnostic targets the one failure mode that would be an implementation defect rather than a paradigm limit — *retrieval found the gold answer and the system hedged anyway*. That failure mode is checkable, because both retrieval baselines log a sample of what they retrieved. The rule, written before the numbers: under 15% of hedges certifies the baseline as-is, over 30% means an over-abstinent prompt, to be adjusted identically for both baselines with both files re-run. Measured, Vector-RAG scored 0% on both datasets and Static GraphRAG 10% and 14%. Both certified, with two caveats recorded rather than smoothed over. The diagnostic samples the top ten of thirty retrieved facts. So it is a lower bound. And Think-on-Graph's 0% is *vacuous* rather than clean, since it logs no retrieved-fact sample and the diagnostic compared against an empty string. As a multi-step navigator it was not the diagnostic's target. But the reading is a null, not a pass. All ten test files completed at exactly 400 records with zero failures and zero duplicate question identifiers.

5.5 Metrics and the Groundedness Judge

5.5.1 Answer Accuracy: Hits@1 and F1

Both metrics are computed over *entity sets* after Unicode NFKC normalisation, case folding, and whitespace stripping — no fuzzy matching, no substring containment. *Hits@1* is 1 if any predicted entity matches any gold entity. That is the convention in this literature, and it is what makes published numbers comparable at all. *F1* is computed per question between the predicted and gold sets as defined in Section 2.1.4, and averaged over questions.

Reporting both is not redundancy. Hits@1 rewards a system that names one correct entity among several wrong ones. On WebQSP the loophole is wide in the tail if not on the typical question: the median question has 1.5 gold answers, but the mean is 7.15 and the largest lists 237. The precision component of F1 is what closes it. Section 5.7.2 shows how far apart the two metrics can sit for one system. That gap is a property of the answer sets a system emits rather than a measurement of any component inside it: removing the verification layer moves F1 by $+0.004$ on WebQSP and -0.009 on CWQ, neither distinguishable from zero (Section 5.12).

Strict matching raises the obvious worry that "St. Petersburg" versus "Saint Petersburg" is manufacturing failures. It was measured rather than assumed: of the full system's wrong answers, 1 of 65 (2%) on WebQSP and 6 of 99 (6%) on CWQ are surface-form near-misses under aggressive normalisation. The failure census of Section 5.13 is therefore reading real errors.

5.5.2 Accounting for Hedges and Abstentions

A *hedge* is an empty predicted entity set: the system declined to assert. Hedges score zero on both accuracy metrics. But hedge rate is reported as its own column. So the abstention–accuracy trade is visible rather than buried — a system can buy precision with abstention, and the reader is given both numbers to see it happen. Hedges are also not epistemically uniform. That is why Section 5.13 keeps them separate from wrong answers throughout: a wrong answer is a reasoning error, whereas a hedge is usually a coverage gap that never produced a committal claim.

One counting convention follows, and governs every hits/hedges/wrong triple in the thesis. *The three counts always partition the whole set.* Nothing is dropped from the denominator. In particular, questions the environment provably cannot answer (Section 3.3.3) are counted as hedges rather than excluded, even though the system had no route to a correct answer. Excluding them would flatter every condition equally and make the counts incomparable with the accuracy figures, which do include them.

5.5.3 Two-Tier Groundedness and Hallucination Rate

Groundedness is measured at two levels. The word *tier* in this thesis refers to this metric and nothing else.

Tier 1, structural, is deterministic, free, applies to every asserted entity in all 4,000 records, and operationalises Section 2.2.2 directly: *an asserted entity is graph-grounded if and only if it exists as a node in the knowledge environment and is connected to at least one of the question's topic entities within four hops.* The entity-level ungrounded rate is the fraction of asserted entities failing this test. The question-level rate is the fraction of answered questions containing at least one such entity. The test is deliberately strict about surface form — a parametric spelling that does not exist as a node is ungrounded *by definition of the environment*. That strictness is the point of defining hallucination relative to an environment at all.

Tier 2, semantic, asks the harder question Tier 1 cannot: is the entity supported *as the answer*? For a stratified sample of 60 asserted answers per system per dataset (600 judgements), the graph's own shortest connecting paths between topic and asserted entity are retrieved. A language model judges whether any one of them expresses the relationship the question asks about. The governing instruction — that *mere connectedness is not support* — is what separates the two tiers. Two design choices are pre-registered and consequential. Judging is against **the environment** rather than each system's own retrieved context. The logs do not preserve per-system evidence uniformly, and environment-judging asks every system the identical question. The metric therefore applies even to the no-retrieval control. And path retrieval takes the three shortest paths. That means a supporting chain can lose to shorter irrelevant ones. Tier 2 rates are accordingly *conservative lower bounds*, uniformly depressed across all systems, and are read as a between-system comparison rather than as absolute support rates.

5.5.4 Efficiency: Tokens, LLM Calls, and Wall-Clock Latency

Tokens and calls are counted by the shared budget meter, identically for every system, and are exact. The cache replays original usage. So they are unaffected by which runs were warm. Wall-clock latency is not, and is computed over **cold-cache records only**. That restriction is repeated wherever latency appears, beginning with the seconds column of Table 5.10. Where a condition's records are entirely cache replays there is no cold-cache mean at all. That is why Table 5.11 drops its seconds column rather than carry one measured on four conditions of five.

5.5.5 The Groundedness Judge and Its Validation

The Tier-2 judge sees a question, one asserted entity, and up to three connecting paths rendered as relation chains with mediator nodes masked. It does not see which system produced the assertion, whether the assertion was scored a hit, the gold answer, or the system's own reasoning.

Its prompt was fixed before any per-system result was inspected. It runs on the same backbone as the systems it judges, an overlap stated as a limitation in Section 5.15.1 rather than claimed as independence from the model family.

One hundred judged items were drawn and written to a sheet stripped of the judge's verdicts, then labelled by the author *blind*: neither the judge's output file nor the answer key was opened until the sheet was saved. Blinding is what makes the resulting statistic a validity check rather than a measure of anchoring. The written annotation guideline fixed during calibration is reproduced in Appendix D.1. Appendix D.10 records how the sample was drawn and labelled. Each of the guideline's eight rules was forced by a case. The last of them — treat a path expressing only a question's identifying descriptor, rather than its actual ask, as unsupported — is the *composition-echo* case that the failure census later finds at scale (Section 5.14.5).

On $n = 100$ items, observed agreement is 85%, with near-symmetric marginals (54 human-supported against 51 judge-supported) and near-symmetric disagreement (9 human-yes/judge-no against 6 human-no/judge-yes). So neither rater is systematically the more lenient. Cohen's κ [38] is

$$\kappa = \frac{0.8500 - 0.5008}{1 - 0.5008} = 0.6995.$$

The pre-registered threshold was $\kappa \geq 0.7$. Rounded to three decimals this reads 0.700 and appears to clear. It does not: it falls short by 0.0005, and it is reported as short. What follows is a matter of weight rather than of discarding. By the conventional bands [39] a κ of 0.70 is substantial agreement. The task is genuinely hard: whether a relation chain really expresses an asked relationship has real grey area for a careful human, as the guideline's list of decided edge cases attests. The Tier-2 numbers are therefore reported with the κ beside them and the miss stated. No conclusion rests on Tier 2 alone — where it is load-bearing, in the between-system semantic-support comparison of Section 5.10, the finding is also supported by the precision column of the main matrix, which is measured without a judge. Tightening the judge prompt and re-judging a fresh sample until the bar was cleared was available and pre-committed. It was not taken, because the honest number was already in hand, and re-running until a pre-registered threshold is met is exactly what pre-registration exists to prevent.

5.6 Ablation Conditions and the Development Set

Four conditions, each a single deviation from the frozen configuration, each isolating one mechanism:

- **no-planner** — decomposition disabled; the whole question becomes one objective. Anchor seeding is untouched, so this varies decomposition and nothing else.

- **no-backtrack** — max_backtracks $= 0$; a configuration change, which correctly alters the budget hash stamped on the records.
- **no-verifier** — verify_claims $=$ false; grounded drafting with no claim extraction or checking.
- **embedding-only** — $\alpha = 1.0$, removing the language-model term from the hybrid scorer while leaving τ at its frozen value.

The reference row is the *unmodified* system restricted to the same questions, obtained by filtering the main-matrix records to the half-set question identifiers rather than by re-running them. That filtering also supplies the matched pairs McNemar's test requires. Comparing against the full 400-question score would be invalid, accuracy on 400 questions and on 198 being different numbers for the same system. Ablations run on stratified halves — 200 WebQSP and 198 CWQ — as the pre-registered response to the benchmark's burn coming in above projection, cutting the ablation *sample* rather than any system or condition. The odd 198 is floor division within each stratum, CWQ's $137/211/49/3$ halving to $68/105/24/1$. The consequence, reduced statistical power, is confronted in Section 5.12.

5.6.1 The Development Set and the α–τ Sweep

The development set is 80 questions, disjoint from both test samples by an assertion in the test-set builder. Seventy-seven were mined from the train and validation splits to span hop counts and to include mediator-heavy and constraint-bearing questions, and certified for answer reachability by the procedure applied to the test samples. The other three are hand-written and, having no gold answer to reach, fall outside both. That is why the coverage report carries 77 rows. The set *contains* the twenty-question smoke set of Section 4.9. That is why identifiers such as smoke-20 appear in the dev-set listing of Appendix C. Seven questions are unanswerable, in two buckets that are not interchangeable: three hand-written ones naming a fabricated entity, so that a correct refusal can be distinguished from a coverage failure, and four ordinary benchmark questions with real gold answers this environment cannot express (Section 3.3.3).

The two free parameters of the navigation loop are the blend weight α in the hybrid scorer (Section 4.4.2) and the low-signal backtracking threshold τ (Section 4.5.3). They were swept jointly over $\alpha \in \{0.3, 0.5, 0.7, 1.0\}$ and $\tau \in \{0.0, 0.20\}$ — eight conditions, each a complete run of all 80 questions, with two additional draft-only conditions to price the verification layer. Each condition is a constructor argument whose values are stamped into every record it writes. So no condition can be mislabelled after the fact. The chosen values, $\alpha = 0.7$ and $\tau = 0.20$, were frozen on the basis of that sweep and never revisited. Section 5.7.1 reports it.

5.6.2 Implementation, Environment, and Reproducibility

The system is implemented in Python against LangGraph for the state machine, the Neo4j Python driver for graph access [21], and sentence-transformers for embeddings [29], with the graph in a local Neo4j instance. All Cypher [22] is templated inside the tool layer and never composed by a language model (Section 4.2). Randomised operations use seed 42 throughout.

Three practices carry more reproducibility weight than the dependency list. Appendix E records them as techniques. *Records self-describe*: every record carries the backbone description, a budget-configuration hash, the full run configuration, and the source commit. That self-description is what made the cache-replay incident of Section 5.3 diagnosable. *Frozen artifacts are committed and regenerable ones are not*, the question sets and result files embodying spent money and non-repeatable runs where the graph import and the indexes are deterministic products of committed scripts. And *code changes are proved harmless by cache identity* rather than by a passing test suite: re-running an answered question and confirming every call is a cache hit proves byte-identical prompts and hence an untouched frozen path. That proof is the certificate behind the ablation reference row.

Two limits belong here rather than in a footnote. The archived results carry neither per-folder documentation nor generated manifests. So file provenance rests on the self-describing records and on this chapter. And one generated summary in the development-set archive predates a fix applied to the runs it summarises. So rescoring that directory yields different numbers than the committed summary. The discrepancy is stated where those numbers are used (Section 5.7.1) rather than resolved by editing a generated file.

5.7 Main Results

This part reports the measurements Section 5.1 specified. It is organised so that the weakest claims are visible next to the strongest: the headline accuracy result is large and significant on every comparison, the structural groundedness result is emphatic but attributes to a different cause than expected, the semantic groundedness result is underpowered, and three of the four ablations detect nothing. All four outcomes are reported at the same volume.

Every test-set number comes from results/phase4/thesis_numbers.json. Section 5.7.1 reports the development sweep and reads the archived phase-3 runs directly, as Chapter 4 does. Confidence intervals are 95% bootstrap intervals over questions with 10,000 resamples at seed 42 [36]. Paired comparisons use the exact-binomial form of McNemar's test [37] on per-question correctness.

Table 5.3: The development-set sweep, $n = 80$. Hits, hedges, and wrong answers partition the set. Verify indicates claim verification enabled. The two draft-only rows disable it. All rows are from the sweep as run, before the answer-entity filter of Section 4.13.4 was corrected. The frozen condition's post-correction figures are given in the text.

α	τ	Verify	Hits	Hedge	Wrong	F1	Tokens	Calls	Bktrk
0.3	0.00	yes	61	12	7	0.728	4,999	7.4	16
0.3	0.20	yes	61	12	7	0.728	5,030	7.5	18
0.5	0.00	yes	62	10	8	0.743	4,874	7.2	16
0.5	0.20	yes	62	10	8	0.743	4,874	7.2	16
0.7	0.00	yes	64	11	5	0.769	4,925	7.3	17
0.7	**0.20**	**yes**	**64**	11	5	0.769	4,925	7.3	17
1.0	0.00	yes	59	15	6	0.713	3,408	4.9	22
1.0	0.20	yes	59	16	5	0.713	3,400	4.9	26
0.5	0.20	no	64	9	7	0.770	4,807	7.1	16
0.7	0.20	no	66	10	4	0.797	4,858	7.2	17

5.7.1 Development-Set Tuning Outcomes

The blend weight α and the low-signal backtracking threshold τ were swept jointly over the 80-question development set (Section 5.6.1). Table 5.3 reports all eight conditions plus the two draft-only conditions that price the verification layer.

Three readings follow, of which only the first is the one the sweep was run to obtain. Appendix E.13 gives the other two: that $\alpha = 0.7$ is the maximum but the curve is not clean, the wrong-answer column peaking at $\alpha = 0.5$ where the system commits more often, and that τ changes almost nothing at the values swept. That flatness is what the signal-maximum formulation of Section 4.5.3 predicts. Both are reported rather than smoothed away. The value $\tau = 0.20$ was frozen on the basis of a sweep in which it changed almost nothing. That is stated plainly rather than presented as a tuned choice.

Verification costs accuracy on this set. Against the frozen condition, the draft-only row scores 66 hits to 64 and F1 0.797 to 0.769. After the answer-entity filter of Section 4.13.4 was corrected — a change made on development data before any test question ran — the frozen condition scores 65 hits, 11 hedges, 4 wrong, F1 0.785. Across those two rows the layer costs one correct answer and removes no wrong ones on this set of 80. That comparison spans the correction, since the draft-only condition was never re-run after it. It is therefore the closest available reading rather than a matched pair. Section 5.12 finds the same thing at $n \approx 200$ on test data. Section 5.15.1 explains why that is the expected result rather than a disappointing one. Only the frozen condition was re-run after the correction. So Table 5.3 is the pre-correction sweep throughout — internally consistent, since all eight conditions ran under the same code. The archived per-question records are authoritative where the committed summary disagrees.

Table 5.4: Main results, $n = 400$ per dataset. Brackets are 95% bootstrap intervals. **P** and **R** are mean per-question precision and recall. **Hedge** is the share of questions on which the system asserted nothing. The Hits@1 ceilings are 97.0% and 99.2% (Table 5.1). Every figure here is measured inside the environment of Chapter 3, whose per-question construction is what puts those ceilings so high. The levels are therefore not comparable with published results over full Freebase, and Section 5.15.1 states why.

Data	System	Hits@1	F1	P	R	Hedge
WebQSP	No-retrieval	0.453 [0.403, 0.500]	0.305 [0.266, 0.345]	0.380	0.303	12.2%
WebQSP	Vector-RAG	0.568 [0.517, 0.618]	0.432 [0.389, 0.475]	0.469	0.460	27.0%
WebQSP	GraphRAG	0.340 [0.292, 0.388]	0.262 [0.224, 0.302]	0.279	0.284	55.8%
WebQSP	Think-on-Graph	0.660 [0.613, 0.705]	0.477 [0.436, 0.518]	0.571	0.480	18.8%
WebQSP	**AGR**	**0.755** [0.713, 0.797]	**0.642** [0.600, 0.684]	**0.697**	**0.653**	**8.2%**
CWQ	No-retrieval	0.307 [0.263, 0.352]	0.279 [0.236, 0.323]	0.285	0.282	38.0%
CWQ	Vector-RAG	0.203 [0.163, 0.242]	0.168 [0.133, 0.204]	0.171	0.189	48.2%
CWQ	GraphRAG	0.205 [0.168, 0.245]	0.183 [0.147, 0.222]	0.187	0.194	60.8%
CWQ	Think-on-Graph	0.435 [0.388, 0.482]	0.378 [0.332, 0.423]	0.403	0.384	22.5%
CWQ	**AGR**	**0.522** [0.475, 0.570]	**0.469** [0.423, 0.514]	**0.484**	**0.481**	23.0%

5.7.2 Main Results on WebQSP

On WebQSP, AGR answers 302 of 400 questions correctly: 75.5% against a reachability ceiling of 97.0%, +9.5 points over Think-on-Graph and +30.2 over the parametric floor. The floor matters. WebQSP is a decade old and plausibly in the backbone's pretraining data. The no-retrieval control confirms it: 45.3% of these questions are answerable from memory alone. Any claim about what retrieval contributes has to be read against 0.453, not against zero.

F1 widens the margin that Hits@1 understates. Against Think-on-Graph the Hits@1 gap is +9.5 points. The F1 gap is +16.5 (0.642 against 0.477). The precision column shows where it comes from (0.697 against 0.571). WebQSP's questions have a mean of seven gold answers. So a system that asserts a long list collects Hits@1 credit for the one entity that matches and pays nothing for the rest. Precision is what charges for it. The two columns together support the sentence this thesis leads with: AGR answers more questions and asserts fewer falsehoods, at an 8.2% hedge rate — the lowest of the five systems. So the precision is not bought with abstention.

Raw hits mislead on the retrieval baselines. The correction is a finding. GraphRAG scores 0.340, below the no-retrieval control's 0.453. That gap invites the conclusion that the implementation is broken. Put the wrong-answer counts beside the hits and the picture inverts. No-retrieval asserts on 351 of 400 questions and is wrong on 170 of them — 51.6% *assertion precision*, the share of the questions a system commits on that it answers correctly. It is a per-question measure. It is also distinct from the per-entity precision of Table 5.4, which is averaged over all 400 questions whether the system committed on them or not. GraphRAG asserts on 177 and is wrong on 41 — 76.8%. The mechanism is the prompt: the retrieval baselines are instructed to answer only from the facts given and to return nothing otherwise. When retrieval

Table 5.5: Paired McNemar tests, AGR against each baseline, on per-question correctness over the same 400 questions. Every comparison rejects.

Dataset	Baseline	AGR only	Baseline only	p
WebQSP	No-retrieval	152	31	2.3×10^{-20}
WebQSP	Vector-RAG	117	42	2.2×10^{-9}
WebQSP	GraphRAG	186	20	6.5×10^{-35}
WebQSP	Think-on-Graph	76	38	4.8×10^{-4}
CWQ	No-retrieval	121	35	2.7×10^{-12}
CWQ	Vector-RAG	151	23	2.8×10^{-24}
CWQ	GraphRAG	148	21	1.0×10^{-24}
CWQ	Think-on-Graph	86	51	3.5×10^{-3}

misses, they therefore abstain, while the no-retrieval control has no such constraint and guesses from memory. On a memorised benchmark, guessing pays. *A grounded system scoring below an ungrounded one on raw hits while committing four times fewer falsehoods* is the contamination story the control was built to expose. It is reported as a result rather than as an anomaly.

5.7.3 Main Results on ComplexWebQuestions

On CWQ, AGR answers 209 of 400: 52.2% against a 99.2% ceiling, +8.7 over Think-on-Graph and +21.5 over the floor. Every system scores lower here. The ordering changes in one important way: *both* static retrieval baselines fall below the parametric control. Vector-RAG drops from 0.568 on WebQSP to 0.203, GraphRAG from 0.340 to 0.205, while no-retrieval falls only from 0.453 to 0.307.

The reason is structural rather than incidental. The two baselines earn that description to different degrees. Vector-RAG's context is a single verbalised triple. One triple cannot contain a chain at any retrieval quality or any radius. That is a paradigm limit Section 5.4 predicted at implementation time. Section 5.9 demonstrates it per stratum. GraphRAG's context is a one-logical-hop neighbourhood (Section 5.4.2). Its CWQ number is therefore the weaker evidence of the two: it confounds the paradigm with the radius. Part of the fall is attributable to a boundary chosen in advance rather than to static retrieval as such. The claim rests on the first baseline.

The number worth leading with on this dataset is assertion precision. AGR and Think-on-Graph commit at effectively the same rate: 308 and 310 of 400 questions answered rather than hedged. Out of those near-identical commitments, AGR is right 209 times and wrong 99, against 174 and 136. That is 67.9% assertion precision against 56.1%: on the harder, deeper benchmark, the system with an explicit plan, a guided scorer, and a claim check finds 35 more answers and asserts 37 fewer falsehoods than beam search on the same graph with the same model, from the same number of commitments — a margin Section 5.8 attributes to the budget rather than to resolution quality.

Table 5.6: The comparison against the agentic baseline, split on whether the shared 25-call cap truncated it. Clipping is read from the per-record budget_exhausted trace flag rather than from a call count, since a run may spend its last permitted call on the step that finishes it. The *combined* rows reproduce the Hits@1 column of Table 5.4, recomputed independently from the raw records. Generated from results/phase4/thesis_numbers.json.

Dataset	Subset	n	ToG	AGR	Δ
WebQSP	ToG ran to completion	283	0.852	0.788	-0.064
	ToG clipped by the cap	117	0.197	0.675	$+0.478$
	combined	400	*0.660*	*0.755*	$\mathbf{+0.095}$
CWQ	ToG ran to completion	224	0.629	0.607	-0.022
	ToG clipped by the cap	176	0.188	0.415	$+0.227$
	combined	400	*0.435*	*0.522*	$\mathbf{+0.087}$

Table 5.5 settles the comparisons that matter. All eight reject, including the two against the agentic baseline — the only comparison in the table where both systems navigate the same graph, with the same model, under the same call budget. The discordant counts make the margin concrete rather than distributional: on CWQ, AGR uniquely solves 86 questions Think-on-Graph misses against 51 the other way. What that budget does to the comparison is the subject of the next section. It changes what the margin means.

5.8 Where the Margin Over the Agentic Baseline Comes From

The shared budget is a control. Like any control, it is doing something. The call cap binds on Think-on-Graph far more often than on AGR: beam search costs width times depth. Section 5.11 records the clip rates. So "same budget" does not mean "same freedom to search". Splitting the 400 questions of each dataset on whether the cap actually cut Think-on-Graph off separates the two things the aggregate confounds: how good its search is, and whether it can afford to run that search to completion.

On the questions it is allowed to finish, Think-on-Graph is ahead on both datasets. It scores 0.852 against AGR's 0.788 on WebQSP and 0.629 against 0.607 on CWQ. The entire aggregate margin comes from the 29.2% and 44.0% of questions where the cap truncates it. There its accuracy falls to 0.197 and 0.188 while AGR's holds at 0.675 and 0.415. This is stated plainly because it is the first thing an unsympathetic reading of Table 5.5 should look for, and because burying it would misrepresent what was measured.

It does not weaken the result. But it does change its content. The finding is not that guided best-first search resolves entities better than language-model-pruned beam search — on the evidence here it does not. The unclipped subset is the fairest available estimate of that. The finding is that *beam search cannot afford to finish under a budget a deployed system would*

actually impose. *Guided search can.* Width times depth is multiplicative and paid on every question. A plan plus a scored frontier is not. A comparison run without a cap would measure search quality in the abstract. That would be a legitimate experiment, and a different one. This thesis fixed the budget in advance (Section 5.4) because cost is a property of the method rather than a nuisance to be controlled away. On that reading the clip rate is not an artefact of the setup. It is the result. Section 5.11 prices the same finding in tokens.

Three qualifications keep the claim honest. The cap of 25 calls is itself a choice. A larger one would move the split — the clip rates are reported so a reader can see how far. The reimplementation's beam width and depth follow the original paper. So its cost profile is the published algorithm's rather than a weakened variant. That fidelity is what makes the cost finding meaningful instead of circular. But the candidate widths *beneath* the beam do not come from the paper: the baseline prunes from 40 relations and 20 neighbours where AGR prunes from 300 and 200. Those cuts bind on about a third of its expansions against almost none of AGR's (Section 5.4.3). That asymmetry cannot explain the clipping — a thinner candidate set makes a step cheaper, not dearer. So the affordability finding stands. It does bear on the sentence before it: 0.852 and 0.629 on the unclipped subset are a *lower bound* on what this baseline's search could resolve given AGR's candidate widths, not an estimate of it. The claim that guided search does not out-resolve beam search is correspondingly the weaker one.

The breadth of AGR's answers is checked rather than assumed. On answered WebQSP questions AGR names 3.37 entities against Think-on-Graph's 2.56. A metric scoring a hit on any match rewards naming more. The check is entity precision over those same answered questions — Table 5.4's measure restricted to the answered subset, not the per-question assertion precision of Section 5.7.3 — where AGR leads 0.760 to 0.703. On CWQ the figures are 1.54 against 1.40 entities and 0.629 against 0.520 precision. Breadth bought without accuracy would show up as precision falling while the entity count rose. It does not.

5.8.1 Positioning Against Published RoG Results

Two facts in this thesis oblige a direct comparison. The knowledge environment of Section 3.1.3 is built from the per-question subgraphs the RoG distribution publishes. Section 2.3.3 surveys RoG [9] as the closest published system in the path-retrieval family. A reader who notices both is entitled to ask how the two systems compare. The aggregate answer does not favour this work.

RoG leads on all four figures. The margin is not sampling noise. AGR's 95% confidence intervals on Hits@1 are $[71.3, 79.7]$ on WebQSP and $[47.5, 57.0]$ on CWQ. RoG's 85.7 and 62.6 fall outside both. That is stated before the qualifications. The qualifications do not reverse it, and a comparison offered only with its excuses attached is not a comparison.

Three differences bound what Table 5.7 establishes. The first is decisive. The other two are ordinary.

Table 5.7: AGR against RoG's published figures, in points. RoG's row is Table 1 of [9], measured over the full test splits (1,628 and 3,531 questions). AGR's row is Table 5.4, measured over the certified 400-question samples of Table 5.1. Those samples are stratified draws from those same splits. The two rows are therefore computed over nested populations but under different protocols. The paragraphs below state what that permits and what it forbids.

	WebQSP		CWQ	
System	**Hits@1**	**F1**	**Hits@1**	**F1**
RoG (published)	85.7	70.8	62.6	56.2
AGR (this work)	75.5	64.2	52.2	46.9

RoG is trained on these benchmarks. AGR is trained on nothing. Section 5.1 of [9] states that its backbone is "instruction finetuned on the training split of WebQSP and CWQ as well as Freebase for 3 epochs". In the distribution this thesis also uses, those training splits hold 2,830 and 16,900 questions. AGR performs no training of any kind: the backbone is frozen, prompted at temperature 0 (Section 5.3), and the system has seen no question from either benchmark before it is evaluated. The comparison is therefore between a supervised system and a zero-shot one. That is a difference in kind rather than in degree. It is the same objection Table 2.2 already raises against every fine-tuned entry in the surveyed literature.

The backbones differ. RoG fine-tunes LLaMA2-Chat-7B. AGR prompts the model named in Section 5.3. Section 2.5 argues that backbone accounts for much of the spread in cross-paper tables. That is precisely why Section 5.1 holds it constant across the five systems it measures. It cannot be held constant against a system whose contribution *is* its fine-tuned weights. So no protocol available here would make this row comparable in the sense Table 5.4 uses the word.

The evaluation populations differ. RoG reports over the full test splits. The figures here are over stratified 400-question samples of the same splits (Section 5.2). The samples preserve the hop-stratum proportions of their populations. So they estimate the same quantity. The intervals quoted above are the estimate's uncertainty. RoG publishes no interval. So the gap can be bounded from one side only.

What the comparison does settle is worth stating positively. It fixes AGR's place in the published landscape: a training-free system reaching 75.5 and 52.2 against a supervised system's 85.7 and 62.6, on the same data, with no in-domain examples. And it locates this thesis's contribution somewhere other than the accuracy column. RoG's answers are read off retrieved paths. So they are faithful by construction. The pre-generation verification question of Section 4.10 does not arise for them (Section 2.3.3). The price is that the plan is generated once, before retrieval, with no mechanism to revise it when it fails to instantiate. AGR's claims are the opposite shape: generated freely and then checked against the graph the agent actually walked. That check is what makes an unsupported assertion detectable at all. Neither the component-level ablation of Section 5.12, with the stratum-dependent decomposition finding it produced, nor the failure census of Section 5.13 has a counterpart in RoG's evaluation. None of them would be settled by

Table 5.8: Hits@1 and F1 per hop stratum. Stratum sizes are those of Table 5.1. WebQSP's h3plus has $n = 4$ and is not interpreted.

Data	System	h1	h2	h3plus	unreach.
WebQSP	No-retrieval	0.43 / 0.29	0.52 / 0.36	0.25 / 0.00	0.17 / 0.17
WebQSP	Vector-RAG	0.82 / 0.63	0.12 / 0.07	0.00 / 0.00	0.08 / 0.08
WebQSP	GraphRAG	0.30 / 0.23	0.44 / 0.35	0.25 / 0.07	0.08 / 0.08
WebQSP	Think-on-Graph	0.63 / 0.45	0.78 / 0.58	0.25 / 0.07	0.17 / 0.17
WebQSP	**AGR**	0.75 / 0.64	**0.84 / 0.73**	0.00 / 0.00	0.08 / 0.08
CWQ	No-retrieval	0.38 / 0.35	0.30 / 0.27	0.14 / 0.14	0.00 / 0.00
CWQ	Vector-RAG	0.33 / 0.27	0.14 / 0.11	0.10 / 0.09	0.67 / 0.67
CWQ	GraphRAG	0.34 / 0.30	0.16 / 0.14	0.02 / 0.02	0.33 / 0.33
CWQ	Think-on-Graph	**0.53** / 0.48	0.37 / 0.32	0.45 / 0.32	0.33 / 0.33
CWQ	**AGR**	0.46 / 0.42	**0.55 / 0.49**	**0.57 / 0.50**	0.33 / 0.33

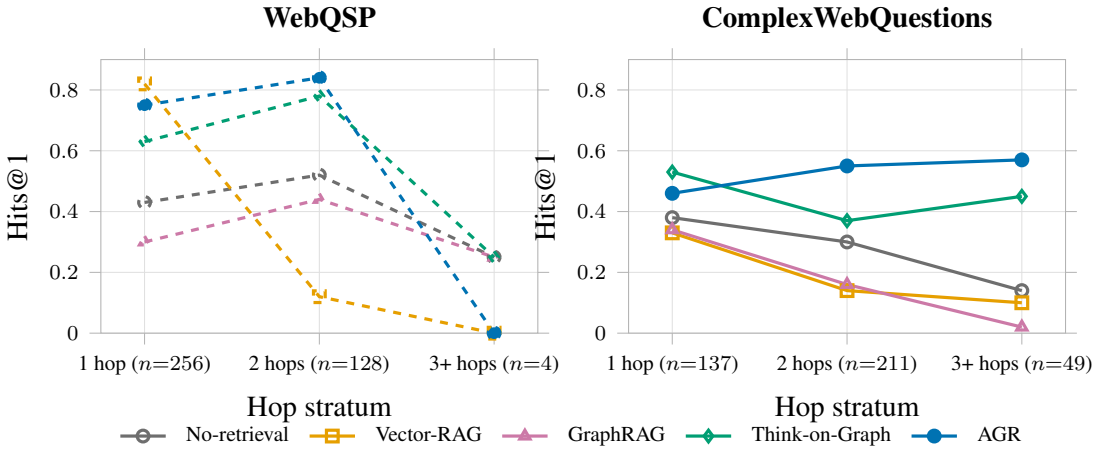

Figure 5.1: Hits@1 across hop strata. The shape is the finding: on ComplexWebQuestions AGR is the only system whose accuracy rises with hop count. Every other profile ends below where it started — three of them monotonically, Think-on-Graph by falling and partly recovering. WebQSP is drawn dashed because its h3plus stratum holds four questions and cannot carry a trend. The stratum sizes are printed on the axis for that reason. The unreachable stratum is omitted here and discussed below, since by construction it measures assertion rather than accuracy. Generated from results/phase4/thesis_numbers.json.

closing the gap in Table 5.7.

5.9 Accuracy Broken Down by Hop Count

Table 5.8 is the direct answer to RQ1. It answers it with a shape rather than with a difference of means — Figure 5.1 is that shape.

On CWQ, AGR's accuracy rises with hop count. Hits@1 goes $0.46 \rightarrow 0.55 \rightarrow 0.57$ across h1, h2, h3plus. F1 goes $0.42 \rightarrow 0.49 \rightarrow 0.50$. It is the only system on that dataset that ends above

where it started. The other four all end below. Three of them decay monotonically: No-retrieval $0.38 \rightarrow 0.30 \rightarrow 0.14$, Vector-RAG $0.33 \rightarrow 0.14 \rightarrow 0.10$, and GraphRAG $0.34 \rightarrow 0.16 \rightarrow 0.02$, though that last row is radius-limited and is read with the qualification given below. Think-on-Graph is the exception to the monotonicity but not to the direction: it falls from 0.53 to 0.37 and partially recovers to 0.45, still 0.08 below its own one-hop score. The distinction is worth keeping — "every other system decays" would be the tidier sentence. It is not what the table says. A system that gets *better* as the required chain gets longer is not merely ahead on average: it is the only one in the table whose profile is consistent with actually traversing the chain. That monotonicity is a stronger form of the multi-hop claim than any aggregate could carry.

Think-on-Graph beats AGR on CWQ's one-hop stratum (0.53 against 0.46, $n = 137$). This is reported rather than buried because it makes the rest of the story more credible, not less. Section 5.12 identifies the mechanism: decomposition costs accuracy on questions that do not need decomposing. CWQ's h1 stratum is full of them.

Vector-RAG's WebQSP profile is the paradigm in two cells. It scores 0.82 at one hop — the best single-hop number anywhere in the table, better than AGR's 0.75 — and 0.12 at two. When one triple suffices, dense retrieval over verbalised triples is excellent. When a chain is required, it has nothing to offer. This is the cell the multi-hop argument rests on. A one-triple context is chain-incapable by construction rather than by configuration.

GraphRAG's WebQSP inversion has two causes. Only one of them was originally named. It scores 0.30 at h1 against 0.44 at h2. Part of that is hub noise: popular one-hop entities have large neighbourhoods. The answer is drowned in the reranked window — or, more exactly, drowned *or absent*, since the retrieval step samples at most 100 edges per topic entity without an ordering, and 55.1% of its topic entities carry more edges than that (Section 5.4.2). Which of the two applies to any given question is not recoverable from the run records. But the inversion also marks the edge of the retrieval radius. The same section scopes that as a bound on this implementation rather than on static retrieval as such. WebQSP's h2 stratum is largely mediator paths, which a one-hop expansion reaches by hopping through the mediator. CWQ's h2, by contrast, is genuine composition, which it cannot reach at all: 0.44 against 0.16 on nominally the same stratum depth. Read GraphRAG's per-stratum row as measuring what its radius covers, not how far static retrieval can see.

The unreachable stratum is a hallucination probe, not an accuracy measurement. By construction no path to gold exists within the hop bound for these questions. So every "hit" is an assertion the environment cannot support that happened to match the label. Vector-RAG scores 0.67 on CWQ's three unreachable questions and no-retrieval 0.17 on WebQSP's twelve. The counts are tiny, and no weight is placed on them. But the direction is exactly what Section 5.10 measures at scale: Hits@1 will happily reward what RQ2 is about the absence of.

Table 5.9: Two-tier groundedness. Tier 1 is deterministic over every asserted entity in all 4,000 records. Tier 2 is a language-model entailment judgement over a stratified sample of 60 assertions per system per dataset. The judge itself was validated separately, at $\kappa = 0.6995$ over 100 paired annotations (Section 5.5.5).

		Tier 1: structural			Tier 2
Data	**System**	**Entities**	**Ungrounded**	**Questions**	**Supported**
WebQSP	No-retrieval	661	179 (27.1%)	37.6%	63.3%
WebQSP	Vector-RAG	1040	38 (3.7%)	6.2%	61.7%
WebQSP	GraphRAG	800	6 (0.8%)	2.8%	55.0%
WebQSP	Think-on-Graph	832	**0 (0.0%)**	**0.0%**	61.7%
WebQSP	AGR	1235	**0 (0.0%)**	**0.0%**	**66.7%**
CWQ	No-retrieval	340	42 (12.4%)	13.7%	36.7%
CWQ	Vector-RAG	289	7 (2.4%)	3.4%	**50.0%**
CWQ	GraphRAG	224	2 (0.9%)	1.3%	36.7%
CWQ	Think-on-Graph	433	**0 (0.0%)**	**0.0%**	43.3%
CWQ	AGR	474	**0 (0.0%)**	**0.0%**	48.3%

5.10 Groundedness and Hallucination Rate Across Systems

5.10.1 Tier 1: Structural Hallucination Is a Property of the Paradigm

The differential is emphatic. The parametric control asserts 661 entities on WebQSP, of which 179 — 27.1% — have no basis in the knowledge environment. And 37.6% of the questions it answers contain at least one such entity. Both navigating systems sit at exactly zero, over 1,235 and 832 asserted entities. Parametric answering hallucinates structurally at double-digit rates. Graph-navigated answering eliminates structural hallucination outright.

But Think-on-Graph reaches zero as well. That changes the attribution. Structural groundedness is a property of the *navigation paradigm* — any system that copies its answer entities out of triples it actually traversed inherits it — not of this thesis's verification layer specifically. The honest claim is therefore narrower and more testable than the one the work set out to make: *both navigating systems achieve zero structural hallucination; they differ at the level of semantic support*, where Think-on-Graph's precision of 0.571 and 0.403 against AGR's 0.697 and 0.484 (Table 5.4) shows it asserting substantially more entities that exist, connect to the topic, and were copied from real triples — and are not the answer. Section 1.3 builds on exactly this gap.

Two readings of the middle band belong on the record. The no-retrieval control's CWQ rate (12.4%) is less than half its WebQSP rate (27.1%) not because it fabricates less on harder questions but because it hedges more — 38.0% against 12.2%. Abstention suppresses the measured hallucination rate. That is the abstention trade appearing in a third metric. And Vector-RAG's 38 ungrounded entities are mostly not fabrications: its index is global. So retrieval can surface facts about a similarly-named subject that exists in the graph but has no connection

to the question's topics. The model then copies faithfully from them. The metric correctly flags those as unanchored to the question. But the mechanism is retrieval error rather than generation error. Structural ungroundedness conflates the two. The distinction is recoverable from whether the entity appears in the system's own retrieved facts.

5.10.2 Tier 2: A Narrow Band and an Underpowered Comparison

Tier 2 asks whether the asserted entity is supported *as the answer*. Under the instruction that mere connectedness is not support, every system lands in a 36.7–66.7% band. AGR is highest on WebQSP at 66.7%. On CWQ it is second: 5.0 points ahead of Think-on-Graph, which is the comparison the rest of this chapter makes, but 1.7 behind Vector-RAG's 50.0%. Its margin over the agentic baseline is 5.0 points on both datasets. Its lead over *every* system is a WebQSP result only.

Neither gap is resolvable at this sample size. With $n = 60$ per cell the 95% binomial interval has a half-width of roughly 12–13 points. So a 5-point difference is inside the noise — as is Vector-RAG's 1.7. No conclusion in this thesis rests on either. What the tier does establish is the level: semantic support is *hard* for every system, including the two that are structurally perfect. Zero structural hallucination coexists with roughly a third to a half of assertions failing a strict semantic-support test. That coexistence is the single most useful thing the metric says. It says it about the paradigm rather than about any one system.

Two properties of the measurement bound its interpretation, both pre-registered. The rates are conservative lower bounds, depressed uniformly across systems. That uniformity preserves the between-system comparison while making the absolute levels untrustworthy. And the judge was validated at $\kappa = 0.6995$ against a pre-registered bar of 0.7 (Section 5.5.5) — substantial agreement, marginally short of the threshold, and a further reason to treat this tier as corroborating rather than decisive.

5.11 Efficiency and Cost

5.11.1 Token and Call Budgets per System

Agency costs roughly an order of magnitude over one-shot retrieval: $4{,}511$ and $6{,}818$ tokens per question against ~ 500, and 11–17 seconds against one to two. That is the price of RQ1's accuracy. It is stated as a price.

The comparison that carries information is against the other agentic system, since the two run under an identical 25-call cap. AGR uses *half* the calls — 6.2 against 12.8 on WebQSP, 8.9 against 18.2 on CWQ — with more tokens per call, because each call carries a richer prompt: a

Table 5.10: Cost per question. Latency is computed over cold-cache records only. Clip rate is the share of questions on which the shared 25-call budget was exhausted mid-search.

Data	System	Tokens	Calls	Seconds	Clipped
WebQSP	No-retrieval	113	1.0	1.2	—
WebQSP	Vector-RAG	527	1.0	1.4	—
WebQSP	GraphRAG	531	1.0	2.1	—
WebQSP	Think-on-Graph	3,615	12.8	12.9	29.2%
WebQSP	AGR	4,511	6.2	11.2	0.0%
CWQ	No-retrieval	118	1.0	1.0	—
CWQ	Vector-RAG	535	1.0	1.4	—
CWQ	GraphRAG	470	1.0	2.2	—
CWQ	Think-on-Graph	5,572	18.2	15.2	44.0%
CWQ	AGR	6,818	8.9	17.2	0.0%

plan, a scored candidate set, a fact window. Fewer, better-informed decisions rather than more, blinder ones.

The clip column is the sharpest form of that. Beam search costs are multiplicative in width and depth. So under the shared budget Think-on-Graph exhausts its calls before finishing on 29.2% of WebQSP questions and 44.0% of CWQ questions, while AGR never does. On nearly half the multi-hop benchmark the baseline is answering from a truncated search. This is a fair constraint (Section 5.4.3): both systems live under the same cap. It is the efficiency finding rather than a caveat on it: under an identical budget, AGR converts its calls into correct answers at roughly twice the rate.

Never reaching the call cap is not the same as running unbounded. The distinction matters. The rest of this thesis restates this result. AGR's own budgets bind, the depth cap of Table 4.3 hardest: the meter refuses a deeper expansion on 25.6% of test questions, against 6.2% for backtracks and 1.9% for repair iterations. Section 4.13 and Section 4.14.1 each narrate a run that ends that way. What the clip column establishes is the narrower and more useful thing — that of the two systems run under one shared call budget, that budget truncates only one of them.

5.11.2 The Accuracy–Cost Frontier

Reading Table 5.4 against Table 5.10, and plotted as Figure 5.2, the frontier depends on which cost is charged. The two costs this thesis records do not agree.

On tokens, the axis of Figure 5.2, Think-on-Graph is not dominated. It buys its lower accuracy at a genuinely lower token price on both datasets — $3{,}615$ against AGR's $4{,}511$ on WebQSP, $5{,}572$ against $7{,}039$ on CWQ — and sits on the Pareto set of both, alongside the no-retrieval control at 113 tokens and, on WebQSP, Vector-RAG at 527. The interior points are the static graph retrievers, which the parametric control beats on both axes at once. Any claim that Think-on-Graph is dominated has to name an axis on which that is true.

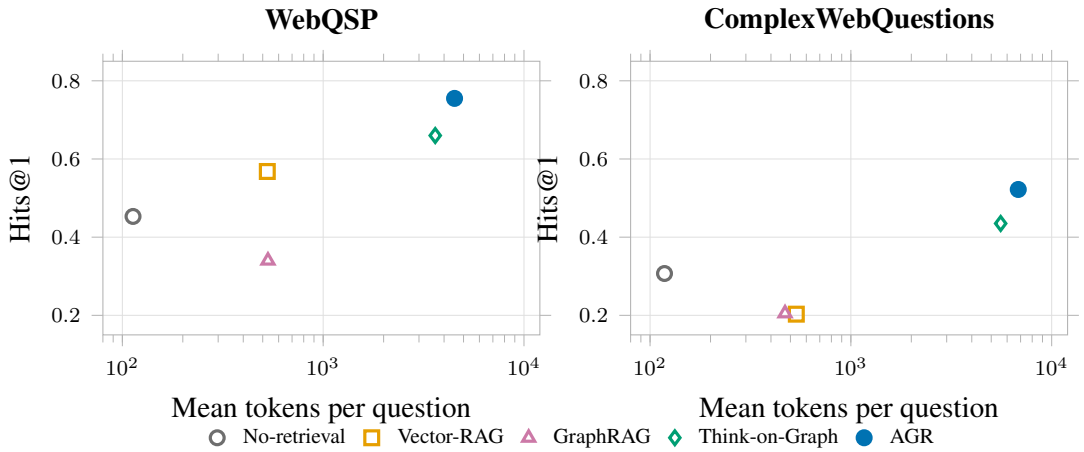

Figure 5.2: Accuracy against token cost, one point per system. The horizontal axis is logarithmic: the agentic systems cost roughly an order of magnitude more than the one-shot ones, and a linear axis would compress every non-agentic system into a single column. Read this figure for the token frontier only. Think-on-Graph sits below *and* to the left of AGR, so it is a frontier point on this axis, not a dominated one. The domination claim holds on calls, which are not plotted and are given in Table 5.10. Generated from results/phase4/thesis_numbers.json.

On calls it is dominated. Calls are what the budget meters (Section 4.7). Think-on-Graph spends 12.8 and 18.2 calls against AGR's 6.2 and 8.9 — lower Hits@1 at roughly double the call cost, on both datasets, with no interior ambiguity. The two costs diverge because AGR spends more than twice as many tokens per call, converting fewer, larger calls into a comparable token total. Which cost binds is a deployment question: token billing separates the two systems very little, whereas a per-call rate or latency budget separates them by a factor of two. It is the call budget that clips Think-on-Graph on 29% and 44% of questions (Section 5.8).

Around that, the corners are unchanged. Vector-RAG on WebQSP is the cheap corner, 0.568 Hits@1 for 527 tokens and 0.82 on the one-hop stratum, which makes it the right choice for shallow questions and useless for deep ones. AGR is the accurate corner on both datasets. And the no-retrieval control holds the cheap end of the token frontier while being off any accuracy-worthy frontier once precision is counted: it remains the cheapest way to be right about half the time on WebQSP, which is the contamination point restated as an economic one.

5.12 Ablation Study

Four single-deviation conditions on stratified halves ($n = 200$ WebQSP, $n = 198$ CWQ), each against the unmodified system restricted to the same questions. Table 5.11 reports the conditions. Table 5.12 summarises the component attribution. Throughout, Δ is *ablated minus reference*. So a positive Δ means removing the component *improved* the metric.

Table 5.11: Ablation conditions on stratified halves. The reference row is the frozen system restricted to the same questions, at zero additional cost. P and R are the per-question means defined in Section 5.5 and reported for the full systems in Table 5.4. They are carried here so that the precision effect of each condition can be read off directly rather than taken on trust from the surrounding prose. Cost is reported in tokens and calls only: latency is omitted because every record in the no-verifier condition is a full cache replay. So a seconds column would not be comparable across conditions.

Data	Condition	Hits@1	F1	P	R	Hedge	Tok.	Calls
WebQSP	reference	0.755 [0.695, 0.810]	0.659 [0.599, 0.717]	0.710	0.664	8.5%	4,562	6.2
WebQSP	no-planner	**0.830** [0.775, 0.880]	**0.742** [0.687, 0.794]	0.788	0.749	5.5%	3,159	4.0
WebQSP	no-backtrack	0.745 [0.685, 0.805]	0.646 [0.586, 0.705]	0.704	0.649	9.5%	4,261	5.8
WebQSP	no-verifier	0.760 [0.700, 0.820]	0.663 [0.604, 0.721]	0.711	0.673	8.0%	4,460	6.1
WebQSP	embedding-only	0.740 [0.680, 0.800]	0.643 [0.583, 0.702]	0.701	0.644	13.5%	3,413	4.4
CWQ	reference	0.495 [0.424, 0.566]	0.445 [0.379, 0.511]	0.461	0.451	23.2%	7,105	9.1
CWQ	no-planner	0.434 [0.369, 0.505]	0.398 [0.333, 0.464]	0.407	0.405	26.8%	5,617	6.7
CWQ	no-backtrack	0.490 [0.419, 0.561]	0.445 [0.379, 0.511]	0.456	0.453	28.3%	5,792	7.4
CWQ	no-verifier	0.490 [0.419, 0.561]	0.436 [0.371, 0.503]	0.451	0.446	20.2%	6,704	8.7
CWQ	embedding-only	0.475 [0.409, 0.545]	0.437 [0.370, 0.502]	0.452	0.444	28.3%	4,925	5.9

5.12.1 Without the Planner: A Stratum-Dependent Effect

This is the one condition that reaches significance. It points the opposite way to the design intent.

On WebQSP, *removing* the planner improves every axis at once: Hits@1 $0.755 \rightarrow 0.830$, F1 $0.659 \rightarrow 0.742$, hedge rate $8.5\% \rightarrow 5.5\%$, tokens $4{,}562 \rightarrow 3{,}159$, calls $6.2 \rightarrow 4.0$. McNemar gives $p = 5.9 \times 10^{-3}$ on 6 against 21 discordant pairs. The per-stratum breakdown locates the damage precisely: h1 rises $0.75 \rightarrow 0.85$ and h2 rises $0.84 \rightarrow 0.89$. So decomposition is hurting simple *and* two-hop questions on this dataset. The planner prompt's first worked example instructs the model not to decompose single-hop questions. The ablation proves that instruction under-triggers, with a p-value.

On CWQ the sign reverses. Significance is not reached: Hits@1 $0.495 \rightarrow 0.434$, 27 against 15 discordant pairs, $p = 0.088$. The stratum table shows a coherent mechanism rather than noise: h1 *improves* ($0.49 \rightarrow 0.54$) while h2 collapses ($0.48 \rightarrow 0.34$) and h3plus moves little ($0.62 \rightarrow 0.54$). That pattern is precisely "decomposition helps genuine composition and hurts questions that do not need it". But it does not clear $\alpha = 0.05$ at $n = 198$. The honest statement carries both halves: *the planner's benefit is stratum-conditional and significant on WebQSP ($p = 0.006$); the CWQ pattern points the same way but does not reach significance at this sample size ($p = 0.088$).*

The finding is therefore not "the planner helps" or "the planner hurts" but that its value is conditional on question structure, and that applying decomposition unconditionally is the defect. Section 5.14.1 reads the 21 and 15 questions where removing the planner flipped a failure into a success and names the mechanisms. Section 6.3 turns them into an adaptive gating proposal that this ablation earned rather than motivated.

5.12.2 Without Backtracking

Hits@1 moves -0.010 on WebQSP and -0.005 on CWQ. F1 moves -0.013 and 0.000. Discordant pairs are 5 against 3 and 6 against 5, giving $p = 0.73$ and $p = 1.00$. There is no detectable effect at this sample size.

Two readings are available. The data does not separate them: backtracking's contribution may be small because the scorer and the verifier already prevent most of the excursions it exists to correct, or $n \approx 200$ may simply be too few questions to resolve a small effect. This is reported as "no detectable effect at $n \approx 200$", never as a confirmed null. The mechanism is not idle — the full runs record 63 backtracks on WebQSP and 199 on CWQ, dominated by evaluator-triggered ones (47 and 170). So the condition removes real activity whose accuracy consequence this experiment cannot measure.

5.12.3 Without the Verification Layer

The cleanest null in the table, and the most informative one. Discordant pairs are 0 against 1 on WebQSP and 1 against 0 on CWQ: across 398 paired questions, removing claim verification changed the correctness of exactly two. Hits@1 moves $+0.005$ and -0.005. F1 moves $+0.004$ and -0.009. Both $p = 1.00$.

This reproduces the development-set result of Section 5.7.1 exactly. It is also what Section 4.10 predicted by construction: on 77 of 80 development questions the layer certified the draft and changed nothing. The repair path never executed. A mechanism that mostly passes cannot move an accuracy metric.

The conclusion this licenses is narrower and sharper than "the verifier helps". It is the one Section 5.15.1 adopts: *claim verification does not measurably change how often AGR is right; it changes how often AGR is right for a reason it can exhibit*. Hits@1 cannot see that distinction. What makes it checkable is the supporting-triple output contract of Section 4.14. The evidence for the layer is that contract and the specimen analyses of Section 5.14.3. It is neither of the two aggregate results it would be natural to reach for instead. Table 5.9 is reached by the agentic baseline as well. That is why Section 5.15.1 attributes it to graph navigation. And the precision column of Table 5.4 does not move when the layer is removed, as the P column of Table 5.11 shows directly. It is not this row either. This row is reported at full length rather than omitted.

5.12.4 Embedding-Only Scoring

Removing the language-model term from the hybrid scorer costs -0.015 and -0.020 Hits@1 and -0.016 and -0.008 F1 — consistently negative on both datasets and both metrics, and consistently short of significance ($p = 0.66$ and $p = 0.48$).

Table 5.12: Component attribution, RQ3. ΔF1 is *ablated minus reference*, so positive means removing the component improved F1. Only the planner condition reaches significance.

Component	WebQSP ΔF1 (p)	CWQ ΔF1 (p)	Verdict
Planner	$+0.083$ (0.006)	-0.047 (0.088)	Stratum-dependent
Backtracking	-0.013 (0.73)	0.000 (1.00)	No detectable effect
Verification	$+0.004$ (1.00)	-0.009 (1.00)	No accuracy effect
α-blend	-0.016 (0.66)	-0.008 (0.48)	Directional, underpowered

Two corroborating details make the direction credible without over-claiming from it. The condition is cheaper than the reference on both datasets — 3,413 against 4,562 and 4,925 against 7,105 tokens, 4.4 against 6.2 and 5.9 against 9.1 calls. So the language-model scoring term is buying accuracy with tokens as designed. And it carries by far the most backtracks, 57 and 143 against the reference's 38 and 108, with the low_score trigger that τ governs roughly tripling in both datasets and accounting for about three quarters of the WebQSP increase and about half of CWQ's.

That second detail was originally read as the τ mechanism of Section 4.5.3 visible in the logs. The reading survives only partly. The signal τ thresholds is not computed identically in the two conditions, for the reason Section 4.5.3 records: at $\alpha = 1$ the two auxiliary maxima are never taken. So σ is a maximum over beam-selected, non-banned edges rather than over every candidate offered. After a backtrack has banned the best relation, σ falls in this condition, where it holds steady in the reference. That divergence produces additional low_score backtracks through an implementation asymmetry rather than through the scoring change under test. The two effects push the same way and cannot be separated without re-running the condition with the auxiliary terms restored — which would no longer be the frozen system.

The honest statement is therefore narrower than the original: removing the model term raises the backtrack count. An unknown but non-trivial share of that rise is an artefact of how σ is computed when the term is disabled. The backtrack count is consequently corroborating rather than load-bearing. That cuts against the ablation's own direction, since extra backtracks make the embedding-only condition look busier than the scoring change alone would. So the defect flatters the component under test rather than the null. Nothing in Table 5.12 depends on it: the accuracy and F1 deltas are computed from answers, not from backtracks. The primary evidence for $\alpha = 0.7$ remains the development sweep of Table 5.3, where the $\alpha = 1.0$ condition lost five hits on 80 questions. This ablation corroborates it at test scale rather than independently establishing it.

5.12.5 Paired Significance Testing and Statistical Power

One of four conditions reaches significance. That is the correct outcome to report, not a disappointing one. A correction for multiple comparisons was not among the pre-registered

decisions of Section 5.1.2. Applying one now would be a choice made after seeing the results. So the uncorrected p-values are reported as computed and read as descriptive rather than as a family-wise claim. Appendix E.16 gives the arithmetic behind the second qualification: at $n \approx 200$ McNemar's test needs a discordance pattern this experiment could not have produced for the effects it was looking for. So three conditions are reported as "no detectable effect at this sample size" rather than as no effect. The distinction is the whole content of Section 5.15.1's entry on underpowered ablations.

5.13 Error Analysis

Section 5.7 measured how often each system fails. This analysis reads the failures and names the mechanism of each one. It is the part of the work no script produced: 259 failure packets — question, gold, answer, plan, explored relations with their scores, backtrack reasons, verifier outcome — read and labelled by hand, one category and one sentence each.

The reading protocol was fixed before it began. For each packet, find where the trajectory *first* left the correct path, and label that point rather than the last visible symptom. When torn, take the earliest plausible cause and record the alternative. That rule is what keeps a run that mis-decomposes, then explores the wrong relation, then hedges from being counted three times under three categories.

A note on the identifiers used throughout this analysis, because they are meant to be looked up. A ComplexWebQuestions identifier is the WebQSP identifier it was derived from followed by a thirty-two character hash. The stem alone does not identify a question: within these samples one stem covers up to twelve distinct CWQ questions. Thirteen stems are themselves complete WebQSP identifiers naming something else entirely. Cases are therefore cited as stem plus the first eight hash characters. That combination is unique within both samples. Appendix C lists every identifier in full.

5.13.1 Annotation Protocol and the Failure Taxonomy

Ten categories carry the census, each naming a distinct architectural locus rather than a symptom, so that a count against a category points at the component that produced it. Appendix D.11 gives the schema in full, together with the open subtype vocabulary beneath it — which drives nothing and exists so that a mechanism can be named before it is understood — and the one naming decision that was got wrong and corrected.

Population, Not Sample The census is a *population*, not a sample: every question the full system did not answer correctly was read and labelled. Three sets are set aside before it is taken, of which only the adjudicated benchmark defects of Section 5.2.4 remove questions from the

Table 5.13: The merged failure histogram, $n = 259$. Wrong and hedge are kept separate. Counts come from scripts/synthesize_census.py over the three label sources of Table D.4. The table body is generated from results/phase4/thesis_numbers.json.

	WebQSP		CWQ		
Category	**wrong**	**hedge**	**wrong**	**hedge**	**Total**
relation_selection	20	6	20	19	**65**
composite_claim	1	—	23	23	**47**
kg_gap	8	4	8	24	**44**
decomposition_error	15	11	10	2	**38**
answer_selection	1	1	8	5	**15**
echo	5	2	4	2	**13**
ambiguous_question	1	1	3	5	**10**
premature_termination	1	4	1	2	**8**
gold_noise	1	—	3	3	**7**
other	—	1	—	5	**6**
verifier_fn	—	3	1	1	**5**
verifier_fp	—	—	—	1	**1**
Total	**53**	**33**	**81**	**92**	**259**

analysis altogether. The other two are of questions that were never failures. Appendix D.8 reconciles the three, which overlap, and shows how the 259 read failures follow. That the census is complete rather than sampled is what licenses the category counts below being read as counts rather than as estimates.

5.13.2 Distribution of Failure Categories Across Datasets

The Shape Flip, and What It Is Not The two datasets fail differently, but not in the way a first reading of Table 5.13 suggests. relation_selection sits at the top on *both* — 26 of WebQSP's 86, level there with decomposition_error, and 39 of CWQ's 173, behind composite_claim alone. So it is not what distinguishes them. Picking the wrong edge is the base rate of graph navigation. It is dataset-independent.

What distinguishes them is a pair of categories that are nearly absent from one and dominant in the other. composite_claim is 1 of 86 on WebQSP and 46 of 173 on CWQ. kg_gap is 12 and 32. In the other direction, decomposition_error is 26 on WebQSP — the tie noted above, the two together accounting for just over 60% of its failures — against 12 on CWQ.

That asymmetry is demonstrable rather than assertable, from two facts already established. First, the stratum distribution of Table 5.1: WebQSP is 64% one-hop with a four-question deep tail. So most of its questions have no composition to get wrong and no constraint to drop, while presenting the planner with exactly the questions that do not need decomposing. That mismatch is also the significant ablation result of Section 5.12, appearing here as a category count. Second, the expressiveness limits certified in Section 3.3.3: the environment carries no date literals, no

Table 5.14: The entity-extraction bug. In each case the drafted answer text names every gold value correctly and the scored entity list holds only the sentence's grammatical subject.

Question ID	Gold	Named in text	Scored entities
WebQTest-1215	6 professions	all 6	[Stephen Covey]
WebQTest-704	6 professions	all 6	[Thor Heyerdahl]
WebQTrn-124_0782789f	4 films	all 4	[Angelina Jolie]

numeric literals, and no ordinals. CWQ's SPARQL templates invoke all three constantly, while WebQSP's naturally-occurring questions rarely do. The failure profile of a system on a dataset is, to this extent, a property of the dataset's construction.

Hedging Where It Should A second reading of Table 5.13 is worth stating as a positive result rather than only as a deficit. On CWQ, kg_gap skews three-to-one toward hedges — 24 against 8 — while on WebQSP it leans the other way, 4 against 8. When CWQ's numeric identifiers, date literals, and ordinal constraints are unreachable, the system much more often declines than asserts something wrong. That is the behaviour the anti-vacuity and grounding provisions of Section 4.11 were written to produce. It is visible in the one place where the environment guarantees the system cannot succeed.

5.14 The Failure Categories

5.14.1 Decomposition, Drafting, and Answer-Selection Errors

Three families share this section because they share a locus: everything downstream of retrieval and upstream of the answer. The reasoning is often complete and correct by the time these failures occur. That timing is what makes them both diagnostic and cheap to fix.

The Entity-Extraction Bug Nine of the 38 decomposition_error cases carry the subtype extraction_bug. They are the most actionable finding in the census. On profession-shaped and "what did X do" questions, the evaluator resolves every correct gold value. The drafted text states them verbatim. answer_entities then collapses to the sentence's grammatical *subject*. Table 5.14 gives three instances read from the raw traces.

Table 5.14 understates how complete the drafts are. For WebQTest-1215, "who was stephen r covey", gold lists Author, Manager, Writer, Consultant, Professor, and Motivational speaker. The drafted answer reads *"Stephen Covey was a Writer, Motivational speaker, Author, Consultant, Professor, and Manager"* — all six, correct, in a different order. answer_entities comes back as ['Stephen Covey']. The navigation found the answers. The verifier certified them. The prose

reports them. What fails is the step that turns answer text into the entity list, which consistently reaches for the subject rather than the predicate values.

The consequence has to be stated as a limitation on Section 5.7, not merely as a bug: scoring reads answer_entities. So every one of these cases is recorded as a wrong answer or a hedge in Table 5.4. The reported accuracy of this system on "list the professions" and "list what X did" question shapes is therefore depressed by an extraction defect independent of any reasoning quality. Nine cases across 259 is not enough to move the headline numbers appreciably. But the correct reading of them is that the measured figure is a floor for that question shape. Section 6.3 lists the fix among the repairs the census earned.

Right Candidate Retrieved, Wrong One Drafted answer_selection (15 cases, 13 of them on CWQ) is the extraction bug's more consequential sibling: the exact gold value demonstrably entered the resolved candidate set. The drafter chose something else. WebQTest-1555 asks what the parliament of Nepal is called. The evaluator resolves Parliament of Nepal itself. The draft names its two component chambers instead. WebQTrn-2784_abedc8ac asks which Tupac Shakur film Malcolm Campbell edited. The trace records resolved=['Nothing but Trouble'], the literal gold value satisfying both constraints. The draft substitutes Gang Related and then hedges on it.

One case in this family was originally filed as a verifier error and is worth the correction in full. It is the kind of mistake this taxonomy exists to prevent. WebQTrn-1294_a4b2006a was labelled verifier_fn on the theory that the verifier had checked a claim about a different candidate than the one the evaluator resolved. The run record refutes that: the draft itself reads "Marlon Brando also played in *Joy*". The verifier rejected exactly what the drafter asserted. Brando was not in *Joy*. De Niro was. The evaluator had resolved him. **The verifier was correct.** The error is upstream, in the answerer. The label was corrected to answer_selection. That case is also the clearest positive demonstration in the census of the verification layer doing its job. Section 5.14.3 opens by pointing back to it.

Failing to Commit: Six Cases of Hedging on a Verified Fact The other category was kept open for mechanisms the schema did not anticipate. All six of its cases turned out to be the same one. It is not in the schema because nothing predicted it: *the evaluator resolves the exact gold value, the verifier returns* grounded, *and the drafted text hedges anyway*. No rejection is involved. So it is not a verifier error at all.

WebQTest-689 states "Spanish Language was spoken in Spain" and then that the first language spoken in Spain could not be determined, in one answer. WebQTrn-125_003c6a04 resolves Memphis, matching gold exactly. The draft says the location could not be determined. WebQTrn-1392_d372995c names both of Eleanor Roosevelt's schools correctly and then declares the school undeterminable. WebQTest-989_4b6636a0 (Battle of Dunkirk), WebQTest-1923_a64ef0f5 (*The Last Song*), and WebQTrn-2721_2d207ed9 (Alois Hitler) are the same shape. In every case the

entity list comes back empty. The question is scored as a hedge.

The contrast with the extraction bug is exact and worth holding: there the text is right and the entity list is wrong; here the text contradicts a fact it has just stated. Six of six instances share the mechanism. That uniformity makes it a defect rather than a distribution.

Context-Stripping: When Decomposition Discards the Question Subtyping is thinner in this category than elsewhere. The shortfall is worth stating before the cases are read. decomposition_error holds 38 failures. The four named subtypes — extraction_bug (9), paraphrase_drift (9), context_stripping (3), and over_decomposition (1) — cover 22 of them, the other 16 being labelled at category level only. The mechanisms below are therefore the mechanisms of the 22, not of all 38. The subtype vocabulary was left open rather than stretched to cover the residue.

paraphrase_drift and over_decomposition are the expected decomposition failures — a rewritten sub-objective scoring worse against the target relation than the raw question text, or a redundant step resolving an already-unambiguous entity. WebQTest-1829 is the clean over-decomposition case: the plan splits "what type of religion did massachusetts have" into ['find Massachusetts', 'find the religion type associated with #1'], burns 14 of 16 calls on three evaluator backtracks, and hedges, while the undecomposed run finds the same fact in three calls with no backtracks.

context_stripping (3 cases) is the mechanism this analysis would feature if it could feature only one. It is a failure of decomposition that decomposition's own design cannot see: splitting a question into steps *removes* a qualifier that appears only in a later clause. So an early sub-objective resolves the wrong sense of an ambiguous term before the disambiguating information ever arrives. In WebQTest-1367, which asks where Glastonbury, England is, the plan's first step anchors on 'find Glastonbury' alone. The explorer resolves toward the identically-named Connecticut town. In WebQTrn-2615_48a6ac56, which asks what instruments the author of *Fela!* plays, the first sub-objective is scored with no mention of instruments and resolves the wrong sense of an ambiguous authorship relation, where the undecomposed run answers all five instruments correctly in fewer calls.

WebQTrn-2570_d63877a4 is the strongest specimen in the census. Its trace shows every link of the mechanism. The question — "This 33rd president was the nation's leader during WW2" — is a single entity under two descriptions. The planner emits ['find the 33rd president', "find the nation's leader during WW2"], with no #1 reference between them, modelling two descriptors of one entity as though they were a chain. Because sub-objective one is the active scoring context, the anchor World War II scores 0.128–0.133 — the noise floor — while presidency-hub relations score 0.46–0.54: decomposition actively *suppressed* the one anchor that could have discriminated. Objective one then never completes (objective_done: false at every round). So objective two never activates. The WW2 constraint is never used at any point in the search. The agent spends its depth budget wandering the presidency hub and drafts a plausible-sounding

president: Woodrow Wilson, the 28th, and nowhere near the war. The undecomposed run answers Harry S. Truman correctly in three calls. "WW2" stayed in the scoring context. Two further observations belong to that trace. No ordinal relation appears anywhere in it. So the "33rd" constraint played no part either — consistent with the environment's lack of ordinal encoding (Section 5.14.4). And the verifier returned grounded on the Wilson claim, which Section 5.14.3 takes up as a boundary case rather than a defect.

Three instances of one mechanism, two of them surfaced by the ablation contrast and one by ordinary reading, is what makes context-stripping a named finding rather than an anecdote. It is the concrete evidence behind the adaptive-gating proposal of Section 6.3. A fourth case belongs beside them and is filed as plain decomposition_error, since no subtype fits it: WebQTest-869, "where is ancient phoenician", whose planner rewrote the sole sub-objective as ['find ancient Phoenician']. So the explorer re-identified the topic instead of locating it. The run answered "Phoenicia" — the question's own subject, grounded, and useless, against an undecomposed run that answered "Ancient Phoenicia was in Lebanon" in three calls. What separates it from the three above is *what* the rewrite discarded: they drop a constraint arriving in a later clause, this one drops the interrogative, the word that says what is being asked for. The result lands in the graph as an echo (Section 5.14.5) — with the ask removed, the nearest well-grounded entity to a topic is the topic. Both losses are visible to a planner that compares the rewrite against the original question, which is why the gating proposal checks the rewrite for the interrogative as well as for redundancy.

5.14.2 Relation-Selection and Navigation Errors

relation_selection is the largest category at 65 and carries no subtypes, because the mechanism is uniform: a wrong relation was explored, or the right one was explored and abandoned. Its interest is not in the individual cases but in the fact that several of them are the *same* gap seen from different angles — one relation confusion reproduced across three unrelated questions, and, most sharply, a CWQ triplet reaching the same entity by three different routes, none of which ever tries the single relation that answers all three. Three independent entry points into the same resolver hitting the identical dead end is the cleanest evidence in the corpus that a relation gap is *systemic* rather than an artefact of one question's phrasing: it is a property of the scorer's ranking over that entity's relation set, not of the question.

premature_termination is small (8 cases) and splits into two mechanisms that must not be merged. Five carry the evaluator subtype and are the *same* systemic pattern — the correct relation is explored with a solid score, as high as 0.731, and the evaluator backtracks away from it and never returns, across entirely unrelated domains. A relation scoring 0.73 being explored and discarded is not a reasoning gap. It reads as a threshold or backtracking-policy defect. It is one of the most directly fixable findings in the census. The other three carry budget and are genuine

exhaustion. Appendix D.7 gives all thirteen with their scores.

5.14.3 Verifier Rejection and Acceptance Errors

Before the errors, the correct rejections. Section 5.14.1 narrates one in full: the verifier rejected a claim the drafter had fabricated about the wrong actor. The layer worked exactly as designed. The 52 questions on which the verifier fired unsupported are not a population of mistakes.

The two error polarities are, however, very unevenly evidenced — five cases against one. That imbalance is not a fact about the verifier. It is a fact about the instrumentation. It is the finding of this section.

Defining the Error Polarities The verifier's positive class is "the claim is supported". A *false positive* is therefore a claim wrongly **accepted**, and a *false negative* a claim wrongly **rejected**. Because the verifier's own outputs are labelled supported and unsupported, "positive" has no stable referent for a reader. The rest of this thesis uses the plain-English terms.

Wrongly Rejected, and Wrongly Accepted Five cases are wrongly rejected, all sharing one mechanism and all carrying the subtype structural_not_semantic: the claim is true and the answer correct. But no single triple states it in the phrasing the drafter used. So a transitively-true or differently-anchored fact is rejected, and the retry loses the answer. Three of the five had already resolved the exact gold answer before the rejection forced the retry. This is the limitation Section 4.15 describes by construction, now observed at census scale. Section 6.3 names the semantic-level check that would address it.

Exactly one clean specimen of wrongful acceptance exists in the whole census. WebQTrn-1597_8adc06ba asserts a Seth MacFarlane connection to *ThunderCats*. The verifier returns grounded. A direct Cypher query confirms no such edge exists in the environment. Its instructive companion is *not* a second defect. WebQTest-1620 asks when President Wilson was in office. The environment exposes inauguration events but no date literal. So the system answers with two inauguration mentions whose entities exist and whose edges are real. The structural check passed *correctly* while the answer is wrong. That contrast is the most useful point in this section: structural grounding is a claim about edge existence, not about answer adequacy. So a verifier that checks whether a claim is *in* the graph cannot, by construction, check whether it is *the answer*. Table 5.9's Tier-2 column measures that gap. Appendix D.6 gives all seven cases with their traces.

A Rate for One Polarity, and a Blank for the Other Three quantities are easy to conflate. Only one is the denominator this polarity needs. The verifier is invoked 828 times over the 800 questions, since a question that repairs is verified again afterwards. Its *first* verdict is

unsupported on 52 questions — 16 of 400 on WebQSP and 36 of 400 on CWQ — of which 13 end grounded after a repair, leaving 761 questions answered under a grounded verdict and 39 under an unsupported one. The five wrongly-rejected cases are drawn from the 52. So that polarity has a denominator in the question unit: roughly one rejected question in ten cost a correct answer. Even that is a lower bound, since a wrongful rejection that did not change the final answer never entered a census that reads only failures. The units matter because the run record stores the *first* verifier entry. So a question repaired into groundedness is filed under unsupported in the committed verifier_outcome field: reading that field as the outcome would put 13 questions on the wrong side of the ledger. Treating the 52 as firings would credit the verifier with 52 of 828 rather than the 52 of 800 the rate is computed against.

Wrongful acceptance has *no logged population at all*. Accepted claims are never persisted (Section 4.8). So the wrongly-accepted cases sit undifferentiated inside the 761 questions answered under a grounded verdict, with nothing marking them. The single specimen surfaced only because the ablation-discordance reading happened to walk that trajectory by hand. What can be given is the size of the space they hide in, since a blank invites the reading that the space is small. The verifier accepted 2,008 of the 2,110 claims it decomposed on test. Neither structural route matches on the relation or the direction. So every acceptance that did not come from the entailment check was made by a relation-blind test — Section 4.12.1 shows the record cannot say how many that is beyond the interval $[39, 2{,}008]$. The two polarities are therefore reported asymmetrically and deliberately: a rate for rejection, an explicit blank for acceptance, and no average of the two, since a combined "verifier error rate" computed from these numbers would be an artefact of what the logs happen to record. Section 5.15.1 treats this as the instrumentation gap it is. Section 6.3 gives the one-line fix.

5.14.4 Knowledge-Graph Gaps: Literals, Temporal Qualifiers, and Ordinals

kg_gap is the third-largest category at 44 and CWQ's single largest hedge category at 24. It invokes the unanswerable-in-environment class defined in Section 3.3.3. Its five subtypes cluster by mechanism. Those five account for 39 of the 44; Appendix D.9 gives them with their specimens. The remaining 5 carry no subtype, because the environment gap in each was specific enough that grouping it with anything else would have been an invention. They are counted in the category total throughout and are not silently dropped.

The distinction between the first four subtypes and the last matters for what follows from them. data_error is a gap in this particular import. ordinal, numeric_literal, date_literal, and temporal_qualifier are gaps in what a triple-and-traversal architecture can express at all, and no larger graph fixes them. They need an operator. That need is why Section 6.3 lists literal and ordinal support alongside the architectural items rather than among the repairs.

5.14.5 The Echo Attractor: Answering with the Topic or an Intermediate Entity

echo names a mechanism that recurs independently of the two mechanical categories above: the system resolves a real, well-grounded entity that sits *near* the correct answer in the graph's structure. The verifier passes it because it genuinely is grounded. The fact is true. It is not what was asked.

Three shapes appear. Their boundary is not always crisp. topic (6 cases) is confusion between entities sharing a name or a tight cluster: WebQTest-1707 asks who plays Lex Luthor on *Smallville* and answers "John Glover plays Lionel Luthor" — a different character in the same fictional cluster, with Lionel, Alexander, and Lucas Luthor variants all surfacing in the trace before the run gives up. granularity (7 cases) is the right entity family at the wrong level: WebQTest-86 answers "Kingdom of Denmark" where gold is "Denmark" — a genuinely distinct node, the formal sovereign-state entity rather than the common-name one. WebQTrn-1405_ced20d84 asks which school founded in 1636 President Kennedy attended, has Harvard College in its very first candidate list, and narrows to Harvard University instead — whose founding-date claim the verifier then correctly rejects. intermediate — stopping at a hop along the way rather than continuing to the target — has no rows of its own in this census. The reading pass filed those cases under granularity. The two are best treated as one family.

The throughline is what makes this a named mechanism rather than a residue category. None of these are cases where the system knew nothing. Every one resolves a real, verifiably-true fact through a real edge. The failure is entirely in *which* true fact gets surfaced — architecturally different from picking a wrong edge and from there being no edge at all. The right edge exists, is used, and lands one node away.

That is also why the echo attractor is invisible to any evaluation treating systems as independent. Section 5.14.6 shows all five systems of the main matrix falling into it together, on the same questions.

5.14.6 Benchmark Defects: Gold Noise and Ambiguous Questions

Two numbers must be kept apart. The consensus pass of Section 5.2.4 *flagged* 58 WebQSP and 47 CWQ questions where at least three of the five systems agreed on the same non-gold answer. Adjudication confirmed 22 and 19 as genuine defects. The gap between those figures is not measurement noise — it is the echo attractor of Section 5.14.5, operating across systems.

Read what the flagged consensus answers have in common. Five systems answer "rick scott" where gold lists his professions, "thor heyerdahl" where gold lists his occupations, "amelia earhart" where gold is Pilot and Writer — the topic entity returned as the answer. Five answer "belgium" where gold is the containing region, "dominican republic" where gold is Greater

Antilles, “san francisco giants” where gold is the 2014 World Series — the composition’s intermediate rather than its final hop. Five answer “maryland” for counties, “nevada” for Las Vegas, “united arab emirates” for Dubai — one level too coarse.

Cross-system consensus against gold is dominated not by label errors but by a shared attractor. That is a finding about evaluation methodology as much as about these systems. A policy of rescoring whenever three systems agree — which is a natural thing to want — would have silently converted this system’s most characteristic failure mode into apparent correctness. The manual adjudication step is what stands between the two. It is the reason the pass is reported with both numbers rather than one.

The Confirmed Defects Appendix D.5 reads the confirmed defects case by case — the twelve WebQSP and seventeen CWQ questions whose gold is wrong, the ten and two whose questions are underdetermined, and where each came from. Two patterns account for most of the CWQ set, both properties of template generation rather than of annotation care: a gold list flattened from a multi-valued relation, and an annotation query that traversed one relation too far.

5.15 Discussion

5.15.1 Findings Against RQ1, RQ2, and RQ3

RQ1 — does agentic navigation improve multi-hop factual accuracy? Yes. The multi-hop qualifier is doing real work. AGR leads every accuracy cell of Table 5.4. All eight paired comparisons reject, including both against the agentic baseline running on the same graph with the same model under the same budget. The stratum profile is the substance of the answer: on CWQ, AGR’s accuracy *rises* with hop count ($0.46 \rightarrow 0.55 \rightarrow 0.57$), the only system that ends above where it started, while Vector-RAG collapses from 0.33 to 0.10 on a context that cannot hold a chain at any radius.

Three boundaries are reported with it. Think-on-Graph hedges marginally less on CWQ, 22.5% against 23.0% — the one cell of Table 5.4 AGR does not lead, and the reason the claim is about the accuracy columns rather than the whole table. It is also better on CWQ’s one-hop stratum, as Vector-RAG is on WebQSP’s. The margin over it is located in the budget: where the shared cap lets it finish it is marginally ahead on both datasets. So the claim defended here is that guided search completes under a realistic budget where beam search does not (Section 5.8). And that comparison is against a systematically narrower baseline, since Think-on-Graph prunes to the published algorithm’s 40 relations and 20 neighbours against AGR’s 300 and 200, binding on roughly a third of its expansions and almost none of AGR’s (Section 5.4.3). The margin is a lower bound on what an equal-width reimplementation would show. The static graph baseline’s per-stratum decay is not offered as evidence here. Its retrieval radius confounds it (Section 5.4.2).

RQ2 — what does pre-generation verification contribute beyond navigation? The answer has three parts that a yes-or-no question would have merged, which is why it was not asked as one.

First, navigation eliminates structural hallucination. The verifier is not what does it. Both navigating systems assert zero ungrounded entities on WebQSP — over $1{,}235$ assertions for AGR and 832 for Think-on-Graph — against 27.1% for parametric answering on the same set. Any system that copies its answer entities out of triples it traversed inherits this. So attributing it to the verification layer would be wrong. Table 5.9 is the evidence that it would be wrong.

Second, the precision advantage belongs to the architecture. This experiment cannot award any of it to the layer. AGR's assertion precision exceeds Think-on-Graph's by 11.8 points on CWQ (67.9% against 56.1%) and its per-question precision by 0.13 and 0.08, at a lower hedge rate on WebQSP. So it is not bought with abstention. That margin is real. What it is not is evidence for the verification layer. The one condition that isolates the layer does not reproduce it. Removing the verifier moves per-question precision by $+0.001$ on WebQSP and -0.010 on CWQ (Table 5.11), a sign flip an order of magnitude below the gap it would have to account for. The mechanism cannot carry the attribution either: entity filtering runs only on the give-up path (Section 4.13.4), which 10 of 400 WebQSP and 29 of 400 CWQ questions entered. A filter touching 2.5% of a test set cannot produce a 0.13 precision margin on it. Two differences the ablation holds fixed remain live as explanations. Neither is the claim check — the groundedness filter on evaluator resolutions (Section 4.5.2), which is navigation rather than verification and runs in every ablation condition, and the drafting prompt's anti-vacuity and grounding provisions, which the baselines' plain prompt lacks (Section 5.4.4). Some share of the precision margin over the agentic baseline may be prompt rather than structure. This work cannot say how large that share is.

Third, verification shows no measurable Hits@1 effect. Two questions out of 398 change when it is removed. This is not a failure of the layer but a statement about what Hits@1 measures: a metric satisfied by one correct entity among several cannot see a mechanism that operates on the others. The layer's value proposition is that an answer arrives with the triples that support it (Section 4.14) — an auditability claim, not an accuracy claim. The honest formulation is that *claim verification does not change how often the system is right; it changes how often the system is right for a reason it can exhibit.*

RQ3 — which components contribute what, at what token cost? One of four reaches significance. The planner is stratum-dependent: removing it improves WebQSP by 0.083 F1 at $p = 0.006$ while saving 31% of tokens and 35% of calls, and costs CWQ 0.047 F1 at $p = 0.088$. That asymmetry, not a pooled average, is the result. It is a finding about how decomposition should be *applied* rather than about whether it works. Backtracking, verification, and the α-blend show no detectable effect at $n \approx 200$, for the different reasons Section 5.12 separates. On cost, the whole apparatus runs at half the language-model calls of the agentic baseline and never

exhausts the call budget that clips the baseline on 44% of CWQ questions.

The three answers do not point the same way. The chapter does not force them to. Navigation is where the accuracy comes from; verification is where the auditability comes from, the precision margin belonging to the architecture as a whole rather than to any one component this ablation could isolate; and decomposition, the component the literature treats as straightforwardly beneficial, is the one component this work can prove is sometimes harmful.

What Agency Buys, and What It Costs Agency buys the deep tail: on CWQ the accuracy of the agentic system *rises* across hop strata while every static system decays. It buys that at roughly an order of magnitude in tokens over one-shot retrieval. Against the other agentic system, it buys the same accuracy for half the language-model calls, never exhausting the call budget that clips the baseline on 44% of multi-hop questions — the difference between a guided search and a blind one, priced. What it costs, beyond tokens, is a longer failure surface. Every mechanism in this census except kg_gap and the benchmark defects is a failure the static baselines cannot have. They have no plan to mis-decompose, no frontier to abandon, and no draft to check. decomposition_error at 38 cases is the clearest instance: a component added to help contributed a failure family of its own. Section 5.12 shows it is a net negative on the shallower benchmark. Agency converts a retrieval problem into a control problem. Control problems fail in more ways.

When Verification Helps and When It Over-Hedges The verification layer's contribution is auditability. Its cost is recall. Both directions are visible in this census at case level — which is the level at which they are established, the aggregate effect of removing the layer being indistinguishable from zero (Section 5.15.1). It helps when a draft asserts something the traversal never supported: the De Niro case of Section 5.14.1, where a fabricated actor was rejected before it could be emitted, and the Vicksburg-president case of Section 5.14.6, where declining to assert was the epistemically correct act and the metric scored it as a loss. It over-hedges when a true claim has no single edge stating it in the drafted phrasing — the five wrongly-rejected cases of Section 5.14.3, three of which had already resolved the exact gold answer before the rejection forced a retry that lost it. That is the same trade the development set measured at Section 5.7.1 and the same one the ablation could not detect at $n \approx 200$. So the claim is narrow: verification does not make this system more accurate. It makes the answers it does give attributable, at a small and measurable cost in recall. It provides no protection at all against the failure mode this census finds most often — an answer that is grounded, connected, true, and not what was asked.

Threats to Validity **The environment is friendlier than the knowledge base it stands in for. This bounds every absolute number.** It is the union of per-question subgraphs the RoG distribution pre-extracted *so as to contain their own question's answer* (Section 3.1.3). Taking the union restores distractor mass but cannot restore what was never extracted. That gap is why

answer reachability is certified at 97.3% and 99.7% (Section 3.3) where a BFS extraction from full Freebase under the same hop bound would not reach those rates. Accuracies measured here are therefore *not* comparable to evaluations over full Freebase. The ceilings should be read as part of the result. The mitigation is real but narrow: every system navigates the identical environment. So the *relative* ordering is sound even where the absolute level is optimistic. Every claim in Section 5.15.1 is relative for this reason.

An identical environment is not identical access to it. One baseline sees less of it. The agentic baseline truncates candidate sets at 40 relations and 20 neighbours against this system's 300 and 200. Those are the reimplemented algorithm's own pruning widths. But the narrower cut binds on about a third of its expansions against effectively none of this system's (Section 5.4.3). The comparison holds the graph constant and does not hold the view of it constant.

Every reported number comes from one run per condition. Run-to-run variance is unmeasured. The confidence intervals and paired tests resample questions. So they bound the sampling variance of the question draw and nothing else. Each system's answer set enters them as fixed. The backbone is not deterministic at the trajectory level even at temperature 0.0 — Section 5.3 measures the frozen model at 8 of 12 probes, about one trajectory in three diverging on a rerun. A divergent trajectory can change an answer, a hedge, a token count and a call count at once. That figure is itself only partly auditable. Only the first of four rounds survives as a committed artifact. So a reader who declines to take the other three on trust should read the stability figure as unquantified rather than as 67%. That substitution makes this threat larger rather than smaller. The margins defended here are large relative to their intervals. The paired tests are computed per question. Both make a rank reversal from decoding noise alone unlikely. But unlikely is the strongest available statement. The experiment that would bound it was not run. At $11.13 against a $15–20 ceiling it was affordable. So this is a gap in the design rather than a budget constraint. A repeated matrix reporting across-run spread is the cheapest correction available (Section 6.3).

Entities are keyed by name, so homonyms are merged. Distinct real-world entities sharing a surface form collapse into one node. Two consequences run in opposite directions. Neither is measured here: a merged node can manufacture a path between things that are not connected, inflating reachability and letting a wrong answer verify as grounded. The same merge can destroy a distinction the question depends on. This is inherent to name-keying rather than a defect of the construction. But it means structural groundedness is a claim about *this* graph's node identity, not about the world.

The instrumentation gap is the most serious of the remaining items. The verifier logs only the claims it rejects, never the ones it accepts with their matching triples. So wrongful acceptance is not self-diagnosing: the single specimen in this census surfaced by manual trace reconstruction. There is no way to know how many others sit inside the 761 grounded questions. One polarity is reported at census scale, the other anecdotally. That asymmetry is a property of the logging

decision rather than of the verifier. Section 6.3 gives the fix.

The judge fell short of its pre-registered bar. Cohen's $\kappa = 0.6995$ against a threshold of 0.7 (Section 5.5.5). Agreement is substantial by conventional reading. The miss is 0.0005. But the Tier-2 numbers are reported as corroborating rather than decisive. The judge also runs on the same backbone as the systems it judges.

Topic entities are given, not linked. Both graph-navigating systems seed from the dataset's annotated topic entities. This isolates navigation from entity linking, which is the intent. But the reported accuracies assume a linking step that a deployed system would have to perform and that would introduce errors of its own.

Ablations are underpowered. Three of four conditions detect nothing at $n \approx 200$. "No detectable effect" is not "no effect" (Section 5.12).

One relabelling was made after seeing the result it affects. A development question originally classified as mediator-heavy was reclassified as unanswerable-in-environment after its outcome was known. Section 4.10 flags it in place. The aggregates were not rescored.

The reported accuracy has an unmeasured floor. The extraction bug of Section 5.14.1 converts correct reasoning into a wrong entity list on a specific question shape. So the measured accuracy on that shape is a lower bound rather than an estimate.

Benchmark defects were adjudicated by the author. The exclusion set of Section 5.2.4 rests on single-annotator judgements made with the consensus signal visible. The scope limits are recorded there. The accuracy tables are not rescored. The defects run in both directions.

Chapter 6

Conclusion

6.1 Summary of Contributions

This thesis asked what an agent gains by navigating a knowledge graph rather than retrieving from it, and what a claim-level check against the traversed subgraph adds on top. The answers are not uniform. The value of the work lies partly in where they diverge.

Section 5.7 reported the measurements. AGR leads every accuracy cell of the main matrix — the agentic baseline hedges marginally less on CWQ. Section 5.15.1 reports that as the one cell it does not lead. Every one of the eight paired comparisons rejects, including both against a reimplementation of the agentic baseline running on the same graph with the same model under the same budget. That last margin is located rather than merely claimed: on the questions where the shared cap lets the baseline finish, the baseline is slightly ahead. The whole margin comes from the 29% and 44% of questions where the cap truncates it (Section 5.8). The finding is therefore about affordability rather than about resolution quality — beam search cannot complete under a budget a deployed system would impose. Guided search can. On ComplexWebQuestions AGR is the only system whose accuracy *rises* with hop count. Every other system ends below where it started — the strongest available form of the multi-hop claim. The agentic baseline is the one that does not simply decay: it falls and then partially recovers. That recovery is weaker than a rise and is reported as such in Section 5.9. AGR runs at half the language-model calls of the agentic baseline and never exhausts the call budget that clips that baseline on 44% of multi-hop questions. Structural hallucination is eliminated: over both datasets, zero ungrounded assertions in $1{,}709$ asserted entities, against 22.1% of the $1{,}001$ the parametric control asserts. And three of four ablations detect nothing. That null result is reported at the same length as the one that does.

Section 5.13 read all 259 remaining failures and named each mechanism. The findings that generalise beyond this system are three. *Context-stripping*: decomposing a question can discard a qualifier that appears only in a later clause. So an early sub-objective resolves the wrong sense

of an ambiguous term before the disambiguating information arrives. In the strongest specimen, decomposition actively suppressed the one anchor that could have discriminated, driving its score to the noise floor. *The echo attractor*: the most common failure of a graph-navigating system is not to invent an entity but to surface a real, well-grounded, verifiably-true entity that sits one node away from the answer. The verifier passes it because it genuinely is grounded. And *shared attractors break consensus-based evaluation*: five independent systems converge on the same wrong answer often enough that a policy of rescoring on majority agreement would have converted this failure mode into apparent correctness.

Four contributions follow from that body of work. Section 1.6 claimed six. The other two are stated above rather than repeated as items here — the echo attractor is the second of the three findings just given. The pre-registered evaluation protocol is reported in the Section 5.1 paragraph, including the threshold it missed.

The framework and its verification layer. An agentic KGQA system that returns its answer paired with the triples supporting it, built so that the supporting evidence is a byproduct of the control flow rather than a post-hoc reconstruction.

A component-level ablation. Four single-deviation conditions with paired significance testing, closing the attribution gap of Section 2.5.1 for this architecture — and finding that only one of the four components has a detectable accuracy effect.

Stratum-dependent decomposition. The literature treats decomposition as straightforwardly beneficial. It is not: removing the planner *improves* WebQSP accuracy by 0.083 F1 at $p = 0.006$ while cutting tokens by 31%, and trends toward harming CWQ. The asymmetry, not the pooled average, is the result. It says that decomposition should be gated on question structure rather than applied unconditionally.

Quantified benchmark defect rates. Fifty-seven questions across the two test samples where the benchmark, not the system, needed correcting — with the two evidence signatures that distinguish a world-fact error from an annotation-pipeline error, and the provenance argument that explains where one class of them comes from.

6.2 Limitations

Section 5.15.1 states the threats to validity in full, against the measurements they bear on. Four of them constrain what may be concluded here more than the rest, and are repeated for that reason.

Wrongful acceptance is unmeasured. The verifier logs the claims it rejects and not the ones it accepts. So one polarity of its error is reported at census scale. The other is reported from a single manually reconstructed specimen. The exposed population is bounded only as $[39, 2{,}008]$ of $2{,}008$ accepted claims (Section 4.15). Nothing measured here narrows it.

Structural grounding is not answer adequacy. Both navigating systems assert zero ungrounded entities on WebQSP. That says the answers are made of the graph's own material. It does not say they answer the question. The echo attractor (Section 5.14.5) is the counter-example. It is the failure this census finds most often.

The evaluation is single-environment, single-backbone, and single-run. Absolute levels are bounded by an environment built from answer-containing subgraphs. Run-to-run variance is unmeasured against a backbone that repeats a trajectory on about two probes in three.

The baseline comparisons are bounded in the baselines' favour, and against them. The agentic baseline prunes from a narrower candidate set than AGR. The static one retrieves a single logical hop with an unordered fanout cap. So the reported margins are lower bounds on what equal-width, equal-radius reimplementations would show (Section 6.3.5). Entity linking is assumed throughout, topic entities being given by the benchmarks.

6.3 Future Work

The census makes it possible to separate proposals backed by a counted, repeated pattern from proposals backed by an argument. Both are listed. They are not the same kind of claim.

6.3.1 Repairs the Census Earned

Three defects each recur with a uniform mechanism across unrelated questions. Each is small. *Entity extraction* (nine instances): the step converting drafted answer text into the scored entity list defaults to the sentence's grammatical subject rather than its predicate values. Fixing it requires telling the claim decomposer which argument of the drafted sentence is the answer — an instruction, not an architecture. *Evaluator abandonment* (five instances): relations scoring as high as 0.73 are explored, backtracked away from, and never revisited, across four unrelated domains. That reads as a threshold or backtracking-policy defect. It would be diagnosed by logging the evaluator's decision against the score of the relation it discards. *Drafting commitment* (six instances, all six of the other category): when the evaluator has resolved the exact gold value and the verifier has returned grounded, the drafter should commit to it rather than stating the fact and declaring it undeterminable in the same sentence.

6.3.2 Logging Accepted Claims to Make Wrongful Acceptance Measurable

Persist each accepted claim together with the triples that matched it. This converts the wrongly-accepted class from an anecdote into a rate with a denominator. It also makes grounded-but-wrong

answers self-diagnosing from the logs rather than by manual trace reconstruction. The same change closes a second gap: the run record currently stores the *number* of supporting triples rather than the triples (Section 4.14). So the output contract holds inside the run. It cannot be checked afterwards from any committed file. It is the highest-value change in this list relative to its cost. It is a logging change.

6.3.3 Operators the Failure Categories Specify

Three additions are specified by categories rather than motivated by them. *Adaptive decomposition gating*: a decomposition step should be gated on whether the rewritten sub-objective preserves the question's actual interrogative — the Phoenician case of Section 5.14.1 lost the word "where" entirely — on whether the step resolves an entity that was already unambiguous, and on whether the sub-objectives reference each other at all, since ones that do not are descriptors of a single entity rather than a chain. *A set-intersection operator*: composite_claim is 47 cases and CWQ's largest category at 46 of them. Its no_set_intersection subtype names the gap precisely — the architecture has no AND-join primitive. So a "find the overlap" sub-objective degrades into another relation-scoring pass over an incomplete first-hop set. *Ordinal and literal-value support*: kg_gap is 44 cases and CWQ's largest hedge category at 24. Two of its subtypes need a minimum-or-maximum-by-value operation and a literal-extraction path rather than a bigger graph.

6.3.4 Semantic-Level Verification

All five wrongly-rejected cases share one mechanism: the structural check requires a single edge asserting the drafted claim. A transitively-true multi-hop fact does not present one. A check that accepts a bounded relation chain expressing the claim would recover those answers. The relation-path formulation of RoG [9] is the natural source for what such a chain may look like. Using it as an acceptance criterion rather than as a retrieval plan is the change proposed. The Tier-2 judge of Section 5.5.3 is a prototype of exactly this judgement. That resemblance suggests folding a bounded form of it into the loop. Its κ of 0.6995 is also a caution about how reliable such a check would be.

6.3.5 Measurements This Thesis Could Not Make

Path fidelity is the third evaluation criterion of the approved proposal, deferred for the reason given in Section 5.1.2: it needs the benchmarks' gold SPARQL relation chains. The distribution used here does not carry them. Reconstructing them would measure whether the system reaches the right answer *by the right path* — a question the groundedness metrics can only approximate, and one that bears directly on the echo attractor.

Two baseline bounds are cheap to lift. Both are re-runs rather than redesigns. *Equalise the candidate widths*: re-run the agentic baseline at AGR's 300 relations and 200 neighbours, leaving beam width, depth, budget, prompts, and scoring untouched. That decides whether the unclipped-subset comparison of Section 5.8 measured search strategy or candidate supply. Both outcomes are usable — though it will also raise the clip rate. Wider candidate sets make each pruning call dearer. So the split must be reported rather than the aggregate. *Widen the static retriever's radius, and order its fanout*: re-run it with neighbours expanded a second time, everything else frozen, and report the same strata, so that its collapse on deep CWQ questions can be attributed to the paradigm rather than to the particular radius chosen. The same pass should impose an ordering on the fanout cap, so that the 100 edges kept per topic entity are a stated selection rather than an arbitrary one (Section 5.4.2). Until then those two limits are confounded. Only their sum is measured.

6.3.6 Beyond This Environment

Four directions extend the work rather than repairing it. *LLM-constructed knowledge graphs* as a controlled comparison would separate what the framework contributes from what a curated environment contributes, by holding the agent fixed while varying the substrate's provenance and noise profile. *Rollout-based value estimation* would supply what Section 2.2.4 explicitly disclaims: the current scorer evaluates each candidate once and never revises it in light of what is found later. Genuine tree search over this same tool API is a well-defined next system rather than a rebranding of this one. *Temporal and multi-source environments* address the failure family that recurs throughout Section 5.13 — questions with a time qualifier the graph cannot express — and raise the question of what a claim check means when sources disagree. And *transfer to high-stakes domains* is where the auditability property, rather than the accuracy, is the point: an answer paired with the triples supporting it is worth more in a setting where a reader must be able to check it than in one where the reader only wants the answer.

6.4 Closing Remark

The result this work would most want carried forward is not the accuracy margin. It is the shape of the gap that remains. A system that traverses a real graph and copies its answers out of real triples can be made to assert nothing ungrounded — that problem is solved. It is solved by the navigation paradigm rather than by any checking layer built on top of it. What remains is an answer that is grounded, connected, and true, and is not the answer to the question that was asked. This thesis's census found that failure more often than any other mechanism. Fact verification against a knowledge graph turns out to be the easier half of the problem; *relevance* verification is the half still open.

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